

INTRINSIC MUON: SPECTRAL OPTIMIZATION ON RIEMANNIAN MATRIX MANIFOLDS

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ABSTRACT. Muon and related norm-constrained matrix optimizers have become central to large-scale learning problems. They are formulated as a linear maximization oracle (LMO) over an ambient matrix-norm ball in unconstrained Euclidean space. However, these do not generalize cleanly to manifold-valued parameters such as low-rank factorizations, orthogonality constraints, or symmetric positive definite (SPD) matrices. Naively restricting the Muon LMO to the tangent space (i) breaks quotient symmetries and (ii) couples the tangent-space constraint with an ambient norm bound, thereby obstructing closed-form solutions on various manifolds of interest. We resolve both issues with a single observation: every Riemannian metric canonically lifts a unitarily invariant Euclidean norm to an intrinsic norm on each tangent space, and the resulting intrinsic norm constrained LMO is symmetry preserving. Building on this, we introduce intrinsic Muon (iMuon), a unified framework that yields closed-form updates on the fixed-rank, SPD, Stiefel, and Grassmann manifolds for any unitarily invariant norm, including the spectral, Frobenius, and nuclear norms. We establish convergence guarantees for both deterministic and stochastic iMuon with rate constants that depend only on the manifold dimension. Notably, on the fixed-rank manifold this constant depends only on the rank, making the rate independent of factor conditioning and removing the runtime factor-rescaling required by prior work. Experiments on LoRA finetuning of LLMs, image classification, and subspace learning illustrate the efficacy of the proposed approach.

1. INTRODUCTION

A growing family of matrix-variate optimizers is unified by a single design principle: at each step, select the update direction that maximizes the inner product with the (stochastic) gradient over a matrix-norm ball, a linear maximization oracle (LMO) [6, 48, 65]. The choice of norm determines the optimizer: the Frobenius norm recovers SGD, the spectral norm yields Muon [6, 27], and a combined spectral-nuclear constraint gives NuMuon [43]. This perspective has driven recent advances in LLM pretraining [26, 42, 61, 70] and low-rank adaptation (LoRA) fine-tuning [17, 23, 46, 52]. Despite their empirical success, the design of such LMO-based optimizers has been mostly confined to unconstrained Euclidean spaces.

However, many modern representation-learning problems are intrinsically manifold-valued: LoRA fine-tuning operates on the fixed-rank manifold [41, 69], orthogonality constraints define the Stiefel manifold, and robust covariance estimation lives on the SPD manifold. Extending norm-constrained matrix optimizers to such Riemannian settings remains relatively unexplored [10, 17]. The seemingly natural fix of restricting the LMO to the tangent space runs into two fundamental obstacles:

- **(O1) Symmetry breaking on quotient manifolds.** The Euclidean spectral norm is not invariant under the symmetry group of a quotient manifold (i.e., manifold arising out of symmetries). For example, on the fixed-rank manifold, equivalent factorizations $X = BA$ that share the same product induce different factor-wise Muon updates, so the update magnitude depends on an arbitrary representative [26].
- **(O2) Coupling of the tangent space and norm constraints obstructs a closed-form solution.** Restricting the spectral-norm LMO to the tangent space couples two constraints, tangent space

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membership and an ambient norm bound, whose joint enforcement generally precludes the Euclidean polar-factor solution and admits no closed form on most manifolds of interest.

Existing literature addresses these issues only partially. Spectron [26] bounds the ambient update by an iterate-dependent factor rescaling computed dynamically. Riemannion [10] relies on an approximate tangent space orthogonalization. Manifold constrained steepest descent (MCSD) [68] drops the tangent constraint and projects back to the manifold instead. However, none of the existing approaches resolves (O1) and (O2) jointly.

In this paper, we propose intrinsic Muon (**iMuon**), a principled and unified framework for spectral, and, more generally, unitarily invariant norm-constrained optimization on Riemannian matrix manifolds (whose points can be represented as matrices) that resolves both (O1) and (O2) simultaneously. The key observation is that the Riemannian metric canonically induces an *intrinsic* norm on each tangent space, with the Frobenius case recovering the standard Riemannian norm and the spectral case giving a manifold-aware analogue of Muon. Our main contributions are:

- **Intrinsic LMO with closed-form solutions on four manifolds (Sections 3 and 4).** We identify two structural properties of the metric—a left-right product form (Proposition 3.1) and singular value invariance of the scaled tangent space (Proposition 3.3)—that together resolve (O1) and (O2) and reduce the intrinsic LMO to a standard Euclidean problem. This yields exact closed-form solutions on the fixed-rank, SPD, Stiefel, and Grassmann manifolds, for various norms including the spectral, Frobenius, and nuclear norms.
- **Unifying viewpoint.** Beyond enabling efficient computation, instantiating these intrinsic LMOs across different norms and geometries provides a generic approach that unifies established algorithms and yields new ones. In the Euclidean special case [6, 27], our proposed LMO recovers standard SGD, Muon, and NuMuon [43] under different norm ball constraints. On the fixed-rank manifold equipped with quotient geometry, the Frobenius instance recovers ScaledGD [40, 69], while the spectral instance is, to our knowledge, the first quotient-Muon optimizer with an exact closed-form tangent space update.
- **Convergence rate (Section 5).** For both deterministic and stochastic iMuon, we prove an $O(1/\sqrt{T})$ rate in the dual intrinsic norm whose constant depends solely on the manifold dimensions (and not on the iterates). On the fixed-rank manifold with the spectral norm, this radius depends only on the rank, yielding a rate that is *independent of factor conditioning*, and, for example, obviates the runtime factor rescaling of Janson et al. [26].
- **Empirical validation (Section 6).** Experiments across LoRA fine-tuning, SPD covariance estimation, and subspace learning confirm the merits of the intrinsic approach in various settings.

2. BACKGROUND AND RELATED WORK

We briefly review concepts of Riemannian geometry for matrix manifolds (Section 2.1), Muon and norm-constrained Euclidean LMO (Section 2.2), and recent attempts at generalizing the latter to low-rank and matrix manifold settings (Section 2.3). Additional background and related works are discussed in Sections I and J and detailed comparisons are provided in Section H.

2.1. Riemannian optimization on matrix manifolds. We consider the problem $\min_{x \in \mathcal{M}} f(x)$ on a Riemannian matrix manifold $\mathcal{M}$ with smooth $f : \mathcal{M} \rightarrow \mathbb{R}$. The metric $g_x : T_x \mathcal{M} \times T_x \mathcal{M} \rightarrow \mathbb{R}$ on the tangent space $T_x \mathcal{M}$ at $x \in \mathcal{M}$ admits a self-adjoint, positive-definite operator form $g_x(\xi, \zeta) = \langle G_x \xi, \zeta \rangle$, with induced Riemannian norm $\|\xi\|_x = \|G_x^{1/2} \xi\|_F$, where $\|\cdot\|_F$ is the Frobenius norm. The Riemannian gradient $\text{grad } f(x) \in T_x \mathcal{M}$ satisfies $g_x(\text{grad } f, \zeta) = \langle \nabla f(x), \zeta \rangle$ for all $\zeta \in T_x \mathcal{M}$. To update parameters, a retraction operator $R_x : T_x \mathcal{M} \rightarrow \mathcal{M}$ generalizes the Euclidean step $x + \eta \xi$ with step size η . On an *embedded submanifold* $\mathcal{M} \subset \mathbb{R}^{p \times q}$, $T_x \mathcal{M}$ is a linear subspace of the ambient space. On a *quotient manifold* $\mathcal{M} = \overline{\mathcal{M}}/\mathcal{G}$, obtained by symmetry invariance $\mathcal{G}$ in the total space $\overline{\mathcal{M}}$, the metric must be $\mathcal{G}$ -invariant for the Riemannian gradient to be well-defined independently of the representative. Please refer to Appendix J and standard references [1, 11] for more details.

2.2. Muon and norm-constrained Euclidean optimizers. For an unconstrained matrix parameter $X \in \mathbb{R}^{p \times q}$ and gradient (or momentum) $M \in \mathbb{R}^{p \times q}$, Muon [6, 13, 27] has the update $X_+ = X - \eta O$, where O is

the solution to:

$$(1) \quad \text{Spectral-norm LMO} \quad O = \arg \max_{\|O\|_2 \leq \tau} \langle O, M \rangle = \tau \text{Ortho}(M),$$

with $\text{Ortho}(M) = UV^\top$ being the polar factor of $M = U\Sigma V^\top$, i.e., the singular value decomposition. Bernstein and Newhouse [6] formalize (1) as steepest descent under the spectral norm. Subsequent works generalize the LMO viewpoint to broader operator norms [48], modular norms across the network [7, 34], additional nuclear-norm budget [43], per-layer and product norms [15], and analyze convergence, preconditioning, and Newton–Schulz approximations of $\text{Ortho}(\cdot)$ [3, 31, 37, 50, 51, 53, 63]. All of these methods reside in unconstrained Euclidean space and our intrinsic LMO (4) (defined later) reduces to them in that special case (Section G).

2.3. Muon variants for structured and manifold settings. A natural attempt to use (1) on a Riemannian manifold $\mathcal{M}$ is to additionally restrict the search to the tangent space $T_x\mathcal{M}$ at $x \in \mathcal{M}$:

$$(2) \quad \xi_{\text{naive}}^* = \arg \max_{\xi \in T_x\mathcal{M}, \|\xi\|_2 \leq \tau} \langle \xi, \nabla f(x) \rangle.$$

This naive lift is the implicit template behind several recent manifold/structured Muon proposals but it inherits the two challenges (O1) and (O2) introduced in Section 1.

LoRA and the fixed-rank manifold. For LoRA-style fine-tuning $X = BA$ with $B \in \mathbb{R}^{m \times r}$, $A \in \mathbb{R}^{r \times n}$, X lies on the fixed-rank manifold $\mathcal{M}_r$, equipped with the $\text{GL}(r)$ symmetry $(B, A) \sim (BN^{-1}, NA)$, where $N \in \text{GL}(r)$ the set of $r \times r$ non-singular matrices. Existing works [40, 55, 69] propose Riemannian preconditioned gradient descent for such settings, which corresponds to the Frobenius instance of our framework on $\mathcal{M}_r$. Recently, [26, 29] study the following factor-wise Muon problem, which independently solves a Euclidean spectral-norm LMO on each factor, i.e.,

$$(3) \quad \dot{B}^* = \arg \max_{\|\dot{B}\|_2 \leq \tau} \langle \dot{B}, \nabla_X f A^\top \rangle, \quad \dot{A}^* = \arg \max_{\|\dot{A}\|_2 \leq \tau} \langle \dot{A}, B^\top \nabla_X f \rangle.$$

Solving (3) requires applying $\text{Ortho}(\cdot)$ to each factor’s gradient (or momentum). However, the resulting ambient update is not invariant under the $\text{GL}(r)$ symmetry, exhibiting (O1). Kang et al. [29] prove a continuous-time convergence for the momentum-free limit, conditional on factor-norm boundedness. On the other hand, [26] bounds the ambient spectral norm of the update by an iterate-dependent rescaling $\rho = \eta/(\|B\|_2 + \|A\|_2 + 1)$ computed at runtime. While this circumvents (O1) numerically, our approach (Section 3) handles (O1) algebraically. Please also refer to Remark 5.4 and Section H.1 for more details.

Embedded manifolds and the coupled tangent space LMO. Restricting (1) to $T_x\mathcal{M}$ couples the spectral-norm constraint to a tangent-space constraint. The resulting LMO (2) generally has no closed form, which is (O2) described earlier. Riemannion [10] approximately solves the tangent-constrained spectral LMO on the embedded fixed-rank manifold, while MCSD/SPeL [68] drops the tangent-space constraint on the Stiefel manifold and projects the ambient solution onto the Stiefel manifold. We discuss these update rules in detail in Sections H.1 and H.3. In contrast, our intrinsic LMO (next section) yields closed-form updates for the fixed-rank and Stiefel manifolds.

3. THE INTRINSIC LMO

We now generalize norm-constrained Euclidean LMOs, e.g. (1), to the geometry-aware manifold setting. We begin by discussing the key idea that the Riemannian metric itself canonically induces the appropriate notion of matrix norm on the tangent space. Later, we propose a novel family of intrinsic norm constrained LMOs on the tangent space of manifolds and analyze it with respect to the challenges (O1) and (O2).

Intrinsic norms on the tangent space. The metric operator $G_x \succ 0$ induces the Riemannian norm $\|\xi\|_x := \|G_x^{1/2}\xi\|_F$, where $\xi \in T_x\mathcal{M}$ and $\|\cdot\|_F$ is the Frobenius norm. This suggests a canonical extension from the Frobenius norm to any unitarily invariant norm φ : for $\xi \in T_x\mathcal{M}$, we define the intrinsic φ -norm by $\|\xi\|_{\varphi,x} := \varphi(G_x^{1/2}\xi)$. In particular, choosing $\varphi = \|\cdot\|_2$, $\|\cdot\|_F$, or $\|\cdot\|_*$ yields the intrinsic spectral, Frobenius, and nuclear norms, respectively. We note that $\|\cdot\|_{\varphi,x}$ is intrinsic in the sense that it is determined entirely by the Riemannian metric. It reduces to the standard matrix norm in the Euclidean case $G_x = I$, while for $\varphi = \|\cdot\|_F$ it recovers the Riemannian norm itself.

3.1. Proposed LMO formulation. Given the intrinsic norm $\|\cdot\|_{\varphi,x}$ introduced above, we propose the update direction at $x \in \mathcal{M}$ for optimizing $\min_{x \in \mathcal{M}} f(x)$ through the following LMO:

$$(4) \quad \xi^* = \arg \max_{\xi \in T_x \mathcal{M}, \varphi(G_x^{1/2} \xi) \leq \tau} g_x(\xi, \text{grad } f),$$

where φ is any unitarily invariant norm and g_x is the metric defined on the tangent space $T_x \mathcal{M}$ at $x \in \mathcal{M}$. The iterate is then updated by retracting along the $-\xi^*$ direction: $x_+ = R_x(-\eta \xi^*)$ with step size $\eta > 0$. Compared with the naive tangent space LMO (2), both the objective $g_x(\xi, \text{grad } f)$ and the feasible set $\{\xi \in T_x \mathcal{M} : \|\xi\|_{\varphi,x} \leq \tau\}$ in the proposed LMO (4) are now intrinsic to the manifold geometry.

Our formulation recovers several familiar special cases. If $\mathcal{M}$ is a Euclidean space with the standard metric, then (4) reduces exactly to the usual Euclidean norm-constrained LMO. If $\varphi = \|\cdot\|_F$ and $\tau = 1$, then (4) returns the normalized Riemannian gradient direction [1, 11]. In the next sections, we show that our LMO formulation simultaneously preserves the quotient symmetries (resolving (O1)) and admits efficient closed-form solutions (resolving (O2)).

3.2. Symmetry preservation on quotient manifolds. We next analyze the proposed intrinsic constraint in (4) when $\mathcal{M} = \bar{\mathcal{M}}/\mathcal{G}$ is a quotient manifold ($\bar{\mathcal{M}}$ being the total space) with a $\mathcal{G}$ -invariant Riemannian metric g_x . In such settings, we show that our intrinsic constraint respects the underlying symmetry if the metric admits a left-right form: $g_x(\xi, \zeta) = \text{tr}(\xi^\top P_x \zeta Q_x)$, or equivalently, the metric operator factors as $G_x \xi = P_x \xi Q_x$, where P_x and Q_x are symmetric positive-definite matrices depending smoothly on x . Then, $G_x^{1/2} \xi = P_x^{1/2} \xi Q_x^{1/2}$ and the intrinsic norm reads $\|\xi\|_{\varphi,x} = \varphi(P_x^{1/2} \xi Q_x^{1/2})$.

In Section 4, we discuss popular quotient manifolds where the metric has the left-right form. It should be noted that the intrinsic norm itself is well-defined for any positive self-adjoint G_x and does not require the left-right factorization. However, on quotient manifolds, this factorization makes the metric scaling explicit and the symmetry behavior of the intrinsic norm transparent. Our next result shows that, under such metrics, the intrinsic norm constraint is $\mathcal{G}$ -invariant.

Proposition 3.1 (Symmetry invariance of the intrinsic constraint). *Suppose $\mathcal{M} = \bar{\mathcal{M}}/\mathcal{G}$ has a $\mathcal{G}$ -invariant left-right metric $g_x(\xi, \zeta) = \text{tr}(\xi^\top P_x \zeta Q_x)$ with $P_x, Q_x \succ 0$. Then $\varphi(G_x^{1/2} \tilde{\xi}) = \varphi(G_x^{1/2} \xi)$ for every unitarily invariant φ norm, where $\tilde{x} = h \cdot x$ and $\tilde{\xi} \in T_{\tilde{x}} \mathcal{M}$ is the corresponding tangent vector under the action of $h \in \mathcal{G}$.*

The proof is given in Section D.1. With respect to the proposed LMO problem (4), the above result shows that on quotient manifolds, both the feasible set $\{\xi : \varphi(G_x^{1/2} \xi) \leq \tau\}$ and the optimal update ξ^* are invariant under the symmetry group $\mathcal{G}$. Thus, our result shows that our intrinsic norm generalizes the Frobenius-norm invariance (that makes the Riemannian gradient well-defined on quotients [1, 11, 39]) to all unitarily invariant norms.

3.3. Efficient and optimal solution to the coupled norm and tangent-space constraints. A key computational challenge in solving (4) efficiently is the simultaneous presence of two constraint sets: $\{\xi : \xi \in T_x \mathcal{M}\}$ and $\{\xi : \varphi(G_x^{1/2} \xi) \leq \tau\}$. We now show that, under certain conditions, these two constraints can be simplified, which allows for obtaining closed-form solutions of (4).

In this regard, we first define the scaled tangent space $\mathcal{Z}_x := \{G_x^{1/2} \eta : \eta \in T_x \mathcal{M}\}$. Since $G_x^{1/2}$ is invertible on $T_x \mathcal{M}$, the map $\xi \mapsto Z := G_x^{1/2} \xi$ is an isomorphism between $T_x \mathcal{M}$ and $\mathcal{Z}_x$. The intrinsic norm may now be rewritten as $\varphi(G_x^{1/2} \xi) = \varphi(Z)$. Moreover, if we define the scaled gradient $H := G_x^{1/2} \text{grad } f$, then the objective of (4) satisfies $g_x(\xi, \text{grad } f) = \langle Z, H \rangle$. To formalize our next result, we also require the following notion.

Definition 3.2. A linear subspace $\mathcal{S} \subseteq \mathbb{R}^{p \times q}$ is *singular value (SV) invariant* if, whenever $U \Sigma V^\top \in \mathcal{S}$ for $p \times p$ and $q \times q$ orthogonal matrices U and V , respectively, and a rectangular diagonal matrix Σ with nonnegative entries, then $U D V^\top \in \mathcal{S}$ for every rectangular diagonal matrix D of the same size with nonnegative entries.

Proposition 3.3 (Closed-form solution). *Let $\mathcal{Z}_x := \{G_x^{1/2} \eta : \eta \in T_x \mathcal{M}\}$. Suppose $\mathcal{Z}_x$ is SV invariant. Then for every unitarily invariant norm φ , the solution of (4) is given by*

$$(5) \quad \xi^* = G_x^{-1/2} Z^*, \text{ where } Z^* = \arg \max_{\varphi(Z) \leq \tau} \langle Z, H \rangle,$$

and $H := G_x^{1/2}(\text{grad } f)$. Moreover, Z^* can be computed as $Z^* = U \text{diag}(z^*)V^\top$, where U and V are the left and the right singular spaces of H (i.e., $H = U \text{diag}(\sigma)V^\top$ is the SVD of H) and z^* solves the vector problem $\max_{\varphi(z) \leq \tau} \langle z, \sigma \rangle = \tau \varphi^\circ(\sigma)$, solvable in closed form for any unitarily invariant φ via norm duality [22] (Section C).

The proof is given in Section D.2. In Section 4, we discuss several manifolds of interest where $\mathcal{Z}_x$ is SV-invariant. In Section P, we also present a broader discussion including cases where SV invariance of $\mathcal{Z}_x$ or the left-right form of the metric fail.

4. APPLICATION OF INTRINSIC LMO (4) ON MATRIX MANIFOLDS

In this section, we discuss the closed-form solutions of the proposed LMO (4) for various manifolds and intrinsic spectral/Frobenius/nuclear norms.

Product of tangent spaces. When the tangent space decomposes as a product $T_x \mathcal{M} = \mathbb{R}^{p_1 \times q_1} \times \dots \times \mathbb{R}^{p_k \times q_k}$ and G_x has the corresponding block structure, then $G_x^{1/2}$ acts component-wise. In such settings, we may choose $\varphi(Z_1, \dots, Z_k)$ as the ℓ_∞ -product norm, i.e., $\varphi(Z_1, \dots, Z_k) := \max_i \varphi(Z_i)$, as it decouples the intrinsic LMO into k independent per-block subproblems:

$$(6) \quad \{Z_i^*\}_{i=1}^k = \arg \max_{\varphi(Z_1, \dots, Z_k) \leq \tau} \sum_i \langle Z_i, H_i \rangle \equiv Z_i^* = \arg \max_{\varphi(Z_i) \leq \tau} \langle Z_i, H_i \rangle, \quad i = 1, \dots, k,$$

each solvable in closed form via Proposition 3.3, since the constraint $\max_i \varphi(Z_i) \leq \tau$ separates into $\varphi(Z_i) \leq \tau$ for each block independently. Interestingly, [15] has also studied ℓ_∞ -based product norms in the context of combining norms across neural network layers.

Fixed-rank manifold. We now consider the set $\mathcal{M}_r$ of rank- r matrices parameterized as $X = BA$ with $B \in \mathbb{R}_*^{m \times r}$, $A \in \mathbb{R}_*^{r \times n}$ (where $\mathbb{R}_*$ denotes full-rank matrices). We note that the factorization $X = BA$ has the quotient symmetry: $(B, A) \sim (BN^{-1}, NA)$ for $N \in \text{GL}(r)$, where $\text{GL}(r)$ denotes the group of $r \times r$ invertible matrices. This results in $\mathcal{M}_r = (\mathbb{R}_*^{m \times r} \times \mathbb{R}_*^{r \times n})/\text{GL}(r)$ and $T_x \mathcal{M}_r = \mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n}$ [1, 41]. If $x = (B, A) \in \mathcal{M}_r$ and $\xi_i = (\dot{B}_i, \dot{A}_i) \in T_x \mathcal{M}_r$ for $i = 1, 2$, then the coupled metric is $g_x(\xi_1, \xi_2) = \text{tr}(\dot{B}_1^\top \dot{B}_2 AA^\top) + \text{tr}(B^\top B \dot{A}_1 \dot{A}_2^\top)$ [39, 40, 69]. We observe that the metric is $\text{GL}(r)$ -invariant with the left-right form (Proposition 3.1) and has a block structure, with $G_x^{1/2} \xi_i = (\dot{B}_i (AA^\top)^{1/2}, (B^\top B)^{1/2} \dot{A}_i)$. Thus, we decouple the LMO into $\dot{B}$ and $\dot{A}$ subproblems (6).

Corollary 4.1 (Intrinsic LMO (4) for fixed-rank manifold). *Let $\mathcal{M} = \mathcal{M}_r$ and at $X = BA \in \mathcal{M}_r$, let $\nabla_X f$ denote the gradient w.r.t. X . Then, with the above discussed $\text{GL}(r)$ -invariant metric and $\varphi = \|\cdot\|_2$, the optimal solution of the proposed LMO (4) is $\xi^* = (\dot{B}^*, \dot{A}^*)$, where $\dot{B}^* = \tau \text{Ortho}((\nabla_B f)(AA^\top)^{-1/2})(AA^\top)^{-1/2}$, $\dot{A}^* = \tau (B^\top B)^{-1/2} \text{Ortho}((B^\top B)^{-1/2} \nabla_A f)$, $\nabla_B f = \nabla_X f A^\top$, and $\nabla_A f = B^\top \nabla_X f$.*

The proof is given in Section D.3. We note that the intrinsic spectral LMO in Corollary 4.1 differs from prior $\mathcal{M}_r$ -Muon constructions in two ways: (i) Factor-wise Muon [26, 29] solves (3) by applying $\text{Ortho}(\cdot)$ to each factor gradient. The resulting ambient update is not $\text{GL}(r)$ -invariant and its spectral norm scales with $\|A\|_2 + \|B\|_2$. Hence, Spectron [26] rescales the solution of (3) during runtime to bound the ambient spectral norm. (ii) Riemannion [10] works in the embedded view with $G_X = I$ and approximates the tangent-constrained LMO by a rank- $2r$ truncated polar followed by tangent space re-projection. Therefore, the spectral-norm bound is satisfied only approximately. Corollary 4.1, in contrast, is the exact maximizer of (4) on each factor block, and its Frobenius instance recovers Riemannian preconditioned gradient for fixed-rank matrices [55, 69]. Section H.1 provides detailed update-rule and norm-bound comparisons.

SPD manifold. The SPD manifold S_{++}^n is the space of all $n \times n$ symmetric positive-definite matrices. At any $X \in S_{++}^n$, $T_X S_{++}^n = \text{Sym}(n)$, the space of $n \times n$ symmetric matrices. Since f is defined on symmetric matrices, the Euclidean gradient $\nabla_X f \in \text{Sym}(n)$ is itself symmetric. Consider S_{++}^n with the affine-invariant metric: $g_X(\xi, \zeta) = \text{tr}(X^{-1} \xi X^{-1} \zeta)$ [8]. Both Propositions 3.1 and 3.3 are satisfied with $G_X^{1/2} \xi = X^{-1/2} \xi X^{-1/2}$ and the Riemannian gradient is $\text{grad } f(X) = X(\nabla_X f)X$. The solution of the proposed LMO (4) with $\varphi = \|\cdot\|_2$ is $\xi^* = \tau X^{1/2} \text{sign}(H) X^{1/2}$, where $H = G_X^{1/2}(\text{grad } f) = X^{1/2}(\nabla_X f)X^{1/2}$ and $\text{sign}(H) = Q \text{diag}(\text{sign}(\lambda_i))Q^\top$ [21] with Q and $\{\lambda_i\}_i$ being eigenvectors and eigenvalues of H , respectively. We also note that the intrinsic Frobenius norm constraint in this setting recovers the natural gradient [2]. Alternative SPD metrics are discussed in Section P.6.

Algorithm 1 Proposed Intrinsic Muon (iMuon) algorithm**Require:** Initial point $x_0 \in \mathcal{M}$, step size $\eta > 0$, norm bound $\tau > 0$, norm φ , iterations T

- 1: **for** $t = 0, 1, \dots, T - 1$ **do**
- 2: Compute Euclidean gradient $\nabla f(x_t)$
- 3: Obtain ξ_t^* by solving intrinsic LMO (4) using $\nabla f(x_t)$ (refer to Section 4 for specific cases)
- 4: Retract: $x_{t+1} = R_{x_t}(-\eta \xi_t^*)$
- 5: **end for**
- 6: **return** x_T

Stiefel manifold. The Stiefel manifold $\text{St}(m, r) = \{X \in \mathbb{R}^{m \times r} : X^\top X = I_r\}$ is the set of $m \times r$ matrices with orthonormal columns with $G_X = I$ (Euclidean metric). The tangent space $T_X \text{St} = \{\xi \in \mathbb{R}^{m \times r} : X^\top \xi + \xi^\top X = 0\}$ decomposes as $\xi = XA + X_\perp B$, where $A \in \text{Skew}(r)$ (set of $r \times r$ skew symmetric matrices), $B \in \mathbb{R}^{(m-r) \times r}$, and $X_\perp \in \mathbb{R}^{m \times (m-r)}$ is an orthonormal basis for the orthogonal complement of $\text{col}(X)$ [1, 11, 16]. Since $G_X = I$, the space $\mathcal{Z}_x = \{G_X^{1/2} \eta : \eta \in T_X \mathcal{M}\}$ is the same as $T_X \mathcal{M}$. As $\mathcal{Z}_x$ is not SV-invariant, Proposition 3.3 cannot be directly applied to $\text{St}(m, r)$. However, we note that any $\xi \in T_x \mathcal{M}$ may be viewed as $\xi = [X, X_\perp][A; B]$. Thus, $T_X \mathcal{M}$ latently decomposes as $[A; B]$. Moreover, for unitarily invariant norms, $\varphi(\xi) \leq \tau = \varphi([A; B]) \leq \tau$. Exploiting the above decomposition with the product viewpoint (6), Proposition 3.3 gives the following result.

Lemma 4.2 (Intrinsic LMO (4) for Stiefel manifold). *Let $\mathcal{M} = \text{St}(m, r)$ and at $X \in \mathcal{M}$, let $G_X = I$. Let $\nabla_X f$ denote the gradient w.r.t. X . Then, with $\varphi = \|\cdot\|_2$, the optimal solution of the proposed LMO (4) is $\xi^* = \tau(X \text{Ortho}(\text{skew}(X^\top \nabla_X f)) + X_\perp \text{Ortho}(X_\perp^\top \nabla_X f))$.*

The proof is given in Section D.4. Recently, [68] studied LMO-based updates in the context of optimization over the Stiefel manifold. However, they sidestep the coupled LMO problem by solving the LMO in the ambient space and projecting the result onto the Stiefel manifold. Thus, the update direction need not lie in $T_x \mathcal{M}$.

Grassmann manifold. The Grassmann manifold $\text{Gr}(m, r)$ is the set of r -dimensional linear subspaces of $\mathbb{R}^m$, represented by an orthonormal basis $X \in \mathbb{R}^{m \times r}$ with $X^\top X = I_r$, up to right multiplication by $O(r)$, i.e., $\text{Gr}(m, r) = \text{St}(m, r)/O(r)$ [11, 16]. However, unlike the Stiefel setting, the tangent space (horizontal space) $T_X \text{Gr} = \{\xi : X^\top \xi = 0\}$ is SV-invariant. Using Proposition 3.3, we get the optimal solution of our intrinsic LMO (4) for $\varphi = \|\cdot\|_2$ as $\xi^* = \tau \text{Ortho}((I - XX^\top) \nabla_X f)$.

Overall. A salient feature of our above analysis across various manifolds is that computing the optimal solution of the proposed problem (4) does not require the computation of the Riemannian gradient. Our closed-form optimal expressions rely only on Euclidean gradients, which are available via back-propagation. Please refer to Section F for proofs and additional results.

5. CONVERGENCE

In this section, we discuss the convergence analysis of our algorithm Intrinsic Muon (**iMuon**), Algorithm 1 in both deterministic and stochastic settings. Our convergence analysis is parametrized by a geometric quantity C_φ , which captures the *size* of the constraint set in (4), defined as follows.

Definition 5.1. The squared Riemannian radius of the φ -ball on $T_x \mathcal{M}$ is $C_\varphi(x) := \max\{\|\xi\|_x^2 : \xi \in T_x \mathcal{M}, \varphi(G_x^{1/2} \xi) \leq 1\}$. Also define $C_\varphi := \sup_{x \in \mathcal{M}} C_\varphi(x)$.

For $\varphi = \|\cdot\|_2$: $C_\varphi = 2r$ on $\mathcal{M}_r$, n on S_{++}^n , $2\lceil r/2 \rceil + \min(m-r, r)$ on $\text{St}(m, r)$, $\min(m-r, r)$ on $\text{Gr}(m, r)$. For $\varphi \in \{\|\cdot\|_F, \|\cdot\|_*\}$, $C_\varphi = O(1)$ on all manifolds. On all four manifolds, C_φ depends only on the manifold dimensions, not on the iterates.

Assumption 5.2 (Retraction- L -smoothness [11, Assumption 4.3]). There exist constants $L > 0$ and $\bar{\eta} > 0$ such that for all $x \in \mathcal{M}$, all $\xi \in T_x \mathcal{M}$, and all $\eta \in (0, \bar{\eta}]$, $f(R_x(-\eta \xi)) \leq f(x) - \eta g_x(\xi, \text{grad } f) + \frac{L\eta^2}{2} \|\xi\|_x^2$.

The next theorem shows that our algorithm iMuon converges at rate $O(1/\sqrt{T})$ with the constant determined by C_φ .

Theorem 5.3 (Deterministic iMuon). *Under Assumption 5.2, with constant step size $\eta = c/(\tau\sqrt{T})$ where $c = \sqrt{2\Delta_0/(LC_\varphi)}$ (assuming $\eta \leq \bar{\eta}$), our Algorithm 1 achieves $\min_{t \leq T} \varphi^\circ(H_t) \leq \sqrt{2LC_\varphi\Delta_0/T}$, where $\Delta_0 = f(x_0) - f^*$, φ° is the dual norm of φ , and $H_t = G_{x_t}^{1/2}(\text{grad } f(x_t))$. With a horizon-free decaying schedule defined as $\eta_t = c/(\tau\sqrt{t+1})$, Algorithm 1 achieves: $\min_{t \leq T} \varphi^\circ(H_t) \leq \frac{\Delta_0(2+\ln T)}{2c(\sqrt{T}-1)}$.*

The proof is provided in Section D.5. We note that the convergence criterion $\varphi^\circ(H_t) \rightarrow 0$ implies $\|\text{grad } f(x_t)\|_{x_t} \rightarrow 0$ (first-order stationarity) as $H_t = G_{x_t}^{1/2}(\text{grad } f(x_t))$ and $G_{x_t}^{1/2} \succ 0$. For $\varphi = \|\cdot\|_F$, the rate recovers standard Riemannian gradient descent with $C_\varphi = 1$. Our key technical ingredient is the uniform bound $\|\xi_t^*\|_{x_t}^2 \leq C_\varphi\tau^2$ (Lemma D.3). We also note manifold specific convergence rates below for constant step size setting.

Remark 5.4 (Fixed-rank). On $\mathcal{M}_r$ with $\varphi = \|\cdot\|_2$, our rate is $O(\sqrt{Lr/T})$, depending only on the rank r and independent of the factor norms $\|B\|, \|A\|$. Existing works propose extrinsic solutions for convergence: [29] establish, in continuous-time setting, that trajectories will strictly either converge or diverge depending on whether they remain bounded (unconditional convergence under ℓ_2 regularization). [26], on the other hand, apply iterate dependent spectral renormalization at runtime.

Remark 5.5 (SPD, Stiefel, Grassmann). Our convergence rates are $O(\sqrt{Ln/T})$ on S_{++}^n , $O(\sqrt{L(r + \min(m-r, r))/T})$ on $\text{St}(m, r)$, and $O(\sqrt{L \min(m-r, r)/T})$ on $\text{Gr}(m, r)$.

In the stochastic setting, stochastic gradients ($\widetilde{\nabla}f$) are available and we appropriately employ the Riemannian stochastic gradient ($\text{grad } f$) in the proposed LMO (4) to obtain the update direction $\tilde{\xi}_t^*$.

Assumption 5.6. There exists $\sigma_\varphi > 0$ such that the stochastic scaled gradient $\tilde{H}_t := G_{x_t}^{1/2}(\widetilde{\text{grad } f_t})$ satisfies $\mathbb{E}[\varphi^\circ(\tilde{H}_t - H_t)^2 | x_0, \dots, x_t] \leq \sigma_\varphi^2$, where $H_t := G_{x_t}^{1/2}(\text{grad } f_t)$ is the true scaled gradient.

For $\varphi = \|\cdot\|_F$, this reduces to the standard Riemannian bounded-variance condition [11, 44].

Theorem 5.7 (Stochastic iMuon). *Under Assumptions 5.2 & 5.6, with step size $\eta_t = c/(\tau\sqrt{t+1})$ where $c = \sqrt{2\Delta_0/(LC_\varphi)}$ (assuming $\eta_t \leq \bar{\eta}$), our Algorithm 1 achieves $\min_{t \leq T} \mathbb{E}[\varphi^\circ(H_t)] \leq \sqrt{2LC_\varphi\Delta_0(1 + \ln T)/T} + 2\sigma_\varphi$, recovering Theorem 5.3 (up to the $\ln T$ factor from the decaying schedule) when $\sigma_\varphi = 0$.*

The proof is provided in Section D.8. The additive $2\sigma_\varphi$ arises from the lack of variance reduction in the gradient estimator and can be removed with gradient averaging or momentum techniques [44, 54], which we leave to future work.

Several prior results arise as instances of our framework (see Section G for details): Euclidean spectral steepest descent [6] ($G_x = I$, $C_\varphi = \min(p, q)$), standard Riemannian GD [11] ($\varphi = \|\cdot\|_F$, $C_\varphi = 1$), factor-wise Muon [29] (factor-norm dependence replaced by $C_\varphi = 2r$), the rank-1 special case of NuMuon [43] ($\varphi = \|\cdot\|_*$, $C_\varphi = 1$), and general Euclidean LMOs [48] ($G_x = I$).

6. EXPERIMENTS

Code availability. Code for reproducing the experiments is available at <https://github.com/1bang118/manifold-intrinsic-muon>.

We evaluate the proposed intrinsic LMO on two manifolds: (i) the fixed-rank manifold $\mathcal{M}_r$ via LoRA fine-tuning of GPT-2 Medium on E2E NLG and Mistral-7B on GLUE and (ii) the SPD manifold S_{++}^n via SPD covariance classification on CIFAR-100. In fine-tuning experiments, we compare our proposed iMuon with factor-wise Euclidean Muon [6, 27] (as in (3)), SGD (factor-wise with Frobenius norm), Scaled GD [69] (preconditioned), and Riemannion [10]. In the SPD setting, we include the Frobenius and nuclear instances of our framework (RGD, iMuon-Nu) and their Euclidean counterparts (EGD, NuMuon). Full details and additional experiments including on Stiefel and Grassmann manifolds are deferred to Sections K to O. LoRA fine-tuning ($\mathcal{M}_r$). We fine-tune GPT-2 Medium on E2E NLG [45] with LoRA rank $r = 4$, and 4-bit quantized Mistral-7B on GLUE [59] with LoRA rank $r = 16$. We follow the LoRA setup of [23], tuning the learning rate per method via grid search.

Table 1 reports the five standard E2E generation metrics, and Table 2 reports the 9 GLUE per-task scores together with the 9-task average, both *without optimizer momentum*: this isolates the LMO direction itself, since momentum strategies differ across methods and compound gradient information across iterations. Our *with-momentum* experiments are discussed in Section K.

TABLE 1. E2E NLG test-set results (GPT-2 Medium, LoRA $r = 4$, no optimizer momentum). iMuon consistently performs better (best is highlighted).

Method	BLEU	NIST	METEOR	ROUGE-L	CIDEr
SGD	66.60	8.54	44.20	68.20	2.32
Scaled GD [69]	69.20	8.71	46.30	70.90	2.48
Factor-wise Muon [26, 29]	70.02	8.81	46.77	71.68	2.53
Riemannion [10]	70.02	8.78	46.79	71.99	2.52
iMuon (ours)	70.74	8.88	46.79	72.14	2.54

TABLE 2. GLUE results with Mistral-7B using 4-bit NF4 quantization and LoRA rank $r = 16$, without optimizer momentum. Best shown in bold.

Method	MNLI	SST-2	MRPC	CoLA	QNLI	QQP	RTE	STS-B	WNLI	Avg.
SGD	88.15	96.10	70.10	55.89	94.22	88.59	50.90	47.64	49.30	71.21
Scaled GD [69]	90.21	96.90	81.62	68.17	94.40	91.15	54.15	90.31	56.34	80.36
Factor-wise Muon [26, 29]	89.77	97.25	82.35	69.32	94.47	88.42	84.12	84.39	45.07	81.68
Riemannion (SGD) [10]	91.55	96.90	82.60	68.90	94.93	87.77	82.67	77.99	60.56	82.65
iMuon (ours)	89.29	97.02	85.54	70.21	94.60	87.86	88.09	88.99	74.65	86.25

On E2E NLG benchmark, iMuon attains the best score on every reported metric, improving by 0.72 BLEU over both factor-wise Muon and Riemannion, with a tie on METEOR task. Our advantage on the GLUE benchmark is more pronounced: iMuon achieves 86.25 average vs. 81.68 for factor-wise Muon (+4.57 points), with consistent gains on the harder, lower-resource tasks (MRPC, CoLA, RTE, WNLI). This is consistent with the discussion in Remark 5.4: factor-wise Muon’s update magnitude scales with $\|A\|_2 + \|B\|_2$ and is sensitive to the factor representative, whereas iMuon’s intrinsic norm absorbs the factor scale algebraically, producing a $GL(r)$ -invariant update with rank-only norm bound.

SPD prototype classification (S_{++}^n). Covariance descriptors are a long-standing SPD representation for image and video recognition, where second-order statistics of feature maps encode appearance and texture cues [24, 56, 57]. We use a Riemannian analogue of nearest-class-mean classification: keep the backbone frozen and learn one SPD prototype $X_c \in S_{++}^{32}$ per class for 20-way coarse-label classification on CIFAR-100 [33], using shrinkage covariance descriptors [35] computed from frozen ResNet-18 `layer3` features [20] and the affine-invariant Riemannian metric on S_{++}^{32} [8, 47]. The training objective is a softmax classifier with logits $-\beta d_{AI}(C_i, X_c)^2$, regularized toward the log-Euclidean [4] class-mean initialization (Section M). We compare all three norm instances of our framework with their Euclidean counterparts: Frobenius (RGD vs. EGD), spectral (iMuon vs. Muon), and nuclear (iMuon-Nu vs. NuMuon, where NuMuon denotes the pure nuclear-norm LMO from [43], i.e., the $\varphi = \|\cdot\|_*$ instance of (1)).

Figure 1 shows test-accuracy trajectories across training for each of the three Euclidean/intrinsic pairs. Two trends are consistent across the panels. First, replacing each Euclidean LMO by its intrinsic counterpart improves test accuracy in every pair: RGD 0.548 ± 0.002 vs. EGD 0.523 ± 0.026 (Frobenius), iMuon 0.475 ± 0.009 vs. Muon 0.407 ± 0.016 (spectral), and iMuon-Nu 0.546 ± 0.000 vs. NuMuon 0.422 ± 0.032 (nuclear), where each entry is the mean test accuracy over 3 seeds at the validation-selected learning rate. For the Frobenius pair this recovers the standard benefit of the affine-invariant (AI) Riemannian gradient on S_{++}^n [8, 47]. The spectral and nuclear pairs extend the same metric-aware construction to LMO directions. Second, the improvement grows as the norm becomes more aggressive: moderate for the Frobenius pair, larger for the spectral pair, and largest for the nuclear pair. We attribute this to the fact that the rank-1 Euclidean update interacts most strongly with the curvature of S_{++}^n and therefore benefits most from the metric scaling $X^{1/2}(\cdot)X^{1/2}$ in our construction. The trajectories also show that the intrinsic methods open these gaps early and maintain them throughout training rather than relying on late-stage convergence behavior.

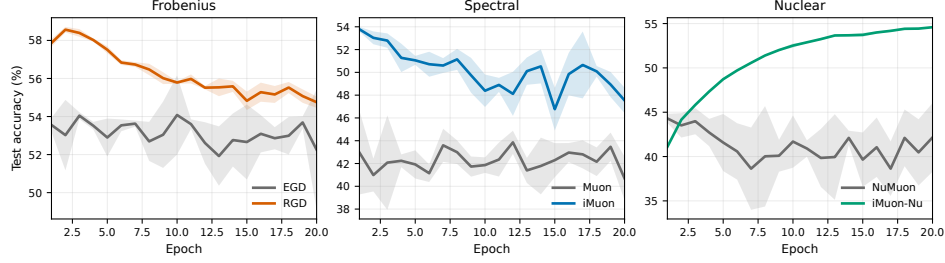


FIGURE 1. SPD classification on CIFAR-100 (S_{++}^{32} , 20 coarse classes). Each panel pairs a Euclidean LMO with its intrinsic counterpart under a common norm: Frobenius (EGD vs. RGD), spectral (Muon vs. iMuon), and nuclear (NuMuon vs. iMuon-Nu). Curves show mean test accuracy with ± 1 std bands over 3 seeds at the validation-selected learning rate. The intrinsic method dominates in every pair, with the gap widening from Frobenius to spectral to nuclear.

7. CONCLUSION

We introduced iMuon, a unified framework for unitarily invariant norm-constrained optimization on Riemannian matrix manifolds. By lifting the ambient norm through the Riemannian metric, iMuon resolves the symmetry breaking (O1) and tangent space and norm constraints coupling (O2) that obstruct prior manifold extensions of Muon, and admits closed-form LMO solutions on the fixed-rank, SPD, Stiefel, and Grassmann manifolds. The resulting convergence rate depends only on the manifold dimension. On the fixed-rank manifold this is independent of factor conditioning, removing the bounded-factor-norm assumption and runtime rescaling required by prior work. Empirically, iMuon performs favorably across LoRA fine-tuning and other learning problems on the fixed-rank, SPD, Stiefel, and Grassmann manifolds, with the benefit of the intrinsic geometry most pronounced when the underlying metric is non-trivial. Future work include momentum and variance-reduced variants, adaptive choice of φ , and extensions to product manifolds such as Tucker low-multilinear-rank manifold. Limitations are discussed in Section B.

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APPENDIX A. BROADER SOCIETAL IMPACT

This paper presents foundational research on Riemannian optimization algorithms. The contribution is methodological and theoretical. The framework is not tied to a particular application or deployment, and our empirical evaluation uses standard publicly available benchmarks for empirical validations.

APPENDIX B. LIMITATIONS

Our framework requires a left-right metric form and singular value invariance of the scaled tangent space, which hold on the four matrix manifolds we treat but not in general (Section P), and on product tangent spaces such as $\mathcal{M}_r$ and $\text{St}(m, r)$ we further use an ℓ_∞ -product norm relaxation of the joint norm.

APPENDIX C. CLOSED-FORM LMO SOLUTIONS VIA NORM DUALITY

We derive the closed-form solution of the intrinsic LMO for a general unitarily invariant norm φ .

C.1. Reduction to a vector problem. The key tool is von Neumann's trace inequality, which reduces the matrix LMO to a vector problem over singular values.

Lemma C.1. *Let $H = U \text{diag}(\sigma_1, \dots, \sigma_{\min(p,q)}) V^\top$ with $\sigma_1 \geq \dots \geq \sigma_{\min(p,q)} \geq 0$. Then $\max_{\varphi(Z) \leq \tau} \langle Z, H \rangle = \tau \varphi^\circ(\sigma)$, where φ° is the dual norm and $\sigma = (\sigma_1, \dots, \sigma_{\min(p,q)})$. The maximizer has the form $Z^* = U \text{diag}(z_1^*, \dots, z_{\min(p,q)}^*) V^\top$, where z^* solves the vector problem $\max_{\varphi(z) \leq \tau} \langle z, \sigma \rangle$.*

Proof. By von Neumann's trace inequality [22, Thm. 7.4.1.1], $\langle Z, H \rangle \leq \sum_i \sigma_i(Z) \sigma_i(H)$ with equality when Z shares the singular vectors of H ; hence $Z^* = U \text{diag}(z^*) V^\top$ for some non-negative z^* with $\varphi(z^*) \leq \tau$, reducing the problem to $\max_{\varphi(z) \leq \tau} \langle z, \sigma \rangle = \tau \varphi^\circ(\sigma)$ by definition of the dual norm. $\square$

C.2. Spectral, Frobenius, and nuclear norms. The vector problem $\max_{\varphi(z) \leq \tau} \langle z, \sigma \rangle$ has closed-form solutions for the three standard norms:

Spectral norm ($\varphi(z) = \|z\|_\infty = \max_i |z_i|$). Each z_i is independently bounded by τ , and $\sigma_i \geq 0$, so $z_i^* = \tau$ for all i . Thus $Z^* = \tau U V^\top = \tau \text{Ortho}(H)$ and the optimal value is $\tau \sum_i \sigma_i = \tau \|H\|_*$. The dual norm is $\varphi^\circ = \|\cdot\|_1$ on singular values, i.e., the nuclear norm.

Frobenius norm ($\varphi(z) = \|z\|_2$). By Cauchy-Schwarz, $z^* = \tau \sigma / \|\sigma\|_2$, giving $Z^* = \tau H / \|H\|_F$. The optimal value is $\tau \|H\|_F$. Self-dual: $\varphi^\circ = \|\cdot\|_F$.

Nuclear norm ($\varphi(z) = \|z\|_1 = \sum_i |z_i|$). Under $\sum_i z_i \leq \tau$ with $z_i \geq 0$, all of the norm bound goes to the largest σ_i : $z^* = \tau e_1$. Thus $Z^* = \tau u_1 v_1^\top$ and the optimal value is $\tau \sigma_1 = \tau \|H\|_2$. Dual: $\varphi^\circ = \|\cdot\|_\infty$, i.e., the spectral norm.

For a general unitarily invariant norm φ , the vector problem $\max_{\varphi(z) \leq \tau} \langle z, \sigma \rangle = \tau \varphi^\circ(\sigma)$ by duality; explicit maximizers beyond the three cases above amount to a finite-dimensional symmetric-gauge convex program.

C.3. Other unitarily invariant norms. The three norms above are the extreme points of the family of unitarily invariant norms on singular values. Here we note three natural interpolations that also admit closed-form LMOs within our framework.

Ky Fan k -norms. The Ky Fan k -norm [22, §3.4] is $\|Z\|_{(k)} := \sum_{i=1}^k \sigma_i(Z)$, the sum of the k largest singular values. It is unitarily invariant for every $k \in \{1, \dots, \min(p, q)\}$, and interpolates between the nuclear norm ($k = 1$) and the spectral norm ($k = \min(p, q)$), since $\|Z\|_{(\min(p,q))} = \|Z\|_*$ and the LMO under $\|\cdot\|_2$ already sets all singular values to τ . The LMO has a clean closed form: the vector problem $\max_{\|z\|_{(k)} \leq \tau} \langle z, \sigma \rangle$ is solved by $z_i^* = \tau$ for $i \leq k$ and $z_i^* = 0$ for $i > k$, giving

$$Z^* = \tau \sum_{i=1}^k u_i v_i^\top,$$

the *rank- k partial polar factor* of H . The dual norm is $\|Z\|_{(k)}^\circ = \max(\|Z\|_2, \|Z\|_*/k)$, and the squared Riemannian radius is $C_\varphi = k$. Thus Theorem 5.3 yields rate $O(\sqrt{Lk\Delta_0/T})$ with convergence in $\|H_t\|_{(k)}^\circ$. Setting k between 1 and r provides a tunable aggressiveness-rank trade-off: $k = 1$ concentrates on the leading singular direction (conservative, $C_\varphi = 1$), while $k = r$ equalizes all r directions (aggressive, $C_\varphi = r$).

Schatten p -norms. The Schatten p -norm $\|Z\|_{S_p} := (\sum_i \sigma_i(Z)^p)^{1/p}$ for $p \in [1, \infty]$ is unitarily invariant and continuously interpolates between the nuclear ($p = 1$), Frobenius ($p = 2$), and spectral ($p = \infty$) norms. The dual norm is the Schatten q -norm with $1/p + 1/q = 1$, so the LMO $\max_{\|z\|_p \leq \tau} \langle z, \sigma \rangle$ is solved by Hölder's inequality:

$$z_i^* = \tau \frac{\sigma_i^{q-1}}{\|\sigma^{q-1}\|_p}, \quad Z^* = U \text{diag}(z^*) V^\top,$$

where σ^{q-1} denotes the vector $(\sigma_1^{q-1}, \dots, \sigma_r^{q-1})$. This is a *power-law reweighting* of the singular values: for $p = 2$ ($q = 2$), $z_i^* \propto \sigma_i$ (proportional, recovering Frobenius); for $p \rightarrow \infty$ ($q \rightarrow 1$), $z_i^* \rightarrow \tau$ (all equal, recovering spectral); for $p = 1$ ($q = \infty$), all budget concentrates on the largest σ_i (recovering nuclear). Intermediate values of p give a “soft Muon” that partially equalizes singular values without the hard cutoff of the Ky Fan norm.

The squared Riemannian radius is

$$C_\varphi = \begin{cases} 1 & p \leq 2, \\ r^{1-2/p} & p \geq 2, \end{cases}$$

where $r = \min(p_0, q_0)$ is the number of singular values (here p_0, q_0 are the matrix dimensions, not the Schatten exponent). For $p \leq 2$, the maximizer of $\|z\|_2^2$ subject to $\|z\|_p \leq 1$ is a unit vector ($C_\varphi = 1$); for $p > 2$, it is the uniform vector $z = (r^{-1/p}, \dots, r^{-1/p})$ giving $C_\varphi = r^{1-2/p}$. Thus Theorem 5.3 yields rate $O(\sqrt{L \cdot \max(1, r^{1-2/p}) \cdot \Delta_0/T})$, smoothly interpolating between the $O(\sqrt{L/T})$ rate of Frobenius/nuclear norms and the $O(\sqrt{Lr/T})$ rate of the spectral norm. In practice, choosing p slightly above 2 gives a mild singular value equalization (“softer” than full Muon) at a reduced effective-dimension cost.

Spectral-nuclear intersection. An alternative interpolation constrains both the spectral and nuclear norms simultaneously:

$$Z^* = \arg \max_{\|Z\|_2 \leq \tau, \|Z\|_* \leq \tau'} \langle Z, H \rangle.$$

Since the objective $\langle Z, H \rangle$ is linear and the constraint set is the intersection of two unitarily invariant norm balls, the problem again reduces to a vector problem over the singular values: $\max_{0 \leq z_i \leq \tau, \sum_i z_i \leq \tau'} \sum_i z_i \sigma_i$. The optimal allocation is greedy: saturate $z_i = \tau$ starting from the largest σ_i until the nuclear budget τ' is exhausted, giving

$$z_i^* = \begin{cases} \tau & i \leq k^*, \\ \tau' - k^* \tau & i = k^* + 1, \\ 0 & i > k^* + 1, \end{cases} \quad k^* = \lfloor \tau'/\tau \rfloor.$$

This is a rank- k^* partial polar factor plus a fractional rank-1 residual. The ratio τ'/τ controls the effective rank: $\tau'/\tau \geq \min(p, q)$ recovers the spectral instance, $\tau'/\tau \leq 1$ recovers the nuclear instance, and intermediate values produce an adaptive-rank update. Unlike the Ky Fan norm, the effective rank here is set by the norm bound ratio rather than a discrete parameter, which may simplify hyperparameter tuning in practice.

APPENDIX D. PROOFS OF MAIN RESULTS

D.1. Proof of Proposition 3.1.

Proof of Proposition 3.1. Since $\mathcal{G}$ acts on the ambient matrix space by multiplication, the induced tangent space action has the form $\xi \mapsto \tilde{\xi} = L_h \xi R_h^\top$ for matrices L_h, R_h depending on x and h . Define $\rho_h(Z) := G_x^{1/2}(L_h(G_x^{-1/2}Z)R_h^\top)$, so that $G_x^{1/2}\tilde{\xi} = \rho_h(G_x^{1/2}\xi)$. Since both $G_x^{1/2}$ and the tangent action are left-right, ρ_h acts as $\rho_h(Z) = AZB^\top$ for $A := P_x^{1/2}L_hP_x^{-1/2}$ and $B := Q_x^{1/2}R_hQ_x^{-1/2}$. (Indeed, expanding $\rho_h(Z) = P_x^{1/2}L_hP_x^{-1/2}ZQ_x^{-1/2}R_h^\top Q_x^{1/2}$ and matching with $AZB^\top$ gives $B^\top = Q_x^{-1/2}R_h^\top Q_x^{1/2}$, hence $B = Q_x^{1/2}R_hQ_x^{-1/2}$.) By $\mathcal{G}$ -invariance of the metric, ρ_h preserves the Frobenius norm: $\|AZB^\top\|_F = \|Z\|_F$ for all Z . This forces $A^\top A = \alpha I$ and $B^\top B = \alpha^{-1}I$ for some $\alpha > 0$, so $A/\sqrt{\alpha}$ and $B/\sqrt{\alpha}$ are orthogonal. Hence $\sigma_i(\rho_h(Z)) = \sigma_i(Z)$ and $\varphi(\rho_h(Z)) = \varphi(Z)$ for every unitarily invariant φ . Applying this with $Z = P_x^{1/2}\xi Q_x^{1/2}$ gives the result. $\square$

D.2. Proof of Proposition 3.3.

Proof of Proposition 3.3. Substitute $Z = G_x^{1/2}\xi$ in (4). Since $g_x(\xi, \text{grad } f) = \langle G_x^{1/2}\xi, G_x^{1/2}(\text{grad } f) \rangle = \langle Z, H \rangle$ (by self-adjointness of $G_x^{1/2}$) and $\varphi(G_x^{1/2}\xi) = \varphi(Z)$, the LMO becomes $\max_{Z \in \mathcal{Z}_x, \varphi(Z) \leq \tau} \langle Z, H \rangle$ with $\xi^* = G_x^{-1/2}Z^*$. Note that $H \in \mathcal{Z}_x$ since $\text{grad } f \in T_x\mathcal{M}$. Write $H = U\Sigma V^\top$ (compact SVD with $\Sigma = \text{diag}(\sigma)$ and $\sigma_1 \geq \dots \geq \sigma_r > 0$). By von Neumann’s trace inequality [22, Thm. 7.4.1.1], $\langle Z, H \rangle \leq \sum_i \sigma_i(Z)\sigma_i(H)$ with equality when Z shares singular vectors with H . Thus every maximizer of the unconstrained problem $\max_{\varphi(Z) \leq \tau} \langle Z, H \rangle$ has the form $Z^* = U \text{diag}(z^*)V^\top$ (Lemma C.1), where z^* solves $\max_{\varphi(z) \leq \tau} \langle z, \sigma \rangle$. Since

$H \in \mathcal{Z}_x$ and $\mathcal{Z}_x$ is singular value invariant, replacing the singular values Σ with $\text{diag}(z^*)$ preserves membership: $Z^* \in \mathcal{Z}_x$. Hence the unconstrained maximizer automatically satisfies the $\mathcal{Z}_x$ -constraint, so Z^* also solves the constrained problem. For the closed form, the vector problem gives $z^* = \arg \max_{\varphi(z) \leq \tau} \langle z, \sigma \rangle$ with optimal value $\tau \varphi^\circ(\sigma)$ by norm duality, yielding $Z^* = U \text{diag}(z^*) V^\top$. $\square$

D.3. Proof of Corollary 4.1.

Proof of Corollary 4.1. On $\mathcal{M}_r$ with the coupled metric, $G_x^{1/2}(\dot{B}, \dot{A}) = (\dot{B}(AA^\top)^{1/2}, (B^\top B)^{1/2} \dot{A})$ and $T_x \mathcal{M}_r = \mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n}$.

Step 1: Left-right form and symmetry (Proposition 3.1). The $\dot{B}$ -block metric is $g^B(\dot{B}_1, \dot{B}_2) = \text{tr}(\dot{B}_1^\top \dot{B}_2 AA^\top)$, which has the left-right form with $P = I_m$ and $Q = AA^\top$. Under $(B, A) \mapsto (BN^{-1}, NA)$, $Q = AA^\top \mapsto NAA^\top N^\top$ and $\dot{B} \mapsto \dot{B}N^{-1}$; the metric is invariant since $\text{tr}((\dot{B}N^{-1})^\top (\dot{B}N^{-1}) NAA^\top N^\top) = \text{tr}(\dot{B}^\top \dot{B} AA^\top)$. By Proposition 3.1, the intrinsic constraint $\varphi(\dot{B}(AA^\top)^{1/2}) \leq \tau$ is $\text{GL}(r)$ -invariant. The $\dot{A}$ -block is symmetric (with $P = B^\top B$, $Q = I_n$).

Step 2: SV-invariance and reduction (Proposition 3.3). The scaled tangent space decomposes as $\mathcal{Z}_x = \mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n}$ (full ambient space on each block), which is trivially singular value invariant (Section E). With the ℓ_∞ -product norm $\max(\varphi(Z_B), \varphi(Z_A)) \leq \tau$, the LMO (4) decouples into independent $\dot{B}$ and $\dot{A}$ subproblems via (6).

Step 3: Closed form for the $\dot{B}$ -block. The objective is $g_x(\xi, \text{grad } f) = \langle \dot{B}A, \nabla_X f \rangle = \langle \dot{B}, \nabla_B f \rangle$ where $\nabla_B f = \nabla_X f A^\top$. Substituting $Z_B = \dot{B}(AA^\top)^{1/2}$ gives objective $\langle Z_B, H_B \rangle$ with $H_B = \nabla_B f (AA^\top)^{-1/2}$. By Proposition 3.3 (with SV-invariance from Step 2), the tangent-space constraint is redundant and the spectral-norm LMO reduces to the unconstrained problem $\max_{\|Z_B\|_2 \leq \tau} \langle Z_B, H_B \rangle$. By Lemma C.1, $Z_B^* = \tau \text{Ortho}(H_B)$, so $\dot{B}^* = \tau \text{Ortho}(\nabla_B f (AA^\top)^{-1/2}) (AA^\top)^{-1/2}$. The $\dot{A}$ -block follows by symmetry: $H_A = (B^\top B)^{-1/2} \nabla_A f$ with $\nabla_A f = B^\top \nabla_X f$, giving $\dot{A}^* = \tau (B^\top B)^{-1/2} \text{Ortho}((B^\top B)^{-1/2} \nabla_A f)$. $\square$

D.4. Proof of Lemma 4.2. The proof requires showing that on each block of the product-norm decoupling, the tangent-space constraint is redundant. For the normal block this follows from SV-invariance (Proposition 3.3); for the skew block we use the following auxiliary result.

Lemma D.1 (Skew-symmetric LMO compatibility). *Let $S \in \text{Skew}(r)$ and let φ be any unitarily invariant norm. Then $\max_{A \in \text{Skew}(r), \varphi(A) \leq \tau} \langle A, S \rangle = \max_{\varphi(Z) \leq \tau} \langle Z, S \rangle$. That is, the $\text{Skew}(r)$ constraint is redundant: the unconstrained maximizer can always be chosen skew-symmetric.*

Proof. The inequality " $\leq$ " is trivial (smaller feasible set). For " $\geq$ ": let Z^* be any maximizer of the right-hand side. Define $\hat{Z} := \frac{1}{2}(Z^* - (Z^*)^\top) \in \text{Skew}(r)$.

Objective preserved. Since $S^\top = -S$: $\langle \hat{Z}, S \rangle = \frac{1}{2} \langle Z^*, S \rangle - \frac{1}{2} \langle (Z^*)^\top, S \rangle = \frac{1}{2} \langle Z^*, S \rangle - \frac{1}{2} \langle Z^*, S^\top \rangle = \frac{1}{2} \langle Z^*, S \rangle + \frac{1}{2} \langle Z^*, S \rangle = \langle Z^*, S \rangle$.

Norm does not increase. Since φ is unitarily invariant, $\varphi(M^\top) = \varphi(M)$ and $\varphi(-M) = \varphi(M)$. By the triangle inequality: $\varphi(\hat{Z}) = \varphi(\frac{Z^* - (Z^*)^\top}{2}) \leq \frac{1}{2} \varphi(Z^*) + \frac{1}{2} \varphi((Z^*)^\top) = \varphi(Z^*) \leq \tau$.

Thus $\hat{Z} \in \text{Skew}(r)$, $\varphi(\hat{Z}) \leq \tau$, and $\langle \hat{Z}, S \rangle = \langle Z^*, S \rangle$, so $\hat{Z}$ is also an optimizer of the left-hand side. $\square$

Proof of Lemma 4.2. On $\text{St}(m, r)$ with $G_X = I$, a tangent vector $\xi \in T_X \text{St}$ decomposes as $\xi = XA + X_\perp B$ with $A \in \text{Skew}(r)$ and $B \in \mathbb{R}^{(m-r) \times r}$. Since $[X \mid X_\perp] \in O(m)$ and φ is unitarily invariant, $\varphi(\xi) = \varphi(\begin{bmatrix} A \\ B \end{bmatrix})$.

Step 1: Product-norm decoupling. Under the ℓ_∞ -product norm $\max(\varphi(A), \varphi(B)) \leq \tau$, the objective separates as $g_X(\xi, \text{grad } f) = \langle A, S \rangle + \langle B, N \rangle$ where $S = \text{skew}(X^\top \nabla_X f) \in \text{Skew}(r)$ and $N = X_\perp^\top \nabla_X f \in \mathbb{R}^{(m-r) \times r}$. By (6), the LMO decouples into independent subproblems:

$$A^* = \arg \max_{A \in \text{Skew}(r), \varphi(A) \leq \tau} \langle A, S \rangle, \quad B^* = \arg \max_{B \in \mathbb{R}^{(m-r) \times r}, \varphi(B) \leq \tau} \langle B, N \rangle.$$

Step 2: Normal block (B). The feasible set $\mathbb{R}^{(m-r) \times r}$ is the full ambient space, hence trivially SV-invariant. By Proposition 3.3 and Lemma C.1, $B^* = \tau \text{Ortho}(N) = \tau \text{Ortho}(X_\perp^\top \nabla_X f)$ for $\varphi = \|\cdot\|_2$.

Step 3: Skew block (A). By Lemma D.1, the constraint $A \in \text{Skew}(r)$ is redundant: the unconstrained maximizer $Z^* = \tau \text{Ortho}(S)$ of $\max_{\varphi(Z) \leq \tau} \langle Z, S \rangle$ can be replaced by its skew-symmetric projection $A^* = \frac{1}{2}(Z^* - (Z^*)^\top)$ without changing the objective value or violating the norm bound. For $\varphi = \|\cdot\|_2$, $A^* = \tau \text{Ortho}(S)$ is already skew-symmetric (since $S \in \text{Skew}(r)$ implies $\text{Ortho}(S) \in \text{Skew}(r)$; see remark below), so no projection is needed.

Assembling: $\xi^* = XA^* + X_\perp B^* = \tau(X \text{Ortho}(\text{skew}(X^\top \nabla_X f)) + X_\perp \text{Ortho}(X_\perp^\top \nabla_X f))$. $\square$

Remark D.2. For even r , $\text{Ortho}(S) = UV^\top$ from the SVD of $S \in \text{Skew}(r)$ is itself skew-symmetric. For odd r , S has a mandatory zero singular value ($\det S = 0$), so $\text{Ortho}(S)$ fills the null direction and is not skew-symmetric. In this case, the skew projection $\hat{Z} = \frac{1}{2}(\text{Ortho}(S) - \text{Ortho}(S)^\top)$ achieves the same objective (the extra direction contributes zero to $\langle \cdot, S \rangle$) with $\varphi(\hat{Z}) \leq \varphi(\text{Ortho}(S)) \leq \tau$. Either way, $A^* \in \text{Skew}(r)$ with optimal value $\tau \varphi^\circ(S)$.

D.5. Proof of Theorem 5.3.

Lemma D.3 (LMO objective value and update norm bound). *Let ξ_t^* be the solution of the intrinsic LMO (4) at x_t with norm bound τ , and let $Z_t^* = G_{x_t}^{1/2} \xi_t^*$ and $H_t = G_{x_t}^{1/2}(\text{grad } f(x_t))$. Then:*

- (i) (Objective value) $g_{x_t}(\xi_t^*, \text{grad } f_t) = \tau \varphi^\circ(H_t)$.
- (ii) (Update norm bound) $\|\xi_t^*\|_{x_t}^2 \leq C_\varphi \tau^2$.

Proof. Part (i). By the change of variables $Z = G_{x_t}^{1/2} \xi$ (Proposition 3.3), the LMO objective equals $\langle Z_t^*, H_t \rangle$. Since SV-invariance makes the tangent-space constraint redundant, Z_t^* is the unconstrained maximizer of $\langle Z, H_t \rangle$ subject to $\varphi(Z) \leq \tau$. By Lemma C.1, this maximum equals $\tau \varphi^\circ(H_t)$.

Part (ii). The Riemannian norm of the update is $\|\xi_t^*\|_{x_t}^2 = \|G_{x_t}^{1/2} \xi_t^*\|_F^2 = \|Z_t^*\|_F^2$. Since Z_t^* is the LMO maximizer, $\varphi(Z_t^*) = \tau$ (the norm constraint is tight because the objective is a non-degenerate linear functional). By Definition 5.1, $C_\varphi(x_t) = \max\{\|Z\|_F^2 : \varphi(Z) \leq 1, Z \in \mathcal{Z}_{x_t}\}$, so for any $Z \in \mathcal{Z}_{x_t}$ with $\varphi(Z) \leq \tau$, we have $\|Z\|_F^2 = \tau^2 \|Z/\tau\|_F^2 \leq \tau^2 C_\varphi(x_t) \leq C_\varphi \tau^2$. Applying this to Z_t^* gives $\|Z_t^*\|_F^2 \leq C_\varphi \tau^2$. $\square$

Proof of Theorem 5.3. By retraction- L -smoothness (Assumption 5.2), with $x_{t+1} = R_{x_t}(-\eta \xi_t^*)$:

$$\begin{aligned} f(x_{t+1}) &\leq f(x_t) - \eta g_{x_t}(\xi_t^*, \text{grad } f_t) + \frac{L\eta^2}{2} \|\xi_t^*\|_{x_t}^2 \\ &= f(x_t) - \eta D_t + \frac{L\eta^2}{2} \|Z_t^*\|_F^2 \\ (7) \quad &\leq f(x_t) - \eta \tau \varphi^\circ(H_t) + \frac{L\eta^2}{2} C_\varphi \tau^2. \end{aligned}$$

Here we used $\langle \xi_t^*, \nabla f_t \rangle = g_{x_t}(\xi_t^*, \text{grad } f_t) = \langle Z_t^*, H_t \rangle = D_t$ and $\|Z_t^*\|_F^2 \leq C_\varphi \tau^2$ (Lemma D.3).

Summing (7) over $t = 0, \dots, T-1$:

$$f(x_T) - f(x_0) \leq -\eta \tau \sum_{t=0}^{T-1} \varphi^\circ(H_t) + \frac{L\eta^2 C_\varphi \tau^2 T}{2}.$$

Rearranging and using $f(x_T) \geq f^*$:

$$\frac{1}{T} \sum_{t=0}^{T-1} \varphi^\circ(H_t) \leq \frac{\Delta_0}{\eta \tau T} + \frac{L\eta C_\varphi \tau}{2},$$

where $\Delta_0 = f(x_0) - f^*$. Setting $\eta = \sqrt{2\Delta_0/(LC_\varphi \tau^2 T)}$:

$$\min_{t \leq T} \varphi^\circ(H_t) \leq \frac{1}{T} \sum_t \varphi^\circ(H_t) \leq \sqrt{\frac{2LC_\varphi \Delta_0}{T}}.$$

$\square$

Proof of schedule (b). With $\eta_t = \sqrt{2\Delta_0/(LC_\varphi)}/(\tau\sqrt{t+1})$, write $c = \sqrt{2\Delta_0/(LC_\varphi)}$ so that $\eta_t = c/(\tau\sqrt{t+1})$ and $Lc^2 C_\varphi/2 = \Delta_0$. The descent step (7) with time-varying η_t gives

$$f(x_{t+1}) \leq f(x_t) - \frac{c \varphi^\circ(H_t)}{\sqrt{t+1}} + \frac{\Delta_0}{t+1}.$$

Summing over $t = 0, \dots, T-1$ and using $f(x_T) \geq f^*$:

$$c \sum_{t=0}^{T-1} \frac{\varphi^\circ(H_t)}{\sqrt{t+1}} \leq \Delta_0 + \Delta_0 \sum_{t=0}^{T-1} \frac{1}{t+1} \leq \Delta_0(2 + \ln T).$$

Since $\min_t \varphi^\circ(H_t) \cdot \sum_t 1/\sqrt{t+1} \leq \sum_t \varphi^\circ(H_t)/\sqrt{t+1}$ and $\sum_{t=0}^{T-1} 1/\sqrt{t+1} \geq 2(\sqrt{T}-1)$:

$$\min_{t \leq T} \varphi^\circ(H_t) \leq \frac{\Delta_0(2 + \ln T)}{2c(\sqrt{T}-1)} = O\left(\sqrt{\frac{LC_\varphi \Delta_0}{T}} \cdot \ln T\right),$$

matching schedule (a) up to a logarithmic factor, without requiring the horizon T in advance.

D.6. Computation of C_φ for the fixed-rank manifold. We compute $C_\varphi = 2r$ for $\mathcal{M}_r$ with $\varphi = \|\cdot\|_2$ (Remark 5.4). The decoupled LMO produces $Z_B^* = \text{Ortho}(H_B) \in \mathbb{R}^{m \times r}$ and $Z_A^* = \text{Ortho}(H_A) \in \mathbb{R}^{r \times n}$, where $H_A := (B^\top B)^{-1/2} \nabla_A f$. Each has rank $\leq r$, so $\|Z_B^*\|_F^2 \leq r$ and $\|Z_A^*\|_F^2 \leq r$. The total Riemannian norm is $\|\xi^*\|_x^2 = \|Z_B^*\|_F^2 + \|Z_A^*\|_F^2 \leq 2r$. Hence $C_\varphi = 2r$, and Theorem 5.3 gives rate $O(\sqrt{Lr/T})$.

D.7. Computation of C_φ for SPD, Stiefel, and Grassmann.

SPD ($C_\varphi = n$). $Z^* = \text{sign}(H) \in \text{Sym}(n)$ has eigenvalues ± 1 , so $\|Z^*\|_F^2 = \text{tr}(I_n) = n$.

Stiefel ($C_\varphi = 2\lfloor r/2 \rfloor + \min(m-r, r)$). With the product-norm decoupling, the skew block gives $Z_S^* = \text{Ortho}(S) \in \text{Skew}(r)$, a skew-orthogonal matrix whose singular values come in equal pairs $(1, 1, \dots)$, so $\|Z_S^*\|_F^2 = 2\lfloor r/2 \rfloor$ (one pair contributes 2 to the squared Frobenius norm; the zero block when r is odd contributes nothing). The normal block gives $Z_N^* = \text{Ortho}(N) \in \mathbb{R}^{(m-r) \times r}$ with rank $\leq \min(m-r, r)$, so $\|Z_N^*\|_F^2 \leq \min(m-r, r)$. The total is $C_\varphi = 2\lfloor r/2 \rfloor + \min(m-r, r)$.

Grassmann ($C_\varphi = \min(m-r, r)$). $Z^* = \text{Ortho}(\hat{H}) \in \mathbb{R}^{m \times r}$ with $\hat{H} = (I - XX^\top) \nabla_X f$ having column space in $\text{col}(X)^\perp \cong \mathbb{R}^{m-r}$. Thus $\text{Ortho}(\hat{H})$ has rank $\leq \min(m-r, r)$, giving $\|Z^*\|_F^2 \leq \min(m-r, r)$.

D.8. Proof of Theorem 5.7.

Proof of Theorem 5.7. The LMO applied to the stochastic gradient $\widetilde{\nabla} f_t$ produces $\tilde{\xi}_t^*$ with scaled variable $\tilde{Z}_t^* = G_{x_t}^{1/2} \tilde{\xi}_t^*$ satisfying $\varphi(\tilde{Z}_t^*) = \tau$. The per-step descent (Assumption 5.2) gives

$$f(x_{t+1}) \leq f(x_t) - \eta_t g_{x_t}(\tilde{\xi}_t^*, \text{grad } f_t) + \frac{L\eta_t^2}{2} C_\varphi \tau^2.$$

The inner product decomposes as

$$g_{x_t}(\tilde{\xi}_t^*, \text{grad } f_t) = \langle \tilde{Z}_t^*, H_t \rangle = \langle \tilde{Z}_t^*, \tilde{H}_t \rangle + \langle \tilde{Z}_t^*, H_t - \tilde{H}_t \rangle.$$

The first term equals $\tau \varphi^\circ(\tilde{H}_t)$ by LMO optimality (Lemma D.3). For the second, Hölder's inequality gives $|\langle \tilde{Z}_t^*, H_t - \tilde{H}_t \rangle| \leq \varphi(\tilde{Z}_t^*) \varphi^\circ(H_t - \tilde{H}_t) = \tau \varphi^\circ(\tilde{H}_t - H_t)$. By the reverse triangle inequality, $\varphi^\circ(\tilde{H}_t) \geq \varphi^\circ(H_t) - \varphi^\circ(\tilde{H}_t - H_t)$. Combining:

$$(8) \quad g_{x_t}(\tilde{\xi}_t^*, \text{grad } f_t) \geq \tau [\varphi^\circ(H_t) - 2\varphi^\circ(\tilde{H}_t - H_t)].$$

Substituting into the descent with $\eta_t = c/(\tau\sqrt{t+1})$:

$$f(x_{t+1}) \leq f(x_t) - \frac{c\varphi^\circ(H_t)}{\sqrt{t+1}} + \frac{2c\varphi^\circ(\tilde{H}_t - H_t)}{\sqrt{t+1}} + \frac{Lc^2C_\varphi}{2(t+1)}.$$

Taking expectations conditional on $x_0, \dots, x_t$ and using $\mathbb{E}[\varphi^\circ(\tilde{H}_t - H_t) | x_0, \dots, x_t] \leq \sigma_\varphi$ (by Jensen and Assumption 5.6), then summing over $t = 0, \dots, T-1$:

$$c \sum_{t=0}^{T-1} \frac{\mathbb{E}[\varphi^\circ(H_t)]}{\sqrt{t+1}} \leq \Delta_0 + 2c\sigma_\varphi \sum_{t=0}^{T-1} \frac{1}{\sqrt{t+1}} + \frac{Lc^2C_\varphi}{2} \sum_{t=0}^{T-1} \frac{1}{t+1}.$$

Using $\sum 1/\sqrt{t+1} \leq 2\sqrt{T}$ and $\sum 1/(t+1) \leq 1 + \ln T$, dividing both sides by $c \sum_{t=0}^{T-1} 1/\sqrt{t+1}$, and noting that the σ_φ term cancels exactly:

$$\min_{t \leq T} \mathbb{E}[\varphi^\circ(H_t)] \leq \frac{\Delta_0}{c \sum_{t=0}^{T-1} 1/\sqrt{t+1}} + 2\sigma_\varphi + \frac{LcC_\varphi(1 + \ln T)}{2 \sum_{t=0}^{T-1} 1/\sqrt{t+1}}.$$

Using $\sum_{t=0}^{T-1} 1/\sqrt{t+1} \geq 2(\sqrt{T}-1)$ in the denominator, and $\sqrt{T}-1 \geq \sqrt{T}/2$ for $T \geq 4$:

$$(9) \quad \min_{t \leq T} \mathbb{E}[\varphi^\circ(H_t)] \leq \frac{1}{\sqrt{T}} \left(\frac{\Delta_0}{c} + \frac{LcC_\varphi(1 + \ln T)}{2} \right) + 2\sigma_\varphi.$$

Setting $c = \sqrt{2\Delta_0/(LC_\varphi)}$ yields $O(\sqrt{LC_\varphi\Delta_0 \ln T/T}) + 2\sigma_\varphi$, recovering Theorem 5.3 (up to log factors) when $\sigma_\varphi = 0$. The additive $2\sigma_\varphi$ floor arises from the lack of variance reduction in the gradient estimator; it can be removed with gradient averaging or momentum techniques [44, 54], which we leave to future work. $\square$

APPENDIX E. SINGULAR VALUE INVARIANCE OF THE SCALED TANGENT SPACES

The closed-form solutions derived in Section 4 all rely on Proposition 3.3, which requires the scaled tangent space $\mathcal{Z}_x = \{G_x^{1/2}\xi : \xi \in T_x\mathcal{M}\}$ to be singular value (SV) invariant (Definition 3.2). Recall that SV-invariance means: if a matrix $U\Sigma V^\top$ lies in $\mathcal{Z}_x$, then so does $UDV^\top$ for every non-negative diagonal D of the same size. When this holds, the tangent space membership constraint $Z \in \mathcal{Z}_x$ in the LMO becomes redundant, the unconstrained maximizer Z^* of $\langle Z, H \rangle$ subject to $\varphi(Z) \leq \tau$ automatically lands in $\mathcal{Z}_x$ (because Z^* shares singular vectors with $H \in \mathcal{Z}_x$, and SV-invariance allows replacing singular values freely). This is what enables the clean closed forms $\text{Ortho}(H)$, $H/\|H\|_F$, and $u_1v_1^\top$ without any projection step.

Below we verify SV-invariance for each manifold treated in Section 4. For the fixed-rank and Grassmann manifolds, $\mathcal{Z}_x$ is a standard matrix space and the verification is immediate. For the SPD manifold, SV-invariance follows from the structure of symmetric matrices. For the Stiefel manifold, the full tangent space is *not* SV-invariant, which is why Section 4 uses the product-norm decoupling into skew and normal blocks; we verify SV-invariance of each block separately.

Fixed-rank manifold $\mathcal{M}_r$. With the coupled metric, $G_x^{1/2}(\dot{B}, \dot{A}) = (\dot{B}(AA^\top)^{1/2}, (B^\top B)^{1/2}\dot{A})$ and the tangent space is $T_x\mathcal{M}_r = \mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n}$. The scaled tangent space decomposes block-wise as $\mathcal{Z}_x = \mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n}$, each block is the full ambient matrix space of that dimension. Full matrix spaces are trivially SV-invariant (every $m \times r$ matrix lies in $\mathbb{R}^{m \times r}$, regardless of its singular values). This is why the LMO on $\mathcal{M}_r$ (Corollary 4.1) reduces to independent unconstrained problems on each factor block.

SPD manifold S_{++}^n . With the affine-invariant metric, $G_X^{1/2}\xi = X^{-1/2}\xi X^{-1/2}$ maps $T_X S_{++}^n = \text{Sym}(n)$ to $\mathcal{Z}_X = \text{Sym}(n)$. To check SV-invariance: a symmetric matrix M has SVD $M = U\Sigma V^\top$ where $V = U \text{diag}(\pm 1)$ (the sign pattern encodes the signs of the eigenvalues). Replacing Σ with any non-negative diagonal D gives $UDV^\top = UD \text{diag}(\pm 1)U^\top$, which is still symmetric. Hence $\text{Sym}(n)$ is SV-invariant, and the LMO on S_{++}^n reduces to the unconstrained problem yielding $\text{sign}(H)$, $H/\|H\|_F$, or rank-1 truncation.

Stiefel manifold $\text{St}(m, r)$ (block-wise). With $G_X = I$, the scaled tangent space coincides with the tangent space itself: $\mathcal{Z}_X = T_X \text{St} = \{XA + X_\perp B : A \in \text{Skew}(r), B \in \mathbb{R}^{(m-r) \times r}\}$. Since $[X \mid X_\perp] \in O(m)$, any $\xi \in T_X \text{St}$ can be written as $[X, X_\perp] \begin{pmatrix} A \\ B \end{pmatrix}$, so $\mathcal{Z}_X$ is isometric to $\{\begin{pmatrix} A \\ B \end{pmatrix} : A \in \text{Skew}(r), B \in \mathbb{R}^{(m-r) \times r}\}$. This space is *not* jointly SV-invariant: if $\begin{pmatrix} A \\ B \end{pmatrix} = U\Sigma V^\top$, replacing Σ with a different D can produce a top block that is no longer skew-symmetric. This is precisely why Section 4 uses the product-norm $\max(\varphi(A), \varphi(B)) \leq \tau$ to decouple the two blocks, and SV-invariance is checked for each block independently:

- *Normal block* $\mathbb{R}^{(m-r) \times r}$: the full ambient space, trivially SV-invariant.
- *Skew block* $\text{Skew}(r)$: a skew-symmetric matrix has real Schur form $S = Q \text{blkdiag}(\sigma_1 J, \sigma_2 J, \dots) Q^\top$ with $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, so its singular values come in equal pairs $(\sigma_1, \sigma_1, \sigma_2, \sigma_2, \dots)$. Replacing σ_i with any $d_i \geq 0$ gives $Q \text{blkdiag}(d_1 J, d_2 J, \dots) Q^\top$, which remains skew-symmetric. Hence $\text{Skew}(r)$ is SV-invariant under this paired replacement.

For $\varphi = \|\cdot\|_2$, the skew-block LMO sets all pairs to $(1, 1)$, producing a skew-orthogonal matrix; for $\varphi = \|\cdot\|_*$, it concentrates on the leading pair, giving a rank-2 skew-symmetric matrix.

Grassmann manifold $\text{Gr}(m, r)$. With $G_X = I$ and $X^\top X = I_r$, the horizontal space is $\mathcal{Z}_X = T_X \text{Gr} = \{Z \in \mathbb{R}^{m \times r} : X^\top Z = 0\}$. If $M = U\Sigma V^\top \in \mathcal{Z}_X$, then $X^\top M = 0$ implies $\text{col}(M) \subseteq \text{col}(X)^\perp$. Since $\dim \text{col}(X)^\perp = m - r \geq \text{rank}(M)$, all left singular vectors U can be chosen in $\text{col}(X)^\perp$. Replacing Σ with any non-negative diagonal D gives $UDV^\top$, whose column space is still in $\text{col}(X)^\perp$, so $X^\top(UDV^\top) = 0$. Hence $T_X \text{Gr}$ is SV-invariant, and no product-norm relaxation is needed on Grassmann, the LMO reduces directly to the unconstrained problem.

APPENDIX F. DERIVATIONS AND IMPLEMENTATION DETAILS

This appendix provides complete derivations of the intrinsic LMO solutions for all three standard norms (spectral, Frobenius, nuclear) on each manifold, along with step-by-step implementation procedures for the spectral instance. The general pipeline from Proposition 3.3 is: (i) scale $H = G_x^{1/2}(\text{grad } f)$, (ii) solve $Z^* = \arg \max_{\varphi(Z) \leq \tau} \langle Z, H \rangle$ in closed form via Section C, (iii) invert $\xi^* = G_x^{-1/2}Z^*$. The three standard norms

give $Z^* = \tau \text{Ortho}(H)$ (spectral), $Z^* = \tau H/\|H\|_F$ (Frobenius), and $Z^* = \tau u_1 v_1^\top$ (nuclear). Throughout, $\nabla_X f$ denotes the Euclidean gradient of f at $X \in \mathcal{M}$.

F.1. Summary of closed-form solutions. Table 3 collects the closed-form intrinsic LMO solutions on four manifolds under three norms.

TABLE 3. Closed-form intrinsic LMO solutions ξ^* (up to the norm bound τ) on four manifolds. $H_B = \nabla_B f (AA^\top)^{-1/2}$ and $H_A = (B^\top B)^{-1/2} \nabla_A f$ are the scaled factor gradients; $H = X^{1/2}(\nabla_X f)X^{1/2}$ is the scaled SPD gradient; $S = \text{skew}(X^\top \nabla_X f)$ and $N = X_\perp^\top \nabla_X f$ are the Stiefel skew and normal blocks; $\hat{H} = (I - XX^\top) \nabla_X f$ is the Grassmann horizontal gradient.

Manifold	$G_x^{1/2}(\xi)$	$\varphi = \ \cdot\ _2$ (spectral)	$\varphi = \ \cdot\ _F$ (Frobenius)	$\varphi = \ \cdot\ _*$ (nuclear)
$\mathcal{M}_r$ ($\dot{B}$ -block)	$\dot{B}(AA^\top)^{1/2}$	$\text{Ortho}(H_B)(AA^\top)^{-1/2}$	$\nabla_B f (AA^\top)^{-1}/\ \cdot\ $	$u_1 v_1^\top (AA^\top)^{-1/2}$
S_{++}^n	$X^{-1/2} \xi X^{-1/2}$	$X^{1/2} \text{sign}(H) X^{1/2}$	$X(\nabla_X f)X/\ \cdot\ _X$	$\text{sign}(\lambda_1) X^{1/2} q_1 q_1^\top X^{1/2}$
$\text{St}(m, r)$	I	$X \text{Ortho}(S) + X_\perp \text{Ortho}(N)$	$X(S/\ S\ _F) + X_\perp(N/\ N\ _F)$	rank-2 skew + rank-1
$\text{Gr}(m, r)$	I	$\text{Ortho}(\hat{H})$	$\hat{H}/\ \hat{H}\ _F$	$u_1 v_1^\top$ of $\hat{H}$

F.2. Computing the polar factor.

Newton–Schulz iterations. All spectral-norm instances require computing the polar factor $\text{Ortho}(M) = UV^\top$ from $M = U\Sigma V^\top$. Following Jordan [27], this can be approximated by Newton–Schulz iterations: $X_0 = M/\|M\|_F$, $X_{k+1} = \frac{3}{2}X_k - \frac{1}{2}X_k X_k^\top X_k$. Each iteration costs $O(pr^2)$ for $M \in \mathbb{R}^{p \times r}$ and converges cubically when $\|X_0\|_2 < \sqrt{3}$, which is ensured by the normalization since $\|X_0\|_2 = \|M\|_2/\|M\|_F \leq 1$. In practice, 5–10 iterations suffice.

The choice of algorithm depends on the shape and size of M :

Tall-skinny or short-wide matrices ($M \in \mathbb{R}^{p \times r}$ with $r \ll p$): dominant case on $\mathcal{M}_r$, $\text{Gr}(m, r)$, and the normal block of $\text{St}(m, r)$. Thin SVD costs $O(pr^2)$; Newton–Schulz avoids the SVD and is GPU-friendly.

Small square matrices ($M \in \mathbb{R}^{r \times r}$): arises for the skew block on $\text{St}(m, r)$. For skew-symmetric S , the real Schur decomposition gives $\text{Ortho}(S)$ directly and preserves skew-symmetry exactly.

Large square matrices ($M \in \mathbb{R}^{n \times n}$): arises on S_{++}^n where the “polar factor” is the matrix sign $\text{sign}(H)$. Newton iterations $X_{k+1} = \frac{1}{2}(X_k + X_k^{-1})$ converge quadratically; eigendecomposition is also standard.

F.3. Fixed-rank manifold $\mathcal{M}_r$.

Derivation (all three norms). On $\mathcal{M}_r$ with the coupled metric, $G_x^{1/2}(\dot{B}, \dot{A}) = (\dot{B}(AA^\top)^{1/2}, (B^\top B)^{1/2} \dot{A})$. The LMO decouples into independent $\dot{B}$ and $\dot{A}$ subproblems via (6). For the $\dot{B}$ -block, the scaled gradient is $H_B = \nabla_B f (AA^\top)^{-1/2}$ where $\nabla_B f = \nabla_X f A^\top$. By Proposition 3.3, Z_B^* is the unconstrained maximizer of $\langle Z_B, H_B \rangle$ subject to $\varphi(Z_B) \leq \tau$, and $\dot{B}^* = Z_B^* (AA^\top)^{-1/2}$. The three norms give:

- *Spectral* ($\varphi = \|\cdot\|_2$): $Z_B^* = \tau \text{Ortho}(H_B)$, so $\dot{B}^* = \tau \text{Ortho}(\nabla_B f (AA^\top)^{-1/2}) (AA^\top)^{-1/2}$.
- *Frobenius* ($\varphi = \|\cdot\|_F$): $Z_B^* = \tau H_B/\|H_B\|_F$, so $\dot{B}^* = \tau \nabla_B f (AA^\top)^{-1/2}/(\|H_B\|_F)$. This is the Riemannian preconditioned gradient of Mishra and Sepulchre [39], Zhang and Pilanci [69], normalized to unit intrinsic Frobenius norm.
- *Nuclear* ($\varphi = \|\cdot\|_*$): $Z_B^* = \tau u_1 v_1^\top$ from the SVD of H_B , so $\dot{B}^* = \tau u_1 v_1^\top (AA^\top)^{-1/2}$ (rank-1 update per block).

The $\dot{A}$ -block follows by symmetry with $H_A = (B^\top B)^{-1/2} \nabla_A f$.

QR-based implementation (spectral norm). We show that $\dot{B} = \tau \text{Ortho}(H_B) (AA^\top)^{-1/2}$ can be computed using only QR factors. Let $A^\top = \hat{Q}_A R_A$ be a thin QR factorization ($\hat{Q}_A \in \mathbb{R}^{n \times r}$ with orthonormal columns, $R_A \in \mathbb{R}^{r \times r}$ upper triangular with positive diagonal); similarly $B = \hat{Q}_B R_B$. The scaled factor gradient is the $m \times r$ matrix

$$H_B = \nabla_B f (AA^\top)^{-1/2} = \nabla_X f A^\top (AA^\top)^{-1/2} = \nabla_X f \hat{Q}_A R_A (AA^\top)^{-1/2}.$$

Writing $Q_A := (AA^\top)^{-1/2} A \in \mathbb{R}^{r \times n}$ (orthonormal rows), we have $H_B = \nabla_X f Q_A^\top$. Since $Q_A^\top$ and $\hat{Q}_A$ span the same column space, $Q_A^\top = \hat{Q}_A \Omega$ for some $\Omega \in O(r)$. Thus $\text{Ortho}(H_B) = \text{Ortho}(\nabla_X f \hat{Q}_A) \Omega$, and on the

quotient manifold the right factor $\Omega(AA^\top)^{-1/2}$ can be replaced by $R_A^{-\top}$ (since both give the same ambient update $\dot{B}$). In practice we compute $\dot{B} = \tau \text{Ortho}(\nabla_X f \hat{Q}_A) R_A^{-\top}$ and $\dot{A} = \tau R_B^{-1} \text{Ortho}(\hat{Q}_B^\top \nabla_X f)$.

Step-by-step procedure ($\dot{B}$ computation; $\dot{A}$ is symmetric):

- (1) **QR factorization:** $A^\top = \hat{Q}_A R_A$; similarly $B = \hat{Q}_B R_B$. $O((m+n)r^2)$.
- (2) **Projected gradient:** $\nabla_X f \hat{Q}_A \in \mathbb{R}^{m \times r}$. $O(mr^2)$.
- (3) **Polar factor:** $\text{Ortho}(\nabla_X f \hat{Q}_A)$ via Newton–Schulz. $O(mr^2)$ per iteration.
- (4) **Triangular solve:** $\dot{B} = \tau \text{Ortho}(\nabla_X f \hat{Q}_A) R_A^{-\top}$. $O(mr^2)$.
- (5) **Retract:** $B_+ = B - \eta \dot{B}$, $A_+ = A - \eta \dot{A}$. $O((m+n)r)$.

Total: $O((m+n)r^2 + r^3)$ per step. No eigendecomposition or matrix square root is needed.

F.4. SPD manifold S_{++}^n .

Derivation (all three norms). With the affine-invariant metric, $G_X^{1/2} \xi = X^{-1/2} \xi X^{-1/2}$ and $\text{grad } f(X) = X(\nabla_X f)X$. The scaled gradient is $H = G_X^{1/2}(\text{grad } f) = X^{1/2}(\nabla_X f)X^{1/2} \in \text{Sym}(n)$. Since $\mathcal{Z}_X = \text{Sym}(n)$ is SV-invariant, Proposition 3.3 applies. The three norms give Z^* and $\xi^* = X^{1/2} Z^* X^{1/2}$:

- *Spectral:* $Z^* = \tau \text{sign}(H) = \tau Q \text{diag}(\text{sign}(\lambda_i)) Q^\top$, so $\xi^* = \tau X^{1/2} \text{sign}(H) X^{1/2}$.
- *Frobenius:* $Z^* = \tau H / \|H\|_F$, so $\xi^* = \tau \text{grad } f / \|\text{grad } f\|_X$ (the normalized natural gradient).
- *Nuclear:* $Z^* = \tau \text{sign}(\lambda_1) q_1 q_1^\top$, so $\xi^* = \tau \text{sign}(\lambda_1) X^{1/2} q_1 q_1^\top X^{1/2}$ (rank-1 symmetric).

All three are $\text{GL}(n)$ -invariant by Proposition 3.1.

Implementation (spectral norm).

- (1) **Matrix square root:** Compute $X^{1/2}$ and $X^{-1/2}$ via eigendecomposition of X . $O(n^3)$.
- (2) **Scaled gradient:** $H = X^{1/2} \nabla_X f X^{1/2}$. $O(n^3)$.
- (3) **Matrix sign:** $\text{sign}(H) = Q \text{diag}(\text{sign}(\lambda_i)) Q^\top$ via eigendecomposition, or Newton iterations $Z_{k+1} = \frac{1}{2}(Z_k + Z_k^{-1})$. $O(n^3)$.
- (4) **Invert:** $\xi^* = \tau X^{1/2} \text{sign}(H) X^{1/2}$. $O(n^3)$.
- (5) **Retract:** $X_+ = X^{1/2} \exp(-\eta X^{-1/2} \xi^* X^{-1/2}) X^{1/2}$. $O(n^3)$.

Total: $O(n^3)$ per step, dominated by eigendecompositions.

F.5. Stiefel manifold $\text{St}(m, r)$.

Derivation (all three norms). With $G_X = I$, the tangent space decomposes as $\xi = XA + X_\perp B$ with $A \in \text{Skew}(r)$, $B \in \mathbb{R}^{(m-r) \times r}$. Under the product norm $\max(\varphi(A), \varphi(B)) \leq \tau$, the LMO decouples into skew and normal blocks with $S = \text{skew}(X^\top \nabla_X f)$ and $N = X_\perp^\top \nabla_X f$.

- *Spectral:* $A^* = \tau \text{Ortho}(S)$ (skew-orthogonal), $B^* = \tau \text{Ortho}(N)$. Result: $\xi^* = \tau(X \text{Ortho}(S) + X_\perp \text{Ortho}(N))$ (Lemma 4.2).
- *Frobenius:* $A^* = \tau S / \|S\|_F$, $B^* = \tau N / \|N\|_F$ (block-wise normalized).
- *Nuclear:* $A^* = \tau$ times the leading conjugate pair of S (rank-2 skew-symmetric with $\|A^*\|_* = \tau$); $B^* = \tau u_1 v_1^\top$ of N (rank-1).

Implementation (spectral norm). $X_\perp \in \mathbb{R}^{m \times (m-r)}$ is never formed explicitly; instead $X_\perp \text{Ortho}(X_\perp^\top \nabla_X f) = \text{Ortho}((I - XX^\top) \nabla_X f)$.

- (1) **Skew gradient:** $P = X^\top \nabla_X f$, then $S = \frac{1}{2}(P - P^\top)$. $O(mr^2)$.
- (2) **Skew polar factor:** $\text{Ortho}(S)$ via real Schur decomposition. $O(r^3)$.
- (3) **Normal gradient:** $\hat{N} = \nabla_X f - X P$. $O(mr^2)$.
- (4) **Normal polar factor:** $\text{Ortho}(\hat{N})$ via Newton–Schulz. $O(mr^2)$.
- (5) **Assemble:** $\xi^* = \tau[X \text{Ortho}(S) + \text{Ortho}(\hat{N})]$. $O(mr)$.
- (6) **Retract:** $X_+ = \text{qf}(X - \eta \xi^*)$. $O(mr^2)$.

Total: $O(mr^2 + r^3)$ per step. Steps 2 and 4 can be parallelized.

F.6. Grassmann manifold $\text{Gr}(m, r)$.

Derivation (all three norms). With $G_X = I$ and $X^\top X = I_r$, the horizontal space $T_X \text{Gr} = \{\xi : X^\top \xi = 0\}$ is SV-invariant. The horizontal gradient is $\hat{H} = (I - XX^\top) \nabla_X f$.

- *Spectral:* $\xi^* = \tau \text{Ortho}(\hat{H})$, orthonormal columns in $\text{col}(X)^\perp$.
- *Frobenius:* $\xi^* = \tau \hat{H} / \|\hat{H}\|_F$, normalized Riemannian gradient on $\text{Gr}(m, r)$ [16].

- *Nuclear*: $\xi^* = \tau u_1 v_1^\top$ from the SVD of $\hat{H}$, rank-1 horizontal update.

Implementation (spectral norm).

- (1) **Horizontal projection**: $\hat{H} = \nabla_X f - X(X^\top \nabla_X f)$. $O(mr^2)$.
- (2) **Polar factor**: $\text{Ortho}(\hat{H})$ via Newton–Schulz. $O(mr^2)$.
- (3) **Assemble**: $\xi^* = \tau \text{Ortho}(\hat{H})$. $O(mr)$.
- (4) **Retract**: $X_+ = \text{qf}(X - \eta \xi^*)$. $O(mr^2)$.

Total: $O(mr^2)$ per step. The simplest instantiation: $G_X = I$ eliminates scaling and inversion.

APPENDIX G. PRIOR CONVERGENCE RESULTS AS SPECIAL CASES

We show how several known rates arise as special cases of Theorems 5.3 and 5.7.

Euclidean spectral steepest descent [6, 13]. Setting $\mathcal{M} = \mathbb{R}^{p \times q}$ (unconstrained), $G_x = I$, and $\varphi = \|\cdot\|_2$ recovers the Euclidean Muon update $\xi^* = \tau \text{Ortho}(\nabla f)$. The use of this spectral-norm steepest-descent operator for deep learning was first proposed by Carlson et al. [13], who also combined it with element-wise adaptive preconditioning; Bernstein and Newhouse [6, Prop. 5] later formalize the connection to steepest descent. The squared Riemannian radius is $C_\varphi = \min(p, q)$ (rank of $\text{Ortho}(\nabla f)$), and Theorem 5.3 gives the nonasymptotic rate $O(\sqrt{L \min(p, q)/T})$.

Standard Riemannian gradient descent [11]. Setting $\varphi = \|\cdot\|_F$ on any manifold gives $\xi^* \propto \text{grad } f / \|\text{grad } f\|_x$ and $C_\varphi = 1$. Theorem 5.3 yields $\min_t \|\text{grad } f(x_t)\|_{x_t} \leq \sqrt{2L\Delta_0/T}$, which is the standard retraction-smooth Riemannian GD rate [11, Cor. 4.9] (with the normalized-gradient step replacing the unnormalized one; the rates coincide up to constants when η is optimized).

Factor-wise Muon on $\mathcal{M}_r$ [29]. Kang et al. [29] analyze factor-wise Muon (SpecGF) on $\mathcal{M}_r$ with $\varphi = \|\cdot\|_2$. Since their orthogonalization operator satisfies $\|T(\cdot)\|_2 = 1$, the ambient update norm satisfies $\|\dot{X}_t\|_F^2 \leq 2\tau^2(\|A_t\|_F^2 + \|B_t\|_F^2)$, which depends on factor conditioning. Accordingly, their main convergence result (Theorem 6.2) is conditional: SpecGF either converges to a global minimum or $\|A_t\|_F^2 + \|B_t\|_F^2 \rightarrow \infty$; unconditional global convergence is established only with ℓ_2 regularization. Our quotient formulation sidesteps this issue entirely: the Gram-root scaling yields a rank-only bound $\|\xi^*\|_x^2 \leq 2r\tau^2$ with $C_\varphi = 2r$ (Remark 5.4, computed in Section D.6), giving a nonasymptotic rate $O(\sqrt{Lr/T})$ without any bounded-factor-norm condition.

NuMuon [43]. Setting $G_x = I$ and $\varphi = \|\cdot\|_*$ gives $\xi^* = \tau u_1 v_1^\top$ (rank-1 update) with $C_\varphi = 1$. Theorem 5.3 gives rate $O(\sqrt{L/T})$ with convergence in $\varphi^\circ = \|\cdot\|_2$ (spectral norm of the gradient). NuMuon [43] uses a joint spectral-and-nuclear-norm constraint ($\|O\|_2 \leq \rho$, $\|O\|_* \leq \tau$), yielding a rank- k update ($k = \lfloor \tau/\rho \rfloor$); their Theorem 3.6 gives a rate with an Lk -dependent term and convergence in $\|\nabla f\|_*$ (nuclear norm of the gradient). Our pure nuclear instance ($k = 1$) recovers the rank-1 special case. On manifolds with $G_x \neq I$, our quotient extensions add symmetry invariance while preserving $C_\varphi = 1$.

General Euclidean LMOs [48]. Pethick et al. [48] study LMOs constrained by general norms (including spectral, RMS, and sign norms) in Euclidean space ($G_x = I$), and propose Scion, a unified algorithmic family. Their spectral-norm instance corresponds to our framework with $G_x = I$ on $\mathbb{R}^{p \times q}$. Our contribution beyond their setting is the non-trivial metric $G_x \neq I$ (which introduces the metric-scaling and inversion steps and the symmetry proposition) and the per-manifold closed forms.

The role of C_φ . Across all these instances, the squared Riemannian radius C_φ is the unifying quantity: it is $\min(p, q)$ for Euclidean Muon, 1 for Frobenius/nuclear norms, $2r$ for quotient-Muon on $\mathcal{M}_r$, and iterate-dependent (scaling with $\|A_t\|_F^2 + \|B_t\|_F^2$) for factor-wise Muon. The spectral norm always gives the largest C_φ but the strongest convergence criterion ($\varphi^\circ = \|\cdot\|_*$); the nuclear norm gives the smallest C_φ but the weakest criterion ($\varphi^\circ = \|\cdot\|_2$).

APPENDIX H. DETAILED COMPARISONS WITH PRIOR METHODS

In the main text (Section 4), we briefly compare our framework to prior methods on the fixed-rank manifold. Here we provide detailed calculations and side-by-side comparisons on all four manifolds.

H.1. Fixed-rank manifold $\mathcal{M}_r$: iMuon vs. Spectron, Riemannion, and factor-wise Muon. We compare three approaches to spectral-norm optimization on the rank- r manifold $\mathcal{M}_r$ with factorization $X = BA$. Throughout, $\nabla_X f$ denotes the Euclidean gradient.

Prior methods. *Factor-wise Muon* applies the standard Euclidean Muon [6, 7, 27] polar factor $\text{Ortho}(\cdot)$ to each factor’s gradient independently: $\dot{B} = \tau \text{Ortho}(\nabla_X f A^\top)$ and $\dot{A} = \tau \text{Ortho}(B^\top \nabla_X f)$. This is simple and requires no Riemannian structure, but the resulting update $\dot{X} = \dot{B}A + B\dot{A}$ has $\|\dot{X}\|_F^2 \propto \|A\|_F^2 + \|B\|_F^2$, which depends on factor conditioning. A closely related theoretical analysis is Kang et al. [29], who study the *continuous-time* spectral gradient flow (SpecGF, the momentum-free limit of factor-wise spectral GD with a smoothed orthogonalization map T_ε) on $L(B, A) = \|BA - Y\|_F^2$ rather than discrete-time factor-wise Muon directly. Their convergence guarantee is stated as a *dichotomy* (Theorem 6.2): for almost every initialization, SpecGF either converges to a global minimum of L , or the factor norms $\|B(t)\|_F^2 + \|A(t)\|_F^2$ diverge to infinity. The standard gradient-flow balancing invariant $A^\top A - BB^\top$ that would otherwise control factor growth is not preserved under SpecGF (Proposition D.10), leaving *unconditional* global convergence without regularization as an explicit open question. Kang et al. [29, Proposition 6.7] establishes an exponential convergence rate *conditional on* the convergent branch and on simple singular values of Y ; [29, Section D.6] closes the gap with ℓ_2 regularization, which keeps factors bounded by construction. Our analysis (Theorem 5.3 and Remark 5.4) operates instead in discrete time on the quotient manifold $\mathcal{M}_r$ with momentum, removes the bounded-factor-norm requirement, and provides a non-asymptotic rate that depends on the manifold dimension only.

Spectron [26] addresses the same factor-growth pathology from a practical angle. Like factor-wise Muon, Spectron orthogonalizes each factor’s momentum via Newton–Schulz iterations, producing $\text{Ortho}(M_A)$ and $\text{Ortho}(M_B)$. The key innovation is a *spectral renormalization* step that adaptively scales the updates to control the composite update norm. Since the ambient update satisfies $\|\dot{X}\|_2 \leq \rho(\|A\|_2 + \|B\|_2 + 1)$ (via submultiplicativity and the triangle inequality), Spectron sets the per-factor constraint radius to $\rho = \eta/(\|A\|_2 + \|B\|_2 + 1)$, yielding factor updates $\dot{A} = \rho \text{Ortho}(M_A)$ and $\dot{B} = \rho \text{Ortho}(M_B)$, which guarantee $\|\dot{X}\|_2 \leq \eta$. The spectral norms $\|A\|_2, \|B\|_2$ are estimated at each iteration via a single power iteration step. This makes training stable and enables native low-rank pretraining that matches dense-model performance.

Spectron and our method address the same instability but through fundamentally different mechanisms. Spectron is an *extrinsic* fix: it applies standard (Euclidean) orthogonalization to each factor and then rescales the result by an iterate-dependent quantity $1/(\|A\|_2 + \|B\|_2 + 1)$ to bound the ambient update. The rescaling shrinks as the factors grow, which is effective in practice but means the effective step size is not a fixed hyperparameter, it depends on the current factor norms. Our method is *intrinsic*: the Gram-root preconditioning $(AA^\top)^{-1/2}$ from the coupled metric absorbs the factor scaling *before* orthogonalization, so the resulting update is $\text{GL}(r)$ -invariant by construction (Proposition 3.1) and the ambient update norm is bounded by $2r\tau^2$ regardless of factor magnitudes (Section D.6), with no power iteration or adaptive rescaling needed.

Riemannion [10] takes an *embedded-manifold* approach. It optimizes X directly on $\mathcal{M}_r = \{X \in \mathbb{R}^{m \times n} : \text{rank}(X) = r\}$, parameterized via a compact SVD $X = A_L \Sigma_r B_R^\top$ with $A_L \in \text{St}(m, r)$, $B_R \in \text{St}(n, r)$, and $\Sigma_r \in \mathbb{R}^{r \times r}$ diagonal positive. At a point $X \in \mathcal{M}_r$, the tangent space $T_X \mathcal{M}_r$ consists of $\text{rank} \leq 2r$ matrices; in SVD coordinates, $\dot{X} = A_L M B_R^\top + U_\perp B_R^\top + A_L V_\perp^\top$ for $M \in \mathbb{R}^{r \times r}$, $U_\perp \perp A_L$, and $V_\perp \perp B_R$. Riemannion maintains a momentum $M_t \in T_X \mathcal{M}_r$ ($\text{rank} \leq 2r$) via Riemannian heavy-ball updates with vector transport. To generalize Muon, it applies a rank- $2r$ orthogonalization $\text{Ortho}_r(M_t)$, replacing only the first $2r$ singular values of M_t with 1 while setting the rest to 0, and then projects back onto the tangent space: $\tilde{M}_t = P_{T_X \mathcal{M}_r}(\text{Ortho}_r(M_t))$. Because Ortho_r preserves the column and row spaces of $M_t \in T_X \mathcal{M}_r$, the result is a good approximation (the authors report singular values in (0.9, 1.1) in practice), but not exact. The iterate is then updated via retraction: $X_{t+1} = R_X(-\eta \tilde{M}_t)$.

We can equivalently describe the entire procedure in the $X = BA$ notation used throughout this section. The embedded tangent space is $T_X \mathcal{M}_r = \{\dot{B}A + B\dot{A} : \dot{B} \in \mathbb{R}^{m \times r}, \dot{A} \in \mathbb{R}^{r \times n}\}$, a subspace of $\text{rank} \leq 2r$ matrices in $\mathbb{R}^{m \times n}$. The orthogonal projection of any $Z \in \mathbb{R}^{m \times n}$ onto $T_X \mathcal{M}_r$ (under the Euclidean metric $G_X = I$) is $P_{T_X \mathcal{M}_r}(Z) = P_B Z + Z P_A - P_B Z P_A$, where $P_B = B(B^\top B)^{-1} B^\top$ projects onto $\text{col}(B)$ and $P_A = A^\top (A A^\top)^{-1} A$ projects onto $\text{row}(A)$. Riemannion’s update in BA language is then: (i) maintain momentum $M_t \in T_X \mathcal{M}_r$ ($\text{rank} \leq 2r$); (ii) orthogonalize $\text{Ortho}_r(M_t)$ (set all $2r$ nonzero singular values to 1); (iii) project back via $\tilde{M}_t = P_B \text{Ortho}_r(M_t) + \text{Ortho}_r(M_t) P_A - P_B \text{Ortho}_r(M_t) P_A$; (iv) retract $X_{t+1} = R_X(-\eta \tilde{M}_t)$. The projection in step (iii) couples B and A through P_B and P_A : the result depends on both factors simultaneously, and there is no block-diagonal decoupling into independent $\dot{B}$ and $\dot{A}$ subproblems. Moreover, the projection perturbs the singular values away from 1, so the spectral-norm constraint $\|\tilde{M}_t\|_2 \leq 1$ is only approximately satisfied.

Because Riemannion works on $\mathcal{M}_r$ directly (not on factors (A, B)), it is reparameterization-invariant: different factorizations of the same X yield the same update. However, it uses the Euclidean metric ($G_X = I$) on the embedded manifold, which provides no preconditioning, the convergence behavior still depends on the conditioning of X (i.e., $\sigma_1(X)/\sigma_r(X)$). To obtain an exact spectral-norm solution, Riemannion would need to solve the tangent-constrained LMO $\max_{S \in T_X \mathcal{M}_r, \|S\|_2 \leq 1} \langle M_t, S \rangle$, which has no closed form because $T_X \mathcal{M}_r$ is not SV-invariant in the embedded viewpoint (Section P.2).

Our method (iMuon) resolves both issues by working on the *quotient* $\mathcal{M}_r = (\mathbb{R}_*^{m \times r} \times \mathbb{R}_*^{r \times n})/\text{GL}(r)$ with the coupled metric. The intrinsic scaling $G_x^{1/2}$ converts the factor gradients into Euclidean coordinates via the Gram-root preconditioning $(AA^\top)^{-1/2}$, and the product-norm decoupling yields independent closed-form LMOs on each factor block. Unlike Riemannion, the spectral-norm constraint is *exactly* satisfied on each block (no approximation), and the coupled metric provides automatic preconditioning that makes the convergence rate independent of factor conditioning ($C_\varphi = 2r$). Unlike Spectron, no power iteration or adaptive rescaling is needed, the $\text{GL}(r)$ -invariance of the intrinsic norm handles factor-scale dependence algebraically rather than numerically.

Side-by-side update rules.

Method	$\dot{B}$	$\dot{A}$
Factor-wise Muon	$\tau \text{Ortho}(\nabla_X f A^\top)$	$\tau \text{Ortho}(B^\top \nabla_X f)$
Spectron	$\rho \text{Ortho}(M_B)$	$\rho \text{Ortho}(M_A)$
Riemannion	$\dot{X} = P_{T_X \mathcal{M}_r}(\text{Ortho}_r(M_t))$, embedded manifold with $G_X = I$	
Ours (iMuon)	$\tau \text{Ortho}(\nabla_X f \hat{Q}_A) R_A^{-\top}$	$\tau R_B^{-1} \text{Ortho}(\hat{Q}_B^\top \nabla_X f)$

where $A^\top = \hat{Q}_A R_A$ and $B = \hat{Q}_B R_B$ are thin QR factorizations, M_A, M_B are per-factor momentum buffers, and $\rho = \eta/(\|A\|_2 + \|B\|_2 + 1)$ is Spectron's adaptive constraint radius. The key differences are:

- *Ortho input*: factor-wise Muon and Spectron both apply Ortho to raw factor gradients (or momenta). Our method applies Ortho to the *scaled* gradient $\nabla_X f \hat{Q}_A$, which projects $\nabla_X f$ onto the row space of A with orthonormalized rows, a different polar direction in general.
- *Norm control mechanism*: factor-wise Muon uses a fixed τ ; Spectron uses the adaptive radius ρ that shrinks as factors grow; Riemannion uses tangent space projection after rank- $2r$ orthogonalization (approximate). Our method uses the Gram-root preconditioning $R_A^{-\top}$ (equivalently $(AA^\top)^{-1/2}$), which absorbs the factor scale *before* orthogonalization.
- *Reparameterization invariance*: our update is $\text{GL}(r)$ -invariant by Proposition 3.1. Riemannion is also reparameterization-invariant (it works on $\mathcal{M}_r$ directly), but violates the spectral-norm constraint after projection. Factor-wise Muon and Spectron are not reparameterization-invariant: different factorizations of the same X yield different updates. Spectron compensates via the adaptive ρ , which bounds the *ambient* update norm but does not make the *direction* invariant.

Ambient update norm comparison. The ambient update $\dot{X} = \dot{B}A + B\dot{A}$ enters the convergence rate through the descent lemma. We compute $\|\dot{X}\|_F^2$ for each method:

Our method. Define $Q_A = (AA^\top)^{-1/2}A \in \mathbb{R}^{r \times n}$, which has orthonormal rows ($Q_A Q_A^\top = I_r$). Then $\dot{B}A = \tau \text{Ortho}(H_B)(AA^\top)^{-1/2}A = \tau \text{Ortho}(H_B)Q_A$. Since $\text{Ortho}(H_B) \in \mathbb{R}^{m \times r}$ has orthonormal columns:

$$\|\dot{B}A\|_F^2 = \tau^2 \text{tr}(Q_A^\top I_r Q_A) = \tau^2 \text{tr}(I_r) = \tau^2 r.$$

Similarly $\|B\dot{A}\|_F^2 = \tau^2 r$. By the triangle inequality: $\|\dot{X}\|_F \leq 2\tau\sqrt{r}$, so $\|\dot{X}\|_F^2 \leq 4\tau^2 r$. This bound depends only on the rank r , not on the factor norms, the Gram root converts A into the partial isometry Q_A , exactly canceling the factor norm dependence.

Factor-wise Muon. $\dot{B}A = \tau \text{Ortho}(\nabla_X f A^\top)A$. Since $\text{Ortho}(\nabla_X f A^\top)$ has orthonormal columns, $\|\dot{B}A\|_F^2 = \tau^2 \|A\|_F^2$. Similarly $\|B\dot{A}\|_F^2 = \tau^2 \|B\|_F^2$. Thus $\|\dot{X}\|_F^2 \leq 2\tau^2(\|A\|_F^2 + \|B\|_F^2)$, *grows with factor norms*. This iterate-dependent bound is the reason the convergence analysis of Kang et al. [29] yields only a conditional asymptotic guarantee (converge or diverge) rather than a nonasymptotic rate; our method removes the dependence entirely.

Riemannion. $\dot{X} = P_{T_X \mathcal{M}_r}(\text{Ortho}_r(M_t))$, where Ortho_r replaces the first $2r$ singular values of M_t with 1. The tangent space projection after rank- $2r$ orthogonalization perturbs singular values (they lie in $(0.9, 1.1)$)

rather than exactly 1), so the spectral-norm constraint $\|\dot{X}\|_2 \leq \tau$ is only approximately satisfied. Moreover, since the Euclidean metric provides no Gram-root scaling, the update norm $\|\dot{X}\|_F^2$ depends on the conditioning of X (σ_1/σ_r), no condition-number-free bound analogous to our $\|\xi^*\|_x^2 \leq 2r\tau^2$ is available.

Spectral-norm control. Our method also controls the spectral norm of the ambient update: $\|\dot{X}\|_2 \leq \|\dot{B}A\|_2 + \|B\dot{A}\|_2 \leq 2\tau$, since $\|\dot{B}A\|_2 = \|\text{Ortho}(H_B)Q_A\|_2 \leq 1 \cdot 1 \cdot \tau = \tau$ (orthonormal columns times orthonormal rows). Factor-wise Muon gives $\|\dot{X}\|_2 \leq \tau(\|A\|_2 + \|B\|_2)$, which can be arbitrarily large.

Nuclear norm ($\varphi = \|\cdot\|_*$). The nuclear instance replaces $\text{Ortho}(H_B)$ with the leading left/right singular vectors $u_1v_1^\top$ of H_B , giving a rank-1 update per factor. The ambient update $\dot{X} = \dot{B}A + B\dot{A}$ has rank ≤ 2 and $\|\dot{X}\|_F^2 \leq 4\tau^2$ (each rank-1 block contributes τ^2 , independent of factor norms). This is the intrinsic analogue of NuMuon [43]: NuMuon constrains the update with both a spectral-norm cap ρ and a nuclear-norm budget τ , yielding a rank- k update ($k = \lfloor \tau/\rho \rfloor$) from the top- k singular vectors of the momentum. In the rank-1 special case ($k = 1$), i.e., $\tau = \rho$, NuMuon produces $u_1v_1^\top$, while our nuclear instance applies the rank-1 truncation to the *scaled* gradient $H_B = \nabla_B f(AA^\top)^{-1/2}$, inheriting the same $\text{GL}(r)$ -invariance and condition-number-free bound as the spectral instance.

H.2. SPD manifold S_{++}^n : spectral LMO vs. natural gradient. On the SPD manifold with the affine-invariant metric, the standard Riemannian optimization method is the *natural gradient* [2], which corresponds to the Frobenius instance ($\varphi = \|\cdot\|_F$) of our framework: $\xi^* \propto \text{grad } f / \|\text{grad } f\|_X$ where $\text{grad } f(X) = X(\nabla_X f)X$. The natural gradient preserves the relative magnitudes of the scaled gradient's eigenvalues, directing more step budget toward larger-eigenvalue directions.

The spectral instance ($\varphi = \|\cdot\|_2$) of our framework produces a qualitatively different update: the matrix-sign $\xi^* = \tau X^{1/2} \text{sign}(H)X^{1/2}$, where $H = X^{1/2}(\nabla_X f)X^{1/2}$. This equalizes all eigenvalues to ± 1 , redistributing the step uniformly across eigendirections, the SPD analogue of how Euclidean Muon replaces the gradient with its polar factor.

The difference is analogous to Muon vs. SGD in Euclidean space:

- The natural gradient preserves the magnitude profile of H 's eigenvalues, large eigenvalues of H receive proportionally larger step components.
- The matrix-sign update equalizes all eigenvalues to ± 1 , redistributing the step uniformly across eigendirections. This is more aggressive (higher squared Riemannian radius $C_\varphi = n$ vs. 1) but drives all eigenvalues of the scaled gradient to zero simultaneously.

The naive alternative, applying Ortho or sign to the Euclidean gradient $\nabla_X f$ directly (without scaling by $X^{-1/2}$), would not be $\text{GL}(n)$ -invariant. For example, $\text{sign}(\nabla_X f) \neq \text{sign}(N^\top(\nabla_X f)N)$ in general under $X \mapsto NXN^\top$, whereas $\text{sign}(H) = \text{sign}(X^{1/2}(\nabla_X f)X^{1/2})$ transforms covariantly because the scaling absorbs the change of basis.

Nuclear norm ($\varphi = \|\cdot\|_*$). The nuclear instance selects the leading eigenpair (q_1, λ_1) of $H = X^{1/2}(\nabla_X f)X^{1/2}$ and produces a rank-1 symmetric update $\xi^* = \tau \text{sign}(\lambda_1) X^{1/2} q_1 q_1^\top X^{1/2}$. This concentrates the entire norm bound on the single most-curved eigendirection of the scaled gradient. Compared to the spectral instance ($C_\varphi = n$, all eigendirections active), the nuclear instance has $C_\varphi = 1$ and converges in the spectral norm of the gradient ($\varphi^\circ = \|\cdot\|_2$), which is weaker but cheaper.

H.3. Stiefel manifold $\text{St}(m, r)$: product norm vs. joint norm. On the Stiefel manifold with $G_X = I$, no prior spectral-norm optimizer exists. The natural baseline is the *joint-norm* LMO: constrain $\varphi(\xi) \leq \tau$ over the full tangent space $T_X \text{St} \subset \mathbb{R}^{m \times r}$. However, as discussed in Section P.3, this tangent space is not SV-invariant, so no closed-form spectral solution exists.

Our framework resolves this via the tangent space decomposition $\xi = XA + X_\perp B$ with the *product norm* $\max(\varphi(A), \varphi(B)) \leq \tau$. This is a deliberate relaxation of the joint norm that enables decoupled, closed-form solutions on each block. We now analyze precisely what this relaxation costs and buys.

What the product norm buys.

- *Closed-form solutions*: the A - and B -blocks decouple, each admitting a direct application of Proposition 3.3 (with the skew-symmetric constraint on A handled by the paired singular value structure).
- *Block-wise control*: the skew block A (which rotates within the current column space) and the normal block B (which moves orthogonally to it) are each independently constrained, preventing one from dominating the other.

What the product norm costs.

- *Relaxation*: the product norm $\max(\varphi(A), \varphi(B)) \leq \tau$ is a relaxation of the joint norm $\varphi\left(\begin{smallmatrix} A \\ B \end{smallmatrix}\right) \leq \tau$. Specifically, for $\varphi = \|\cdot\|_2$: $\max(\|A\|_2, \|B\|_2) \leq \left\|\begin{smallmatrix} A \\ B \end{smallmatrix}\right\|_2 \leq \sqrt{2} \max(\|A\|_2, \|B\|_2)$. Thus $\max(\|A\|_2, \|B\|_2) \leq \tau$ is a relaxation of $\|\xi\|_2 \leq \tau$, it permits tangent vectors with $\|\xi\|_2$ up to $\sqrt{2}\tau$.
- *Frobenius discrepancy*: for $\varphi = \|\cdot\|_F$, the product-norm solution normalizes A and B independently (block-wise), while the joint-norm solution normalizes ξ jointly, preserving the relative magnitudes $\|S\|_F / \|N\|_F$. The two coincide only when $\|S\|_F = \|N\|_F$.

The squared Riemannian radius reflects this: $C_\varphi = 2\lfloor r/2 \rfloor + \min(m-r, r)$ counts both blocks, and is larger than the C_φ from a hypothetical joint-norm LMO (which the framework cannot compute in closed form).

Nuclear norm ($\varphi = \|\cdot\|_*$). The nuclear instance concentrates on the leading singular direction of each block: a rank-2 skew-symmetric matrix A^* from the leading conjugate pair of $S = \text{skew}(X^\top \nabla_X f)$ (with $\|A^*\|_* = \tau$, $\|A^*\|_F^2 = 2\tau^2$), and a rank-1 matrix $B^* = \tau u_1 v_1^\top$ from the normal block $X_\perp^\top \nabla_X f$. The total squared Riemannian radius is $C_\varphi = 1 + 1 = 2$ (one from each block), compared to $2\lfloor r/2 \rfloor + \min(m-r, r)$ for the spectral norm. The product-norm decoupling is especially natural here: the skew and normal nuclear updates are geometrically independent (one rotates within the current column space, the other adds a new direction).

Comparison with MCSD/SPEL [68]. MCSD (Manifold Constrained Steepest Descent) is a recent single-loop framework for norm-constrained optimization on embedded submanifolds. Its Stiefel specialization, SPEL, computes the spectral-norm LMO of the Riemannian gradient $\nabla_{\mathcal{M}} f(X_t) = P_{T_{X_t} \text{St}}(\nabla f(X_t))$ in the *ambient* Euclidean space, $d_t = -\text{Ortho}(\nabla_{\mathcal{M}} f(X_t))$, and then projects back to $\text{St}(m, r)$ via the polar map: $X_{t+1} = \text{Ortho}(X_t + \eta_t d_t) = \text{Ortho}(X_t - \eta_t \text{Ortho}(\nabla_{\mathcal{M}} f(X_t)))$. A defining design choice of MCSD is that the update direction d_t *need not* lie in $T_X \text{St}$; manifold feasibility is restored solely by the polar projection step, which is what makes MCSD single-loop.

Both SPEL and our Stiefel LMO (Lemma 4.2) avoid solving the intractable joint-norm tangent space problem $\max_{\xi \in T_X \text{St}, \|\xi\|_2 \leq \tau} \langle \xi, \nabla f \rangle$, but they use different relaxation strategies:

- *Our approach* decomposes the tangent space into skew and normal blocks with a product norm $\max(\|A\|_2, \|B\|_2) \leq \tau$, solving each block’s LMO in closed form. The update remains in $T_X \text{St}$ by construction, and each block receives independent spectral control.
- *SPEL* drops the tangent space membership constraint, solving the LMO over all of $\mathbb{R}^{m \times r}$, and compensates with the polar manifold projection. This yields a simpler per-iteration formula (one Ortho of an $m \times r$ matrix to obtain the LMO direction, plus one Ortho for the polar projection) but couples the skew (rotation within $\text{col}(X)$) and normal (extension into $\text{col}(X)^\perp$) contributions into a single ambient direction. The MCSD framework also extends beyond Stiefel to other embedded submanifolds (Grassmann, sphere) but does not natively handle quotient geometries such as $\mathcal{M}_r$ with the coupled metric.

H.4. Grassmann manifold $\text{Gr}(m, r)$: connections to subspace tracking. The Grassmann manifold $\text{Gr}(m, r)$ represents r -dimensional subspaces of $\mathbb{R}^m$. With $G_X = I$ and $X^\top X = I_r$, the horizontal space $\{\xi : X^\top \xi = 0\}$ is SV-invariant, so Proposition 3.3 applies directly and no product-norm relaxation is needed. The three norm instances of our framework produce geometrically distinct updates that connect to known methods in subspace estimation:

- *Spectral* ($\varphi = \|\cdot\|_2$): $\xi^* = \tau \text{Ortho}((I - XX^\top) \nabla_X f)$, an $m \times r$ matrix with orthonormal columns in $\text{colspan}(X)^\perp$, updating all $\min(m-r, r)$ subspace directions simultaneously.
- *Frobenius* ($\varphi = \|\cdot\|_F$): $\xi^* = \tau (I - XX^\top) \nabla_X f / \|(I - XX^\top) \nabla_X f\|_F$, the normalized Riemannian gradient on $\text{Gr}(m, r)$ [16], whose discrete updates are closely related to Oja’s subspace rule.
- *Nuclear* ($\varphi = \|\cdot\|_*$): $\xi^* = \tau u_1 v_1^\top$ from the SVD of $(I - XX^\top) \nabla_X f$, a rank-1 horizontal update ($C_\varphi = 1$).

The nuclear instance is closely related to GROUSE (Grassmannian Rank-One Update Subspace Estimation) [5], which performs online subspace tracking via rank-1 updates on $\text{Gr}(m, r)$. Both produce a rank-1 tangent vector in the horizontal space; the difference is that GROUSE derives its direction from a *single observation’s residual* (streaming setting with one sample at a time), whereas our LMO uses the *leading singular vectors of the full horizontal gradient* $(I - XX^\top) \nabla_X f$ (batch or mini-batch). Geometrically, each

rank-1 update adds one new direction $u_1 \perp \text{colspan}(X)$ while downweighting one direction v_1 within the current subspace. Since $G_X = I$ on $\text{Gr}(m, r)$, no scaling is needed, the Euclidean and intrinsic nuclear norms coincide, and the nuclear iMuon on $\text{Gr}(m, r)$ is simply NuMuon restricted to the horizontal space.

APPENDIX I. BACKGROUND ON RIEMANNIAN OPTIMIZATION

This appendix provides a more thorough discussion of the background material summarized in Section 1, aimed at readers less familiar with Riemannian optimization on matrix manifolds.

I.1. Riemannian optimization: key concepts. Riemannian optimization [1, 11] generalizes gradient-based methods to curved spaces (manifolds) by replacing Euclidean gradients with Riemannian gradients and straight-line updates with retractions. The key ingredients are:

- A *tangent space* $T_x\mathcal{M}$: the set of valid update directions at $x \in \mathcal{M}$.
- A *Riemannian metric* g_x : a smoothly varying inner product on $T_x\mathcal{M}$ that measures lengths and angles. The Riemannian gradient $\text{grad } f(x)$ is the unique tangent vector satisfying $g_x(\text{grad } f, \zeta) = Df(x)[\zeta]$ for all $\zeta \in T_x\mathcal{M}$.
- A *retraction* $R_x : T_x\mathcal{M} \rightarrow \mathcal{M}$: a map that moves from x along a tangent direction back onto $\mathcal{M}$, generalizing the Euclidean update $x + \eta\xi$.

Standard convergence analyses [11] assume *retraction- L -smoothness*: $f(R_x(\xi)) \leq f(x) + Df(x)[\xi] + \frac{L}{2}\|\xi\|_x^2$. Under this condition, Riemannian gradient descent achieves an $O(\sqrt{L/T})$ rate to an approximate first-order stationary point, matching the Euclidean rate.

I.2. Quotient manifolds and scale invariance. Many structured parameterizations involve redundancies. For example, a rank- r matrix $X = BA$ can be factored as (B, A) or (BN^{-1}, NA) for any invertible $N \in \text{GL}(r)$, these are different points in parameter space representing the same X . The mathematically clean way to handle this is the *quotient manifold*: define an equivalence relation $(B, A) \sim (BN^{-1}, NA)$ and work on the quotient space $\mathcal{M}_r = (\mathbb{R}_*^{m \times r} \times \mathbb{R}_*^{r \times n})/\text{GL}(r)$, whose points are equivalence classes $[(B, A)]$. The tangent space of the total space $\overline{\mathcal{M}}$ at a representative x splits into a *vertical space* $\mathcal{V}_x$ (directions along the symmetry orbit, which do not change the equivalence class) and a complementary *horizontal space* $\mathcal{H}_x$ (directions that represent genuine motion on the quotient); only horizontal tangent vectors carry information about the quotient geometry.

For optimization on a quotient manifold to be well-defined, the Riemannian metric must be *invariant* under the equivalence group: $g_{(BN^{-1}, NA)}(\cdot, \cdot) = g_{(B, A)}(\cdot, \cdot)$ (after appropriate pushforward of tangent vectors). If the metric is not invariant, the Riemannian gradient depends on which representative (B, A) of the equivalence class is used, a mathematical inconsistency that manifests as training instability in practice [26].

The *coupled metric* of Mishra and Sepulchre [39], Mishra et al. [41] achieves invariance by weighting each factor by the *other* factor's Gram matrix: $g(\dot{B}, \dot{A}) = \text{tr}(\dot{B}^\top \dot{B} AA^\top) + \text{tr}(B^\top B \dot{A} \dot{A}^\top)$. Under $(B, A) \mapsto (BN^{-1}, NA)$, the cross-Gram weighting transforms as $AA^\top \mapsto NAA^\top N^\top$, which exactly compensates the change $\dot{B} \mapsto \dot{B}N^{-1}$, preserving lengths. The resulting Riemannian gradient has the form $\nabla_B f (AA^\top)^{-1}$ (for the B -factor), which can be recognized as a preconditioned gradient that automatically equalizes the contribution of large and small singular directions of A .

APPENDIX J. EXTENDED RELATED WORK

J.1. Riemannian Frank–Wolfe methods. Weber and Sra [62] study Riemannian Frank–Wolfe methods for optimization over geodesically convex constraint sets on manifolds. Their LMO operates over a *feasible set on the manifold* to keep iterates within a constraint (e.g., a geodesic ball or a convex subset of the manifold). Our LMO is structurally different: it constrains a norm in the *tangent space* to select the step direction, then retracts onto the manifold. This is a trust-region-like norm bound on the update magnitude rather than feasibility of the iterates. The two approaches are complementary; ours targets unconstrained manifold optimization with spectral control of the step.

J.2. Euclidean spectral optimizers and LMO. The idea of using spectral-norm (Schatten- ∞) steepest descent for matrix-valued parameters in deep learning predates the recent Muon line of work. Carlson et al. [13] observe that the objective function in feedforward neural networks admits a tighter majorization bound under the Schatten- ∞ norm than under the Frobenius norm, and derive the corresponding steepest-descent operator, which is precisely the orthogonal polar factor $\text{Ortho}(\cdot)$ used by Muon. They further combine this non-Euclidean gradient with element-wise adaptive learning rates (RMSprop, ADAgrad), yielding preconditioned spectral descent algorithms (RMSspectral, ADAspectral) that demonstrate speedups on RBMs, feedforward nets, and CNNs. Our framework can be seen as replacing their element-wise diagonal preconditioning with a principled Riemannian metric G_x that is determined by the manifold geometry: where Carlson et al. scale each matrix entry independently, the intrinsic norm $\varphi(G_x^{1/2}\xi)$ scales the entire tangent vector via the metric operator, simultaneously achieving preconditioning, symmetry invariance on quotient manifolds, and closed-form solutions.

More recently, Bernstein and Newhouse [6] formalize the connection between the polar factor and steepest descent under the spectral norm, and [48] (Scion) generalize the LMO viewpoint to a broad family of induced *operator* norms in Euclidean space, including the spectral norm, $\text{sign} / \ell_\infty$, ColNorm, and RMS-to-RMS norms, which goes beyond the unitarily invariant family used in our framework. Our intrinsic LMO (4) recovers all of these unitarily invariant instances as the special case $G_x = I$ on $\mathbb{R}^{p \times q}$, and extends them to non-trivial metrics and manifold-structured parameters.

J.3. Muon extensions for low-rank/LoRA finetuning. A first wave of works ports Muon-like spectral control to LoRA-style and natively low-rank training. Riemannion [10] works on the embedded fixed-rank manifold $\mathcal{M}_r \subset \mathbb{R}^{m \times n}$ with the ambient Frobenius metric, projects the Euclidean gradient to $T_X \mathcal{M}_r$, and approximates a tangent space spectral LMO by replacing all $\leq 2r$ singular values of the tangent space momentum with 1 via a tangent space ‘Ortho_r’ operator and re-projecting; the spectral-norm constraint is satisfied only approximately (singular values of the resulting tangent space step lie in (0.9, 1.1) in their experiments). Kang et al. [29] prove a continuous-time convergence dichotomy for SpecGF on $\|BA - Y\|_F^2$ (see Section H.1). Janson et al. [26] (Spectron) renormalizes the orthogonalized step of each factor by $\rho = \eta / (\|A\|_2 + \|B\|_2 + 1)$, with $\|A\|_2, \|B\|_2$ estimated by a single power iteration per step. Park et al. [46] (Stiefel-LoRA) restrict LoRA’s B factor to the Stiefel manifold with Riemannian SGD/AdamW. NuMuon [43] adds a nuclear-norm budget to Muon’s spectral LMO, yielding $\text{rank-}\lfloor \kappa/\tau \rfloor$ updates suitable for compressible LLMs. Wang et al. [61] factorize the EMA momentum buffer to compress optimizer-state memory in both Adam-style and Muon-style optimizers. Schotthöfer et al. [52] provide a geometric framework for momentum-based low-rank training. None of these works simultaneously (i) preserves quotient symmetries on $\mathcal{M}_r$, (ii) gives an exact closed-form tangent space spectral LMO, and (iii) attains a condition-number-free rate, the three properties of our framework specialized to $\mathcal{M}_r$.

J.4. Convergence theory of Muon. Discrete-time convergence guarantees for Muon and its variants have appeared in a sequence of works: Li and Hong [36] provide a first convergence note; Sato et al. [51] establish convergence and a critical batch-size analysis under both Nesterov and weight-decay variants; Chen et al. [14], Kim and Oh [31], Shen et al. [53] treat Muon convergence under various smoothness assumptions, and Ma et al. [37] analyze preconditioning benefits of spectral orthogonalization. Riabinin et al. [50] (Gluon) bridge theory and practice in LMO-based optimizers. Amsel et al. [3] study optimal polynomial approximations of the matrix sign function used inside Muon. Our Theorems 5.3 and 5.7 cover both deterministic and stochastic settings on Riemannian matrix manifolds, with the squared Riemannian radius C_φ as the unifying complexity parameter, and recover the Euclidean Muon rate of Bernstein and Newhouse [6], Shen et al. [53] as the $G_x = I$ special case (Section G).

J.5. Riemannian and manifold aware optimizers for LLM. A complementary line of work injects manifold-derived primitives into LLM-scale pretraining without strictly constraining iterates to a manifold. Mano [17] projects momentum onto the tangent space of the Oblique manifold and applies alternating column-wise and row-wise normalization across odd/even iterations, which imposes only a soft manifold constraint on the update direction (no retraction is applied to the iterate), so the parameter trajectory itself remains Euclidean. SSO [65] constrains weights to the spectral sphere $\mathcal{S}_R = \{X : \|X\|_2 = R\}$ and runs steepest descent under the spectral norm via bisection on a single Lagrange multiplier; we discuss this case (where SV-invariance fails) in Section P.5. Yang et al. [66, 67] provide the spectral / μ P framework that

motivates much of this line. These works are largely orthogonal to our setting, which targets problems where the parameters genuinely live on a Riemannian manifold (LoRA-style fixed-rank factorizations, SPD covariance descriptors, subspaces). Within that setting, Mishra and Sepulchre [39], Mishra et al. [41], Tong et al. [55], Zhang and Pilanci [69] are the closest direct precursors of our Frobenius-norm instance on $\mathcal{M}_r$.

J.6. Stability of low-rank training. Janson et al. [26] provide a concrete diagnosis: when training neural networks with natively low-rank factorized weights $X = AB^\top$ using standard optimizers, the scaling invariance $X = (\lambda A)(B/\lambda)^\top$ permits unbounded spectral-norm growth of the weight update ΔX , triggering training divergence. They propose Spectron, which orthogonalizes the factor gradients (via Newton–Schulz) and adaptively sets a local constraint radius $\rho = \eta/(\|A\|_2 + \|B\|_2 + 1)$ so that the composite update satisfies $\|\Delta X\|_2 \leq \eta$. Our framework resolves the same instability by a different mechanism: the scaled constraint $\varphi(G_x^{1/2}\xi) \leq \tau$ is automatically $\text{GL}(r)$ -invariant (Proposition 3.1), so the update direction and magnitude are independent of the factorization without requiring explicit spectral-norm estimation. The Gram-root preconditioning $(AA^\top)^{-1/2}$ that emerges from our framework is closely related to Scaled Gradient Descent (ScaledGD) [55], which achieves condition-number-free rates for low-rank matrix estimation by a similar scaling of the factors.

APPENDIX K. EXPERIMENTAL DETAILS FOR LORA FINETUNING

Compared methods. Each LoRA adapter parameterizes the trainable update as $X = BA$, where $B \in \mathbb{R}^{m \times r}$ and $A \in \mathbb{R}^{r \times n}$, while the pretrained weight is kept fixed. This representation exposes the fixed-rank structure used by the geometry-aware methods. We compare Euclidean, Riemannian-preconditioned, and spectral-update methods under this common LoRA parameterization.

SGD and AdamW [69] update the LoRA factors directly as ordinary Euclidean parameters. Scaled GD and Scaled AdamW [40, 55, 69] use the $\text{GL}(r)$ -invariant preconditioning induced by the quotient geometry of $\mathcal{M}_r$: Scaled GD corresponds to the Frobenius-norm instance of the intrinsic LMO, which recovers the Riemannian gradient direction, while Scaled AdamW uses the same geometry-aware preconditioning inside an AdamW-style update. Factor-wise Muon [6, 26, 27, 29] applies the Euclidean spectral LMO factor-wise as in (3). In contrast, iMuon is the spectral-norm instance of our intrinsic LMO on $\mathcal{M}_r$ (Corollary 4.1). We also include Riemannion [10], reproduced from the authors’ released code, as an embedded fixed-rank Muon baseline with approximate tangent space orthogonalization.

Momentum implementation. For the Adam family methods, we use standard Adam first and second moment update. For iMuon and Factor-wise Muon, we use the *Nesterov* SGD style momentum update. For a LoRA pair (B_t, A_t) with $W_t = B_t A_t$ at iteration t , let

$$G_{B,t} = \nabla_W \mathcal{L}(W_t) A_t^\top, \quad G_{A,t} = B_t^\top \nabla_W \mathcal{L}(W_t).$$

denote the Euclidean gradient of LoRA factors. Factor-wise momentum buffers are

$$M_{B,t} = \beta M_{B,t-1} + G_{B,t}, \quad M_{A,t} = \beta M_{A,t-1} + G_{A,t},$$

with $\beta = 0.95$. Quantities passed to spectral step are

$$\tilde{G}_{B,t} = G_{B,t} + \beta M_{B,t}, \quad \tilde{G}_{A,t} = G_{A,t} + \beta M_{A,t}.$$

Factor-wise Muon applies spectral LMO to these individual factors whereas iMuon forms the momentum smoothed ambient matrix direction

$$\widehat{M}_t = \tilde{G}_{B,t} A_t + B_t \tilde{G}_{A,t},$$

on which metric scaled LMO is applied.

Compute Resource. All the experiments are run on Nvidia GPU clusters using L40, A40, A100, RTX 6000 GPUs depending on availability of the resources. The E2E NLG experiments use single-GPU training. For GLUE, all tasks except MNLI and QQP use single-GPU training, while MNLI and QQP use two-GPU DDP. No hardware specific tuning is performed and the same training code path, optimizer settings are used across GPU types.

We additionally compare the per-batch runtime of factor-wise Muon and iMuon over one E2E NLG epoch at batch size 8. We profile each batch into three stages: forward pass, backward pass, and optimizer step. The optimizer-step overhead motivates a CholeskyQR variant of iMuon, which replaces the thin QR by a Cholesky-based factorization of $AA^\top$. Table 4 reports per-stage means and standard deviations for factor-wise Muon, iMuon (PyTorch thin-QR), and iMuon (CholeskyQR)

TABLE 4. Per-batch runtime profile on E2E NLG for one training epoch with batch size 8. Timings are reported in milliseconds as mean $\pm$ standard deviation over training batches.

Stage	Factor-wise Muon	iMuon (QR)	iMuon (CholeskyQR)
Forward pass	135.97 $\pm$ 1.24	133.45 $\pm$ 1.30	135.48 $\pm$ 1.42
Backward pass	158.00 $\pm$ 0.34	157.41 $\pm$ 0.31	157.45 $\pm$ 0.28
Optimizer step	3.22 $\pm$ 0.09	13.08 $\pm$ 0.14	5.49 $\pm$ 0.13
Total per batch	297.20 $\pm$ 1.21	303.95 $\pm$ 1.26	298.41 $\pm$ 1.38

TABLE 5. E2E NLG results with GPT-2 Medium and LoRA rank $r = 4$. For each method, we report the run with the best test-set BLEU. The remaining metrics are computed from the same run. Bold entries denote the best value in each block.

Method	BLEU	NIST	METEOR	ROUGE-L	CIDEr
<i>Without momentum</i>					
SGD	66.60	8.54	44.20	68.20	2.32
Scaled GD	69.20	8.71	46.30	70.90	2.48
Factor-wise Muon	70.02	8.81	46.77	71.68	2.53
Riemannion (SGD)	70.02	8.78	46.79	71.99	2.52
iMuon	70.74	8.88	46.79	72.14	2.54
<i>With momentum</i>					
AdamW	68.90	8.69	46.50	71.30	2.51
Scaled AdamW	69.60	8.77	46.60	71.80	2.52
Factor-wise Muon	70.26	8.83	46.76	71.75	2.52
Riemannion (Adam)	70.21	8.83	46.55	71.87	2.53
iMuon	70.36	8.82	46.75	71.92	2.53

K.1. E2E NLG.

Setup. We fine-tune GPT-2 Medium (355M parameters) on the E2E NLG Challenge [45], a table-to-text generation task with approximately 42K training and 4.7K test examples in the restaurant domain. LoRA adapters are applied to the query and value projections within the merged `c_attn` linear layer, following LoRA [23] and Riemannian LoRA [69]. The LoRA configuration uses rank $r = 4$, scaling $\alpha = 32$, and dropout 0.1. Training runs for 5 epochs with batch size 8, a linear learning-rate schedule with 500 warmup steps, weight decay 0.01, label smoothing 0.1, and gradient clipping is disabled. Generation uses beam search with beam size 10, length penalty 0.8, and no-repeat 4-gram decoding. All runs use seed 110, which is also the one used by riemannianLoRA. We compute BLEU, NIST, METEOR, ROUGE-L, and CIDEr using the standard `e2e-metrics` package.

Learning rates are tuned separately for each method. We start from the common grid

$$\{5 \times 10^{-5}, 10^{-4}, 5 \times 10^{-4}, 10^{-3}\},$$

and perform additional boundary checks when the best run lies at the edge of this grid. Following the E2E reporting protocol, we report the best test-set BLEU run from the resulting sweep, with the remaining metrics computed from the same run. Full per-method grids are reported in the individual tables below. Results. Table 5 reports the best test-set BLEU run for each method, with the remaining metrics are computed from the same run. iMuon achieves the highest BLEU in both the no-momentum and momentum blocks, improving over factor-wise Muon under the same LoRA setup. The improvement is more pronounced without momentum, while the momentum-based methods are closer to each other. The secondary generation metrics are broadly comparable across the spectral-update methods, indicating that the BLEU gains do not come from a degradation in the other reported metrics. Overall, the E2E results support the benefit of replacing the Euclidean factor-wise spectral update with the intrinsic fixed-rank spectral update, without changing the model, adapter configuration, or decoding protocol.

TABLE 6. Robustness of E2E NLG results (without momentum) across seeds. Values show results for Seed 42.

Method	BLEU	NIST	METEOR	ROUGE-L	CIDEr
Factor-wise Muon	70.01	8.81	46.68	71.56	2.54
Riemannion (SGD)	69.92	8.82	46.68	71.28	2.51
iMuon	69.58	8.74	46.84	71.65	2.50

TABLE 7. E2E NLG learning-rate sweep without momentum. The selected learning rate for each method is marked in bold. For Riemannion (SGD), low-LR points $\{5 \times 10^{-5}, 10^{-4}\}$ are omitted because pilot runs showed the useful range was at substantially larger learning rates.

Method	LR	BLEU	NIST	METEOR	ROUGE-L	CIDEr
Factor-wise Muon	5×10^{-5}	64.22	8.33	42.46	67.49	2.21
	10^{-4}	65.31	8.42	43.40	67.85	2.30
	5×10^{-4}	69.09	8.73	46.20	71.17	2.48
	10^{-3}	70.02	8.81	46.77	71.68	2.53
Riemannion (SGD)	5×10^{-4}	65.05	8.40	43.30	68.05	2.29
	10^{-3}	66.08	8.48	43.88	68.07	2.36
	5×10^{-3}	68.60	8.70	45.84	70.49	2.45
	10^{-2}	69.40	8.75	46.28	71.18	2.47
	5×10^{-2}	70.02	8.78	46.79	71.99	2.52
iMuon	5×10^{-5}	64.25	8.09	41.32	67.62	2.07
	10^{-4}	65.08	8.42	42.56	68.16	2.20
	5×10^{-4}	67.24	8.62	44.24	68.67	2.32
	10^{-3}	68.80	8.71	45.72	70.64	2.44
	5×10^{-3}	70.74	8.88	46.79	72.14	2.54

We conduct a seed sensitivity analysis on the E2E NLG task to verify the robustness of factor-wise Muon, Riemannion (SGD) and iMuon under seed variation. Table 6 reports scores for the without-momentum setting using Seed 42, utilizing the best learning rates identified in Table 5. Performance remains stable across metrics, with iMuon showing negligible variance in measures like METEOR / CIDEr.

Tables 7 and 8 report the full learning-rate sweeps used to select the runs in Table 5. We use $\{5 \times 10^{-5}, 10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$ as the base grid and add local or boundary probes when pilot runs indicate improvement near the best base-grid point. The selected learning rate and the corresponding BLEU score are marked in bold.

K.2. GLUE.

Setup. We evaluate LoRA finetuning on GLUE benchmark, following the setup of Mistral-7B in [69]. We fine-tune a 4-bit quantized Mistral-7B-v0.1 model whose frozen base weights are stored in NF4 and use bfloat16 compute. LoRA adapters use rank $r = 16$, scaling $\alpha = 16$, dropout 0.05, and no bias, and are applied to five target modules: `q_proj`, `k_proj`, `v_proj`, `o_proj`, and `gate_proj`. Riemannion uses its doubled-rank PEFT parameterization with rank $r = 32$ and $\alpha = 32$, corresponding to effective rank 16. The doubling is used for its tangent-space implementation rather than as a rank-32 forward adapter.

Training uses per-device batch size 8 for up to 5 epochs, with two-GPU DDP for MNLI and QQP and single-GPU training for the remaining seven tasks. We use a linear learning-rate schedule with no warmup, weight decay 0.01, and maximum gradient norm 1. AdamW-family methods use $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-6}$. For runs evaluated in this work, early stopping is enabled with patience 10, and evaluation is performed every 500 steps; we select the best checkpoint by the validation metric used in the training script: accuracy for most tasks, Matthews correlation for CoLA, and Pearson correlation for STS-B. All runs use seed 42.

TABLE 8. E2E NLG learning-rate sweep with momentum. The selected learning rate for each method is marked in bold.

Method	LR	BLEU	NIST	METEOR	ROUGE-L	CIDEr
Factor-wise Muon	5×10^{-5}	68.53	8.67	45.78	70.80	2.43
	10^{-4}	69.57	8.77	46.40	71.53	2.50
	5×10^{-4}	69.97	8.81	46.70	71.44	2.53
	7×10^{-4}	70.26	8.83	46.76	71.75	2.52
	10^{-3}	70.22	8.82	46.61	71.93	2.53
Riemannion (Adam)	5×10^{-5}	61.80	7.57	39.83	65.24	1.95
	10^{-4}	64.59	8.16	41.17	67.11	2.10
	5×10^{-4}	65.23	8.38	43.87	67.82	2.31
	10^{-3}	66.46	8.54	44.50	67.90	2.34
	5×10^{-3}	69.27	8.79	46.03	70.12	2.47
	10^{-2}	69.58	8.79	46.57	70.85	2.52
	5×10^{-2}	70.21	8.83	46.55	71.87	2.53
iMuon	5×10^{-5}	65.00	8.33	42.07	67.54	2.17
	10^{-4}	67.55	8.59	44.22	69.09	2.34
	5×10^{-4}	70.32	8.81	46.60	71.97	2.52
	10^{-3}	70.36	8.82	46.75	71.92	2.53
	5×10^{-3}	70.20	8.81	46.75	71.56	2.52

We evaluate on the nine GLUE tasks: MNLI, SST-2, MRPC, CoLA, QNLI, QQP, RTE, STS-B, and WNLI. We report task-specific metrics following the GLUE convention: accuracy for classification tasks, Matthews correlation for CoLA, and Pearson correlation for STS-B. The reported average is the unweighted average across the nine tasks.

Results. Table 9 reports the GLUE results for 4-bit Mistral-7B fine-tuning with LoRA rank $r = 16$. We split the comparison into two groups: SGD-style methods without momentum and methods that use momentum or adaptive preconditioning. In the non-momentum block, iMuon consistently improves over factor-wise Muon, outperforming it on most tasks and comparable on others. Consequently, iMuon achieves the highest average score in this block, suggesting that applying the spectral update intrinsically to the fixed-rank LoRA update is more effective than applying Muon separately to the two LoRA factors.

In the momentum/adaptive block, Scaled AdamW obtains the highest average score. However, iMuon remains close to the best method, with an average score within one point of Scaled AdamW. These results suggest that iMuon is stronger in the direct SGD-style spectral comparison, while remaining competitive with strong adaptive baselines.

Learning-rate selection and stability. The Mistral-7B GLUE setting is sensitive to the learning rate, both across tasks and across optimizers. We therefore select the learning rate separately for each optimizer–task pair using validation performance, rather than using a single global learning rate for all GLUE tasks. This is consistent with the task-specific learning-rate selection reported by Zhang and Pilanci [69] for 4-bit Mistral-7B GLUE fine-tuning.

For factor-wise Muon and iMuon, the main grid is $\{5 \times 10^{-5}, 10^{-4}\}$, with a small local extension only when pilot runs indicated that the best result lay outside the base grid. Riemannion requires a different LR scale: the Adam variant requires lower learning rates on some tasks, while the SGD variant underfits at the factor-wise Muon/iMuon LR scale and only becomes effective at learning rates around 10^{-3} – 10^{-2} .

APPENDIX L. LEARNING ON FIXED-RANK MANIFOLD

Common LMO stepsize convention. All LMO based methods use constraint radius $\tau = 1$. The reported learning rate scales the resulting unit-radius direction. When Spectron is included, we use its adaptive extrinsic normalization separately.

L.1. Matrix completion: synthetic experiments.

TABLE 9. GLUE results with 4-bit Mistral-7B and LoRA rank $r = 16$. The average is the unweighted average across the nine tasks. Bold entries denote the best value in each block.

Method	MNLI	SST-2	MRPC	CoLA	QNLI	QQP	RTE	STS-B	WNLI	Avg.
<i>Without momentum</i>										
SGD	88.15	96.10	70.10	55.89	94.22	88.59	50.90	47.64	49.30	71.21
Scaled GD	90.21	96.90	81.62	68.17	94.40	91.15	54.15	90.31	56.34	80.36
Factor-wise Muon	89.77	97.25	82.35	69.32	94.47	88.42	84.12	84.39	45.07	81.68
Riemannion (SGD)	91.55	96.90	82.60	68.90	94.93	87.77	82.67	77.99	60.56	82.65
iMuon	89.29	97.02	85.54	70.21	94.60	87.86	88.09	88.99	74.65	86.25
<i>With momentum</i>										
AdamW	89.86	96.79	88.48	71.05	94.42	91.24	90.61	90.42	81.69	88.28
Scaled AdamW	90.68	97.25	89.46	71.30	94.67	92.22	91.34	91.10	83.10	89.01
Factor-wise Muon	91.29	96.90	89.71	71.45	94.97	91.74	89.89	92.18	78.87	88.56
Riemannion (Adam)	91.21	96.44	79.90	67.00	94.55	90.36	80.87	62.13	83.10	82.84
iMuon	89.93	97.13	88.97	70.32	93.45	91.55	89.53	92.37	81.69	88.33

TABLE 10. Learning-rate grids and selected learning rates for the GLUE entries evaluated in this work. The “LR grid” column reports the grid swept for each optimizer, with task-specific extensions noted where applicable; the task columns report the learning rate selected by validation performance. The SGD, Scaled GD, AdamW, and Scaled AdamW rows in Table 9 are taken from Zhang and Pilanci [69] and are therefore not included here.

Method	LR grid	MNLI	SST-2	MRPC	CoLA	QNLI	QQP	RTE	STS-B	WNLI
<i>Without momentum</i>										
Factor-wise Muon	$\{5 \times 10^{-5}, 10^{-4}\}$	5×10^{-5}	5×10^{-5}	10^{-4}	10^{-4}	5×10^{-5}	5×10^{-5}	10^{-4}	10^{-4}	10^{-4}
Riemannion (SGD)	$\{5 \times 10^{-5}, 10^{-4}, 5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}, 5 \times 10^{-3}, 10^{-2}, 5 \times 10^{-2}\}$	5×10^{-3}	5×10^{-3}	5×10^{-3}	10^{-2}	5×10^{-3}	10^{-2}	5×10^{-3}	2×10^{-3}	10^{-2}
iMuon	$\{5 \times 10^{-5}, 10^{-4}\}$	5×10^{-5}	5×10^{-5}	10^{-4}	10^{-4}	5×10^{-5}	10^{-4}	10^{-4}	10^{-4}	10^{-4}
<i>With momentum</i>										
Factor-wise Muon	$\{5 \times 10^{-5}, 10^{-4}\}$	5×10^{-5}	5×10^{-5}	10^{-4}	10^{-4}	5×10^{-5}	5×10^{-5}	10^{-4}	10^{-4}	10^{-4}
Riemannion (Adam)	$\{10^{-5}, 2 \times 10^{-5}, 3 \times 10^{-5}, 4 \times 10^{-5}, 5 \times 10^{-5}, 10^{-4}, 5 \times 10^{-4}\}$ $\{5 \times 10^{-5}, 10^{-4}\}$;	10^{-4}	4×10^{-5}	10^{-4}	10^{-4}	5×10^{-5}	10^{-4}	10^{-4}	5×10^{-5}	5×10^{-4}
iMuon	RTE/WNLI also probed at $2 \times 10^{-4}, 3 \times 10^{-4}$	5×10^{-5}	5×10^{-5}	10^{-4}	5×10^{-5}	5×10^{-5}	5×10^{-5}	10^{-4}	10^{-4}	2×10^{-4}

Setup. We study synthetic fixed-rank matrix completion at large scale, following the standard sampled-entry low-rank recovery setup [12, 30, 49] and the quotient fixed-rank factorization used in Riemannian matrix completion [39–41, 58]. For each trial, we generate a rank- r ground-truth matrix $X^* \in \mathbb{R}^{m \times n}$, with $m = n = 5000$ and $r = 10$. The nonzero singular values are ordered as $\sigma_1 \geq \dots \geq \sigma_r > 0$, and we control the target conditioning by setting $\kappa = \sigma_1(X^*)/\sigma_r(X^*)$. Given an observed index set Ω , we optimize

$$\min_{X=BA} f(X) = \frac{1}{2|\Omega|} \|\mathcal{P}_\Omega(X - Y)\|_F^2,$$

where $B \in \mathbb{R}^{m \times r}$, $A \in \mathbb{R}^{r \times n}$, and Y contains the observed entries. In the noiseless case, $Y_{ij} = X_{ij}^*$ for $(i, j) \in \Omega$. In the noisy case, $Y_{ij} = X_{ij}^* + \varepsilon_{ij}$, with Gaussian observation noise scaled as described below. We use an oversampling ratio s , meaning that $|\Omega| = sr(m + n)$, and all reported runs use $s = 10$. We set $\alpha = 1$, so no artificial factor rescaling is applied, and initialize all methods from the rank- r SVD of the zero-filled observed matrix $\mathcal{P}_\Omega(Y)$. The stepsize follows $\eta_t = \eta_0/\sqrt{t}$, where η_0 is the selected initial stepsize.

Methods. We compare norm-matched Euclidean and intrinsic updates:

- **Frobenius norm:** The intrinsic LMO recovers standard (normalized) Riemannian gradient descent, denoted by **RGD**, compared with (normalized) Euclidean gradient descent on the chosen representative, denoted by **EGD**.

- **Spectral norm:** **iMuon** denotes the intrinsic spectral norm LMO induced by the Riemannian metric, while factor-wise **Muon** denotes the corresponding Euclidean spectral norm LMO applied to the same representative as in (3).
- **Nuclear norm:** **iMuon-Nu** denotes the intrinsic nuclear norm LMO. **NuMuon** is its Euclidean nuclear norm counterpart, following the nuclear norm constrained Muon terminology of Mohaghegh Dolatabadi et al. [43]. In this section, we use only the nuclear norm constraint so that the comparison is norm matched to iMuon-Nu.

Each comparison is made within the same norm family, isolating the effect of using the intrinsic metric induced geometry rather than the ambient Euclidean geometry.

Dataset. We vary the target condition number over $\kappa \in \{1, 10, 100\}$. To match the matrix-completion convention of reporting noise by its variance or standard deviation, we parameterize the observation noise by the relative entrywise scale

$$\rho = \sigma_\varepsilon / \text{rms}_\Omega(X^*),$$

where $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is added only on observed entries and $\text{rms}_\Omega(X^*)$ is the root mean square magnitude of the sampled clean entries. We report the noiseless case $\rho = 0$, a mild noise setting $\rho = 0.05$, and two additional noise-stress settings $\rho = 0.10$ and $\rho = 0.50$. For each (κ, ρ) setting and each seed, we generate a fresh rank-10 target matrix X^* , sample the observation set Ω , and add observation noise when applicable. Results are averaged over seeds $\{0, 1, 2\}$.

Results. Figure 2 plots final recovery error against the relative noise level for each condition number. Frobenius and nuclear intrinsic updates improve over their Euclidean counterparts for low and moderate noise. The spectral pair is tied for $\kappa = 1$ and $\kappa = 10$, and favors Muon when $\kappa = 100$. At $\rho = 0.50$, observation noise dominates and the ordering becomes mixed.

L.2. Fixed-rank CIFAR-100 Classification Heads. The task is to assign each CIFAR-100 image to one of 100 semantic categories using frozen ResNet-18 features [20, 33]. A low-rank linear head maps these features to class scores while making the class weight vectors share a small set of latent visual directions, so the learned classifier is the fixed-rank parameter $X = BA$. We consider two regimes: a representative-sensitivity diagnostic and a balanced spectral comparison with Spectron [26].

For both regimes, given frozen features $h_i \in \mathbb{R}^{512}$ and labels y_i , we train a rank- r classifier $X = BA$ and bias $b \in \mathbb{R}^{100}$ by minimizing

$$\min_{B, A, b} \frac{1}{N} \sum_{i=1}^N \ell_{\text{CE}}(h_i^\top BA + b, y_i) + \frac{\lambda}{2} \|BA\|_F^2 + \frac{\lambda_b}{2} \|b\|_2^2.$$

We use $\lambda = \lambda_b = 10^{-4}$. The rank r , initialization, and learning-rate grids are specified below.

L.2.1. Representative-sensitivity diagnostic.

Setup. The representative-sensitivity diagnostic uses a rank-10 head,

$$X = BA, \quad B \in \mathbb{R}^{512 \times 10}, \quad A \in \mathbb{R}^{10 \times 100}.$$

To test representative sensitivity, we keep the initial product fixed and rescale only the factors,

$$(B_0, A_0) = (\alpha \tilde{B}_0, \alpha^{-1} \tilde{A}_0), \quad \alpha = 10^3.$$

Thus $B_0 A_0$ is unchanged up to numerical precision, while the two factors are imbalanced. This is a real supervised task with a deliberately chosen representative of the same initial classifier.

Methods. We compare the three norm-matched Euclidean/intrinsic pairs used above. The Euclidean methods use the grid

$$\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}.$$

For the intrinsic methods, we use local grids,

$$\text{RGD} : \{0.1, 0.3, 1.0\}, \quad \text{iMuon} : \{0.03, 0.1, 0.3\}, \quad \text{iMuon-Nu} : \{0.3, 1.0, 3.0\}.$$

The learning rate is selected by validation accuracy for each method.

Dataset. We use the standard CIFAR-100 train/test split, reserve 10% of the training set for validation, extract frozen ResNet-18 features, and apply feature ℓ_2 -normalization. We train for 50 epochs with batch size 32 and average over seeds $\{0, 1, 2\}$.

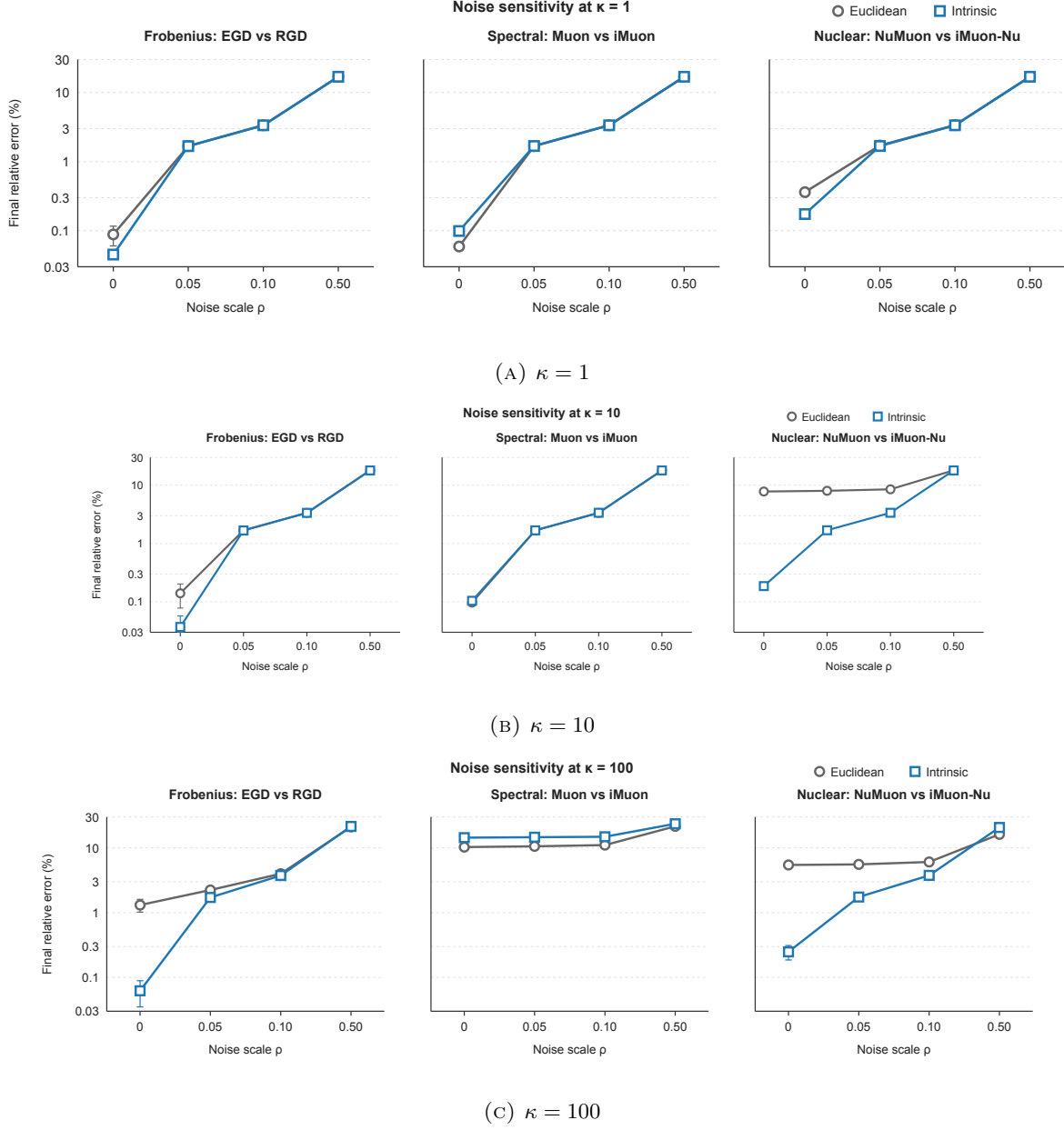


FIGURE 2. Noise sensitivity in large-scale synthetic fixed-rank matrix completion. Each row fixes the condition number and plots final relative recovery error as the relative observed-entry noise scale ρ varies. The three panels in each row compare the Frobenius, spectral, and nuclear norm pairs. The y-axis is logarithmic, and lower is better.

Results. Table 11 reports validation-selected test accuracy. Under the same rescaled initial product, the intrinsic methods reach 33.9% to 34.3%, while their Euclidean counterparts range from 6.8% to 10.0%.

L.2.2. Balanced spectral comparison with Spectron.

Setup. The balanced spectral comparison uses a rank-40 head:

$$X = BA, \quad B \in \mathbb{R}^{512 \times 40}, \quad A \in \mathbb{R}^{40 \times 100},$$

TABLE 11. Representative-sensitivity diagnostic on a frozen-feature CIFAR-100 rank-10 classification head with $\alpha = 10^3$. We report mean test accuracy $\pm$ standard deviation over three seeds. Boldface marks the higher-accuracy method within each Euclidean/intrinsic pair.

Norm	Euclidean	Intrinsic	Euclidean acc.	Intrinsic acc.
Frobenius	EGD (10^{-5})	RGD (0.1)	0.075 ± 0.007	0.342 ± 0.004
Spectral	Muon (10^{-5})	iMuon (0.03)	0.068 ± 0.005	0.339 ± 0.003
Nuclear	NuMuon (10^{-4})	iMuon-Nu (0.3)	0.100 ± 0.003	0.343 ± 0.004

with $\alpha = 1$. For each seed, all methods use the same train/validation split and the same rank-40 class-mean-SVD initialization of $B_0 A_0$. This regime compares spectral fixed-rank updates without artificial factor rescaling.

Methods. We compare Muon, Spectron [26], and iMuon. Muon applies factor-wise Euclidean spectral LMO updates to B and A [6, 27]. Spectron orthogonalizes the factor gradients and rescales the update by $1/(\|B\|_2 + \|A\|_2 + 1)$. We run Spectron without momentum to match the optimizer protocol used here. iMuon uses the coupled fixed-rank metric and the intrinsic spectral LMO from Corollary 4.1. We train for 50 epochs with batch size 2048, use feature ℓ_2 -normalization, and select the learning rate by validation accuracy. The learning-rate grids are

$$\begin{aligned} \text{Muon} &: \{0.03, 0.05, 0.07, 0.10, 0.15\}, \\ \text{Spectron} &: \{0.5, 0.7, 1.0, 1.5, 2.0\}, \\ \text{iMuon} &: \{0.05, 0.07, 0.10, 0.15, 0.18, 0.20, 0.25\}. \end{aligned}$$

Dataset. We use the standard CIFAR-100 train/test split and reserve 10% of the training split for validation. Results are averaged over seeds $\{0, 1, 2\}$.

Results. Table 12 reports validation-selected test accuracy. iMuon obtains the highest mean accuracy, 53.93%, compared with 53.82% for factor-wise Muon and 53.92% for Spectron. The gaps are small, so we view this as a balanced comparison with fixed-rank spectral baselines rather than a large separation. Figure 3 shows the corresponding training trajectories. The three methods reach similar training objectives, while iMuon remains competitive in test accuracy throughout training.

TABLE 12. Frozen-feature CIFAR-100 classification with a rank-40 fixed-rank head. We report validation-selected test accuracy over three seeds. Higher is better.

Method	Selected LR	Test acc. (%)
Factor-wise Muon	0.05	53.82 ± 0.02
Spectron	0.5	53.92 ± 0.09
iMuon	0.18	53.93 ± 0.09

L.3. MovieLens-1M.

Setup. We also test MovieLens-1M [19], a standard real-data matrix completion benchmark. The data contain 6040 users, 3952 movies, and 1,000,209 observed ratings. Following standard recommender preprocessing [32], we subtract a global mean, user biases, and movie biases. We then fit the residual ratings with a rank-5 fixed-rank factorization $X = BA$. On the training entries Ω_{train} , the objective is

$$f(B, A) = \frac{1}{2|\Omega_{\text{train}}|} \sum_{(u,i) \in \Omega_{\text{train}}} ((BA)_{ui} - R_{ui})^2 + \frac{\lambda}{2} (\|B\|_F^2 + \|A\|_F^2),$$

where R is the residual rating matrix after bias correction. All methods start from the rank-5 SVD of the zero filled observed residual matrix. We use $\lambda = 1$.

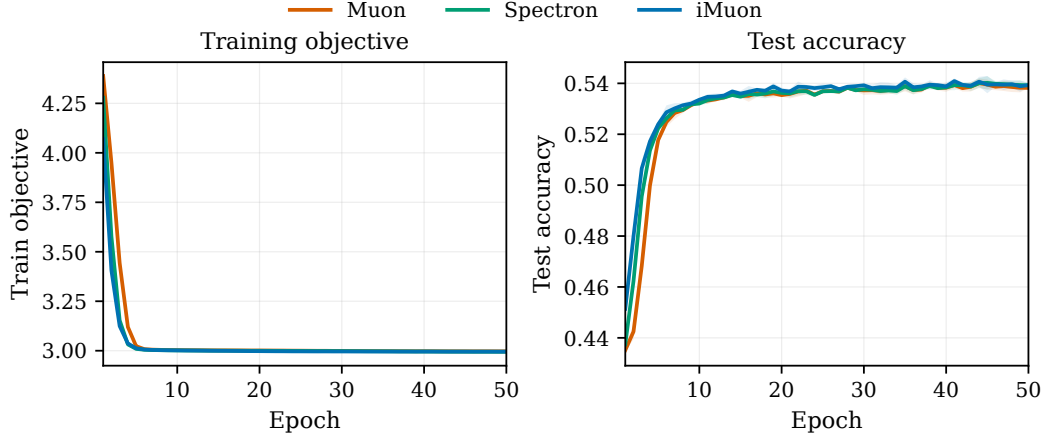


FIGURE 3. Training trajectories for the balanced fixed-rank CIFAR-100 rank-head comparison. Curves show means over three seeds with standard-deviation bands. The three methods reach similar training objectives, while iMuon remains competitive in test accuracy.

TABLE 13. MovieLens-1M fixed-rank matrix completion with observed SVD initialization, rank 5, factor ℓ_2 regularization $\lambda = 1$, and $\eta_t = \eta_0/\sqrt{t}$. We report mean test RMSE $\pm$ standard deviation over three seeds. Lower is better. Boldface marks the lowest RMSE within each norm group.

Norm	Method	Selected η_0	Test RMSE
Frobenius	EGD	5	0.8694 ± 0.0023
	RGD	500	0.8691 ± 0.0021
Spectral	Muon	20	0.8693 ± 0.0019
	Spectron	700	0.8691 ± 0.0022
	iMuon	300	0.8692 ± 0.0018
Nuclear	NuMuon	7	0.8729 ± 0.0026
	iMuon-Nu	100	0.8715 ± 0.0023

Methods. We compare the same norm-matched Euclidean/intrinsic pairs as above and add Spectron [26] to the spectral group. Spectron is run without momentum, matching the optimizer protocol used in this section. Each method uses the decaying schedule $\eta_t = \eta_0/\sqrt{t}$, with η_0 selected by validation RMSE from the same grid

$$\{0.3, 0.5, 1, 1.5, 2, 3, 5, 7, 10, 15, 20, 30, 50, 75, 100, 150, 200, 300, 500, 700, 1000\}.$$

The values of η_0 are not directly comparable across methods because the update directions have different normalizations.

Dataset. We use a random 20% test split and hold out 10% of the remaining observations for validation. Validation RMSE is used for model selection. Results are averaged over seeds $\{0, 1, 2\}$.

Results. Table 13 reports validation-selected test RMSE. RGD and iMuon-Nu improve over their Euclidean counterparts. In the spectral group, Spectron gives the lowest mean RMSE, while iMuon remains close and improves over factor-wise Muon. But these margins are small.

APPENDIX M. SPD PROTOTYPE LEARNING FOR COVARIANCE-BASED IMAGE CLASSIFICATION

Table 14 gives the SPD update directions for the six methods: EGD, Muon, and NuMuon use the Euclidean column, while RGD, iMuon, and iMuon-Nu use the intrinsic column. Let $G = \nabla_X f$ and $H = X^{1/2}GX^{1/2}$. The table reports the LMO maximizer ξ^* . The optimizer retracts along $-\eta\xi^*$.

TABLE 14. Euclidean and intrinsic LMO directions for the SPD prototype experiment. Here (μ_1, u_1) and (λ_1, q_1) are eigenpairs of G and H , respectively, associated with the largest absolute eigenvalue, and $\text{sign}(\cdot)$ denotes the matrix sign function.

Norm	Euclidean ξ^*	Intrinsic ξ^*
Frobenius	$G/\ G\ _F$	$XGX/\ H\ _F$
Spectral	$\text{sign}(G)$	$X^{1/2}\text{sign}(H)X^{1/2}$
Nuclear	$\text{sign}(\mu_1)u_1u_1^\top$	$\text{sign}(\lambda_1)X^{1/2}q_1q_1^\top X^{1/2}$

Setup. We consider supervised prototype classification from SPD covariance descriptors. Each example is a covariance matrix $C_i \in S_{++}^{32}$ with coarse label $y_i \in \{1, \dots, K\}$, where $K = 20$ for CIFAR-100 coarse labels. We learn one SPD prototype $X_c \in S_{++}^{32}$ per class, so the optimization variable is $X = (X_1, \dots, X_K) \in (S_{++}^{32})^K$.

Given prototypes $X_1, \dots, X_K$, we form logits from negative squared AI distance

$$z_{ic} = -\beta d_{\text{AI}}(C_i, X_c)^2, \quad d_{\text{AI}}(C, X)^2 = \left\| \log \left(X^{-1/2} C X^{-1/2} \right) \right\|_F^2.$$

We minimize the regularized cross-entropy objective

$$f(X) = \frac{1}{N} \sum_{i=1}^N \ell_{\text{CE}}(z_i, y_i) + \lambda \sum_{c=1}^K d_{\text{AI}}(X_c, \bar{X}_c)^2,$$

where $\bar{X}_c$ is the log-Euclidean class-mean initialization for class c . We use the AI metric from the main text, initialize prototypes by log-Euclidean class means, set the logit scale to $\beta = 8.0$, and use $\lambda = 10^{-3}$. We train for 20 epochs with training batch size 64 and evaluation batch size 512, and report results over seeds $\{0, 1, 2\}$.

Methods. We use the same norm-matched comparison structure: RGD vs. EGD for the Frobenius norm, iMuon vs. Muon for the spectral norm, and iMuon-Nu vs. NuMuon for the nuclear norm. All methods are tuned over the shared learning-rate grid

$$\{10^{-3}, 3 \cdot 10^{-3}, 10^{-2}, 3 \cdot 10^{-2}, 10^{-1}\}.$$

The selected values are 0.01 for RGD and iMuon, and 0.001 for iMuon-Nu, EGD, Muon, and NuMuon. Model selection uses validation accuracy.

Dataset. We use CIFAR-100 coarse labels (20 classes) [33] together with frozen ResNet-18 visual features [20]. We extract the `layer3` feature map for each image, keep the backbone fixed, and project each spatial token from the backbone channel dimension to a 32-dimensional space using a fixed random projector with seed 17. We then build a shrinkage regularized covariance descriptor $C_i \in S_{++}^{32}$ [24, 35, 57] for each image using covariance shrinkage 0.1 and diagonal stabilization $\varepsilon = 10^{-4}$. We use the standard CIFAR-100 train/test split and reserve 10% of the training set as a validation split for learning-rate selection.

Results. Table 15 reports validation-selected test accuracy for the three norm-matched Euclidean/intrinsic pairs. Each intrinsic method improves over its Euclidean counterpart. The Frobenius gap is positive (RGD 0.548 ± 0.002 vs. EGD 0.523 ± 0.026), the spectral gap is positive (iMuon 0.475 ± 0.009 vs. Muon 0.407 ± 0.016), and the largest endpoint gap is for the nuclear pair (iMuon-Nu 0.546 ± 0.000 vs. NuMuon 0.422 ± 0.032).

Figure 4 reports a companion convergence plot for the classification term. It uses the same covariance features and norm-matched pairs, removes the prototype anchoring term, and uses a smaller CPU split. The intrinsic methods reach lower training cross entropy or higher test accuracy in each pair. Endpoint comparisons use Table 15.

APPENDIX N. GRASSMANN FRÉCHET SUBSPACE LEARNING FOR VIDEO FACE IDENTIFICATION

Setup. We consider video-based face identification on YouTube Faces [64]. Each video clip is encoded as a k -dimensional subspace $S_i \in \text{Gr}(64, k)$, and each identity class has a learned Grassmann prototype $X_c \in \text{Gr}(64, k)$. For each class c , we optimize the Fréchet prototype objective

$$\min_{X_c \in \text{Gr}(64, k)} \frac{1}{|\mathcal{I}_c|} \sum_{i \in \mathcal{I}_c} d_{\text{Gr}}(X_c, S_i)^2,$$

TABLE 15. SPD covariance prototype classification on CIFAR-100 coarse labels. We report mean test accuracy $\pm$ standard deviation over three seeds. Boldface marks the higher accuracy method within each pair.

Method	Selected LR	Mean test acc.
RGD	0.01	0.548 $\pm$ 0.002
EGD	0.001	0.523 $\pm$ 0.026
iMuon	0.01	0.475 $\pm$ 0.009
Muon	0.001	0.407 $\pm$ 0.016
iMuon-Nu	0.001	0.546 $\pm$ 0.000
NuMuon	0.001	0.422 $\pm$ 0.032

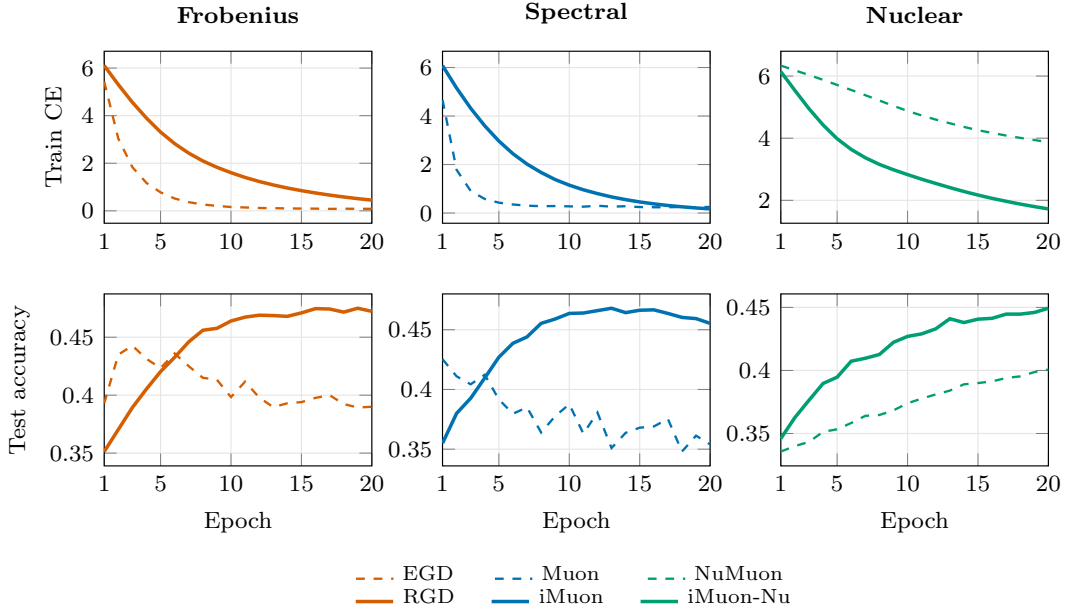


FIGURE 4. SPD convergence plot on frozen covariance features. The top row reports training cross-entropy and the bottom row reports test accuracy. Columns compare the Frobenius, spectral, and nuclear norm pairs. This variant omits the prototype anchoring term, so the accuracy values are not directly comparable with Table 15.

where $\mathcal{I}_c$ indexes the training clips of class c , and d_{Gr} is the geodesic distance induced by principal angles. A test clip is classified by the nearest learned prototype under d_{Gr} .

We use the canonical Grassmann metric. With a basis $X \in \mathbb{R}^{64 \times k}$ for a subspace, tangent vectors lie in the horizontal space $\{\xi : X^\top \xi = 0\}$, where $G_X = I$. Hence $\varphi(G_X^{1/2} \xi) \leq \tau$ reduces to the Euclidean constraint $\varphi(\xi) \leq \tau$. This boundary case checks that the intrinsic LMO recovers the projected Euclidean update when the metric scaling is identity.

Methods. We optimize the three norm-matched pairs for 100 iterations, using the squared-geodesic loss with scale $c_{Gr} = 0.25$ and $\varepsilon = 10^{-6}$. We also report GDA [18] and PML [25] as reference methods. GDA learns a discriminative Grassmann projection, while PML learns a projection metric for subspaces.

Dataset. We use YouTube Faces clips [64] represented as k -dimensional subspaces of a 64-dimensional feature space, with $k \in \{3, 5, 8, 10\}$. We use the standard Grassmann geometry [1, 16] and evaluate over 5 random train/test splits with training fraction 0.7.

Results. Table 16 reports classification accuracy on YouTube Faces. Across all k , each intrinsic method matches its Euclidean counterpart within the displayed precision. This is expected because the intrinsic and Euclidean LMO feasible sets are the same on the horizontal space, so they give the same spectral

TABLE 16. Grassmann Fréchet prototype learning on YouTube Faces. We report mean classification accuracy $\pm$ standard deviation over 5 splits. Boldface marks the highest accuracy in each row. Since $G_X = I$ on the horizontal space, each intrinsic method coincides with its Euclidean projected counterpart.

k	RGD	iMuon	iMuon-Nu	EGD	Muon	NuMuon	GDA	PML
3	0.136 $\pm$ 0.017	0.147 $\pm$ 0.012	0.138 $\pm$ 0.018	0.136 $\pm$ 0.017	0.147 $\pm$ 0.012	0.138 $\pm$ 0.018	0.087 $\pm$ 0.012	0.160 $\pm$ 0.046
5	0.219 $\pm$ 0.037	0.206 $\pm$ 0.050	0.211 $\pm$ 0.038	0.219 $\pm$ 0.037	0.206 $\pm$ 0.050	0.211 $\pm$ 0.038	0.094 $\pm$ 0.028	0.179 $\pm$ 0.031
8	0.232 $\pm$ 0.016	0.238 $\pm$ 0.042	0.238 $\pm$ 0.034	0.232 $\pm$ 0.016	0.238 $\pm$ 0.042	0.238 $\pm$ 0.034	0.166 $\pm$ 0.022	0.194 $\pm$ 0.031
10	0.262 $\pm$ 0.016	0.264 $\pm$ 0.038	0.266 $\pm$ 0.034	0.262 $\pm$ 0.016	0.264 $\pm$ 0.038	0.266 $\pm$ 0.034	0.164 $\pm$ 0.032	0.213 $\pm$ 0.011

projections. The two implementations differ only in what is projected. For example, the intrinsic method projects $(I - XX^\top)\nabla_X f$ whereas the Euclidean method projects $\nabla_X f$.

APPENDIX O. STIEFEL SUB-CENTER PROTOTYPE LEARNING FOR FROZEN-FEATURE IMAGE CLASSIFICATION

Setup. We consider supervised classification with class prototypes constrained to the Stiefel manifold [1, 16]. Given frozen image features $h_i \in \mathbb{R}^{512}$ and labels $y_i \in \{1, \dots, C\}$, with $C = 100$ for CIFAR-100, we learn $q = 4$ prototypes per class, giving $Q = Cq$ prototypes in total. The prototypes are the columns of

$$X = [x_{1,1}, \dots, x_{1,q}, \dots, x_{C,1}, \dots, x_{C,q}] \in \text{St}(512, Q), \quad X^\top X = I_Q.$$

For class c , the classifier score is the best matching prototype score

$$s_c(h_i, X) = \max_{1 \leq j \leq q} h_i^\top x_{c,j}.$$

We use an additive margin softmax objective [60]. For the target class y_i , the score is shifted to $s_{y_i}(h_i, X) - m$ before multiplying all class scores by a logit scale γ . The optimization problem is

$$\min_{X \in \text{St}(512, Q)} \frac{1}{N} \sum_{i=1}^N \ell_{\text{CE}}(\gamma(s_1(h_i, X), \dots, s_{y_i}(h_i, X) - m, \dots, s_C(h_i, X)), y_i).$$

We use margin $m = 0.5$, and logit scale $\gamma = 64$. The Stiefel constraint enforces global orthogonality among all class prototypes. This gives a real supervised task for comparing intrinsic and extrinsic spectral updates on $\text{St}(512, Q)$, $Q = 400$.

Methods. We compare RGD, iMuon, and SPEL. RGD is normalized Riemannian gradient descent under the canonical embedded Stiefel geometry. iMuon is the spectral norm instance of our intrinsic LMO, using the closed form tangent decomposition from the main text. SPEL is the Stiefel specialization of manifold constrained steepest descent [68]. It forms a spectral steepest descent direction from the Riemannian gradient using extrinsic normalization, then retracts to the manifold. The updates are discussed in Section H.3. All methods use polar retraction. Learning rates are selected by validation accuracy from $\{0.02, 0.03, 0.05, 0.1\}$. Dataset. We use CIFAR-100 [33] with frozen ResNet-18 features [20]. Features are ℓ_2 -normalized before training the Stiefel classifier. Each seed uses 50 training examples, 10 validation examples, and 30 test examples per class. We train for 30 epochs with batch size 64, evaluation batch size 4096, and select the learning rate by validation accuracy.

Results. Table 17 reports test accuracy and test cross entropy (CE), the average negative log likelihood under the additive margin softmax classifier. iMuon has the best mean accuracy and the lowest CE. It improves over RGD by 1.1 percentage points and is slightly above SPEL. Figure 5 shows that iMuon and SPEL reduce the training objective faster than RGD, with iMuon slightly higher in mean test accuracy near the end of training.

APPENDIX P. ADDITIONAL MANIFOLDS AND METRICS

The LMO framework of Section 3 is stated for general Riemannian matrix manifolds. In the main text we instantiate it on four manifolds: three quotient manifolds $(\mathcal{M}_r, S_{++}^n, \text{Gr}(m, r))$ and the Stiefel manifold $\text{St}(m, r)$. Here we discuss broader applicability and several instructive boundary cases: (i) general conditions for closed-form solutions, (ii) the fixed-rank manifold from the *embedded* viewpoint, why the joint spectral-norm constraint on $\dot{X}$ does not decouple, and how the quotient formulation resolves this, (iii) the

TABLE 17. Stiefel subcenter prototype classification on frozen CIFAR-100 features. We train a $q = 4$ prototype per class additive margin classifier on $\text{St}(512, 400)$, select the learning rate by validation accuracy for each seed, and report mean test accuracy and test cross entropy. CE denotes average negative log likelihood. Higher accuracy and lower CE are better.

Method	Selected LR	Test acc.	Test CE
RGD	0.1, 0.1, 0.1	0.542 ± 0.013	1.853 ± 0.026
iMuon	0.03, 0.02, 0.02	0.553 ± 0.006	1.839 ± 0.024
SPEL	0.03, 0.03, 0.05	0.545 ± 0.012	1.869 ± 0.066

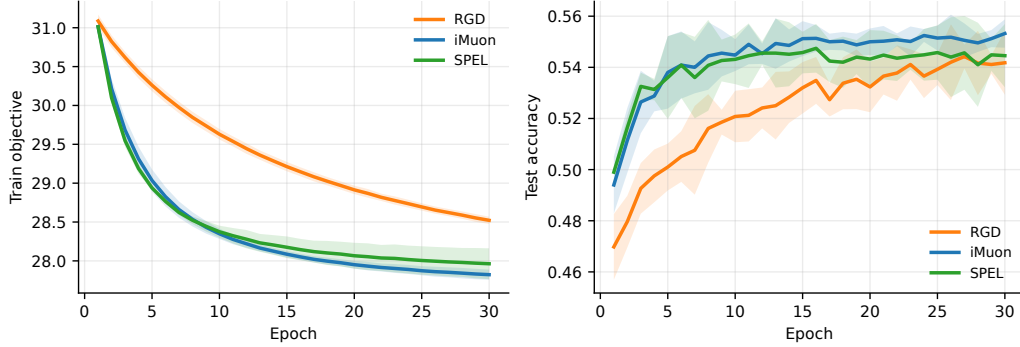


FIGURE 5. Stiefel subcenter prototype classification trajectories. The left panel reports the training objective and the right panel reports test accuracy over epochs. Curves show means over three seeds with standard deviation bands.

Stiefel manifold from the *embedded submanifold* viewpoint, why the naive approach fails and how the block decomposition of Section 4 resolves it, (iv) the spectrahedron (trace-one fixed-rank PSD manifold), a case where SV-invariance fails and the clean reduction breaks down, (v) the spectral sphere $\{X : \|X\|_2 = R\}$, another non-SV-invariant example, recently used in the SSO optimizer [65], and (vi) alternative SPD metrics (Log-Euclidean and Bures–Wasserstein).

P.1. General applicability. The LMO (4) is well-defined on *any* Riemannian manifold $\mathcal{M}$ with a metric operator G_x : it is simply a convex optimization over the tangent space. The two structural properties that enable closed-form solutions and symmetry preservation are:

- (1) **Left-right structure** (Proposition 3.1): the scaled map $\rho_g = G_x^{1/2} \circ D_x g \circ G_x^{-1/2}$ acts as $Z \mapsto Q_g Z R_g^\top$ for orthogonal Q_g, R_g . This ensures $\varphi(\rho_g Z) = \varphi(Z)$ for unitarily invariant φ .
- (2) **singular value invariance** (Proposition 3.3): the scaled tangent space $\mathcal{Z}_x$ is SV-invariant (Definition 3.2), so the tangent-space constraint is redundant and the LMO reduces to a standard Euclidean problem with closed-form solutions.

For *embedded submanifolds* (no quotient symmetry), the symmetry condition (1) is not required. Condition (2) alone suffices for closed-form solutions. If $\mathcal{Z}_x$ is *not* SV-invariant, the LMO remains a convex problem but requires a constrained solver rather than Ortho or $H/\|H\|_F$.

P.2. Fixed-rank manifold: embedded viewpoint. In Section 4 we work on the quotient $\mathcal{M}_r = (\mathbb{R}_*^{m \times r} \times \mathbb{R}_*^{r \times n})/\text{GL}(r)$ with the coupled metric (Section 4), which gives $G_x^{1/2} \neq I$ and decouples the LMO into independent $\dot{B}$ and $\dot{A}$ subproblems. Here we discuss the alternative *embedded* viewpoint, working directly on $\{W \in \mathbb{R}^{m \times n} : \text{rank}(W) = r\}$ with $G_x = I$, and explain why it does not yield the same clean decoupled solutions.

The embedded LMO.. On the embedded rank- r manifold, the tangent space at $X = BA$ is $T_X \mathcal{M}_r = \{\dot{B}A + B\dot{A} : \dot{B} \in \mathbb{R}^{m \times r}, \dot{A} \in \mathbb{R}^{r \times n}\}$. With $G_x = I$, the LMO becomes

$$\dot{X}^* = \arg \max_{\dot{X} \in T_X \mathcal{M}_r, \varphi(\dot{X}) \leq \tau} \langle \dot{X}, \nabla_X f \rangle.$$

For $\varphi = \|\cdot\|_2$, the unconstrained maximizer over all of $\mathbb{R}^{m \times n}$ is $\text{Ortho}(\nabla_X f)$, which has rank $\min(m, n)$ and generally does not lie in $T_X \mathcal{M}_r$ (a subspace of rank $\leq 2r$). Riemannion [10] addresses this by first computing a rank- $2r$ orthogonalization $\text{Ortho}_r(M_t)$ of the momentum (which lies in $T_X \mathcal{M}_r$), and then projecting the result back onto $T_X \mathcal{M}_r$. This tangent space projection changes the singular values, so the spectral-norm constraint is only approximately satisfied.

Failure of SV-invariance. The scaled tangent space $\mathcal{Z}_X = T_X \mathcal{M}_r = \{\dot{B}A + B\dot{A}\} \subset \mathbb{R}^{m \times n}$ is *not* singular value invariant: given $\dot{X} = U\Sigma V^\top \in T_X \mathcal{M}_r$, replacing Σ with a different non-negative diagonal D yields $UDV^\top$ which generally does not decompose as $\dot{B}A + B\dot{A}$ for any $(\dot{B}, \dot{A})$. Thus Proposition 3.3 does not apply, and the constrained LMO requires solving $\max_{\dot{X} \in T_X \mathcal{M}_r, \|\dot{X}\|_2 \leq \tau} \langle \dot{X}, \nabla_X f \rangle$, a constrained problem with no clean closed form.

Why the quotient formulation resolves this. On the quotient, each factor block lives in a full ambient space ($\mathbb{R}^{m \times r}$ for $\dot{B}$, $\mathbb{R}^{r \times n}$ for $\dot{A}$), which is trivially SV-invariant. The product-norm decoupling $\max(\varphi(Z_B), \varphi(Z_A)) \leq \tau$ then gives two independent LMOs, each with a clean closed form. The price is twofold: (a) the product norm is a relaxation of the joint norm $\varphi(\dot{X}) \leq \tau$ (specifically, $\|\dot{X}\|_2 \leq 2\tau$ rather than τ), and (b) the non-trivial metric $G_x^{1/2} \neq I$ must be inverted (the Gram-root inversion). But this price buys symmetry invariance (Proposition 3.1) and a condition-number-free bound $\|\xi^*\|_x^2 \leq 2r\tau^2$ (Section D.6).

Frobenius norm. For $\varphi = \|\cdot\|_F$, the embedded LMO does have a closed form: $\dot{X}^* \propto P_{T_X}(\nabla_X f)$, the tangent space projection of the gradient. This coincides with the quotient solution $\dot{X} = \dot{B}A + B\dot{A}$ for the Frobenius norm, since both recover the Riemannian gradient (up to normalization). The distinction between embedded and quotient viewpoints matters only for $\varphi \neq \|\cdot\|_F$.

P.3. Stiefel manifold: embedded viewpoint. In Section 4 we derived exact closed-form LMO solutions on $\text{St}(m, r)$ via the tangent space decomposition $T_X \text{St} = X \cdot \text{Skew}(r) \oplus X_\perp \cdot \mathbb{R}^{(m-r) \times r}$ with product-norm decoupling. Here we discuss the alternative *embedded* viewpoint, treating $T_X \text{St}$ as a single subspace of $\mathbb{R}^{m \times r}$, and explain why it does not yield clean closed-form solutions.

The embedded LMO.. Since $G_X = I$ (Euclidean metric), the LMO (4) on $\text{St}(m, r)$ is

$$\xi^* = \arg \max_{\xi \in T_X \text{St}, \varphi(\xi) \leq \tau} \langle \xi, \nabla_X f \rangle$$

as $g_x(\xi, \text{grad } f) = \langle \xi, \nabla_X f \rangle$. For $\varphi = \|\cdot\|_F$, the solution is the projected gradient $P_{T_X}(\nabla_X f)$ normalized to unit Frobenius norm, which is straightforward. For $\varphi = \|\cdot\|_2$, the unconstrained maximizer over all of $\mathbb{R}^{m \times r}$ is $\text{Ortho}(\nabla_X f)$, but this generally does *not* lie in $T_X \text{St} = \{\xi : X^\top \xi + \xi^\top X = 0\}$. A tangent space projection of $\text{Ortho}(\nabla_X f)$ is possible but changes the singular values, so the result is no longer the polar factor, and the spectral-norm constraint $\|\xi\|_2 \leq \tau$ is not tight in general.

Why the decomposition does not help the embedded (joint-norm) LMO.. One might hope that the tangent space decomposition $\xi = XA + X_\perp B$ could also solve the embedded LMO under the *joint* norm $\varphi(\xi) = \varphi\left(\begin{smallmatrix} A \\ B \end{smallmatrix}\right) \leq \tau$. However, this constraint couples A and B : for $\varphi = \|\cdot\|_2$, the unconstrained maximizer $\text{Ortho}\left(\begin{smallmatrix} S \\ N \end{smallmatrix}\right)$ produces a stacked matrix whose top $r \times r$ block is generally *not* skew-symmetric. Unlike the product-norm case $\max(\varphi(A), \varphi(B)) \leq \tau$ (which decouples), the joint constraint requires solving a constrained problem where the skew-symmetry of A interacts with the spectral-norm bound, no clean closed form exists. For $\varphi = \|\cdot\|_F$, the distinction vanishes: $\|\xi\|_F^2 = \|A\|_F^2 + \|B\|_F^2$ and the unconstrained solution $\xi^* \propto P_{T_X}(\nabla_X f) = XS + X_\perp N$ automatically has $A = S \in \text{Skew}(r)$, so both the joint and product norms give the same answer. Failure of SV-invariance. The root cause is that $T_X \text{St} \subset \mathbb{R}^{m \times r}$ is *not* singular value invariant (Definition 3.2): given $\xi = U\Sigma V^\top \in T_X \text{St}$, replacing Σ with a different non-negative diagonal D yields $UDV^\top$ which generally violates $X^\top(UDV^\top) + (UDV^\top)^\top X = 0$. Thus Proposition 3.3 does not apply to the full tangent space, and the constrained LMO does not reduce to the unconstrained problem. Solving the constrained problem directly requires a Lagrange multiplier for the symmetry constraint, which does not admit a clean closed form.

Resolution via block decomposition. The product-norm decomposition of Section 4 resolves this by splitting $T_X \text{St} = X \cdot \text{Skew}(r) \oplus X_\perp \cdot \mathbb{R}^{(m-r) \times r}$ and applying the LMO independently to each block. The normal block $\mathbb{R}^{(m-r) \times r}$ is trivially SV-invariant, and the skew-symmetric block $\text{Skew}(r)$ is SV-invariant under paired singular value replacement (skew-symmetric matrices have equal-pair singular values, and the standard closed-form solutions of Proposition 3.3 respect this pairing).

P.4. Spectrahedron (trace-one fixed-rank PSD manifold).

Manifold and quotient model. The fixed-rank trace-one PSD manifold [28] is

$$\mathcal{M}_{r,1} := \{X \in \mathbb{R}^{m \times m} : X \succeq 0, \text{rank}(X) = r, \text{tr}(X) = 1\}.$$

Writing $X = YY^\top$ with $Y \in \mathbb{R}_*^{m \times r}$ and $\|Y\|_F = 1$, one obtains the quotient representation

$$\mathcal{M}_{r,1} \cong \overline{\mathcal{M}}_{r,1}/O(r), \quad \overline{\mathcal{M}}_{r,1} := \{Y \in \mathbb{R}_*^{m \times r} : \|Y\|_F = 1\},$$

where Y and YQ represent the same X for every $Q \in O(r)$. This is the natural PSD analogue of the fixed-rank quotient [28], with the additional trace-one constraint enforced by $\|Y\|_F = 1$.

Tangent and horizontal spaces. Let $R := Y^\top Y \succ 0$ and write a compact QR factorization $Y = UR^{1/2}$ with $U \in \text{St}(m, r)$. The tangent space of the total space is

$$T_Y \overline{\mathcal{M}}_{r,1} = \{\eta \in \mathbb{R}^{m \times r} : \text{tr}(Y^\top \eta) = 0\}.$$

The vertical space is $\mathcal{V}_Y = \{Y\Omega : \Omega \in \text{Skew}(r)\}$. With the Euclidean metric on the total space, the horizontal space is therefore

$$\mathcal{H}_Y = \{\eta \in T_Y \overline{\mathcal{M}}_{r,1} : Y^\top \eta \text{ is symmetric}\} = \{\eta : Y^\top \eta \in \text{Sym}_0(r)\},$$

where $\text{Sym}_0(r) := \{S \in \text{Sym}(r) : \text{tr}(S) = 0\}$. Every horizontal vector decomposes uniquely as

$$\eta = UR^{-1/2}S + U_\perp B, \quad S \in \text{Sym}_0(r), \quad B \in \mathbb{R}^{(m-r) \times r},$$

with $U_\perp$ an orthogonal complement of U .

Metric and scaling. With the quotient metric induced by the Euclidean metric upstairs,

$$g_Y(\eta_1, \eta_2) = \text{tr}(\eta_1^\top \eta_2) = \text{tr}(R^{-1}S_1 R^{-1}S_2) + \text{tr}(B_1^\top B_2).$$

Equivalently, after the change of variables

$$Z_S := R^{-1/2}S R^{-1/2}, \quad Z_B := B,$$

the norm becomes Euclidean:

$$\|\eta\|_Y^2 = \|Z_S\|_F^2 + \|Z_B\|_F^2.$$

Thus the scaled tangent space is not the full product space $\text{Sym}(r) \times \mathbb{R}^{(m-r) \times r}$, but rather

$$\mathcal{Z}_Y = \{(Z_S, Z_B) : R^{1/2}Z_S R^{1/2} \in \text{Sym}_0(r), Z_B \in \mathbb{R}^{(m-r) \times r}\}.$$

Why the clean reduction fails. The key obstruction is the trace-free constraint on the symmetric block. Unless R is a scalar multiple of the identity, the set $\{Z_S : R^{1/2}Z_S R^{1/2} \in \text{Sym}_0(r)\}$ is not singular value invariant in the sense of Definition 3.2. Indeed, if $R = I$ and $Z_S = \text{diag}(1, -1, 0, \dots, 0)$, then $Z_S \in \mathcal{Z}_Y$ (symmetric, trace zero). Its SVD is $Z_S = U\Sigma V^\top$ with $\Sigma = \text{diag}(1, 1, 0, \dots, 0)$ and $V = \text{diag}(1, -1, 1, \dots, 1)$. Choosing $D = \text{diag}(1, 0, \dots, 0)$ gives $UDV^\top = \text{diag}(1, 0, \dots, 0)$, which has trace 1 $\neq 0$ and thus $\notin \mathcal{Z}_Y$. Therefore Proposition 3.3 does not apply: the unconstrained Euclidean LMO over all matrices no longer coincides with the tangent-constrained problem.

Consequences for the three norms. For the Frobenius norm, the LMO remains clean because orthogonal projection onto the horizontal space is explicit. If $\nabla_X f \in \text{Sym}(m)$ denotes the Euclidean gradient with respect to $X = YY^\top$, the lifted objective is

$$\langle \eta, \nabla_Y f \rangle, \quad \nabla_Y f = 2(\nabla_X f)Y,$$

and the Frobenius-norm solution is simply the normalized horizontal projection of $\nabla_Y f$. This is the quotient analogue of standard Riemannian steepest descent.

For the spectral and nuclear norms, however, no analogue of the fixed-rank/SPD/Grassmann closed forms emerges. The problem is still a convex LMO on a linear space, so it is well posed and can be solved numerically, but the maximizer is constrained by the coupled condition $R^{1/2}Z_S R^{1/2} \in \text{Sym}_0(r)$ rather than by singular values alone. As a result, one should not expect a simple formula based only on $\text{Ortho}(H)$, a matrix sign, or a rank-1 truncation of the scaled gradient.

Takeaway. The trace-one fixed-rank PSD manifold fits the intrinsic-LMO framework and retains the explicit Frobenius instance, but the coupling $R^{1/2}Z_S R^{1/2} \in \text{Sym}_0(r)$ breaks singular value invariance, so the spectral and nuclear cases no longer admit the closed forms of Section 4.

P.5. Spectral sphere. The spectral sphere $\mathcal{S}_R := \{X \in \mathbb{R}^{m \times n} : \|X\|_2 = R\}$ constrains only the *largest* singular value of X to equal R . The remaining singular values are free to take any values in $[0, R]$. When the top singular value is simple (a generic condition), the manifold is smooth with tangent space

$$T_X \mathcal{S}_R = \{\xi \in \mathbb{R}^{m \times n} : \langle u_1 v_1^\top, \xi \rangle = 0\},$$

where (u_1, v_1) are the leading left and right singular vectors of X , and the gradient of the spectral norm is $\nabla_X \|X\|_2 = u_1 v_1^\top$.

Failure of SV-invariance. The tangent space is the hyperplane $\{\xi : \langle u_1 v_1^\top, \xi \rangle = 0\}$, defined by an inner product with a *data-dependent* rank-1 matrix. This is not SV-invariant: given $\xi = U \Sigma V^\top \in T_X \mathcal{S}_R$, replacing Σ with a different non-negative diagonal D yields $UDV^\top$ whose inner product $\langle u_1 v_1^\top, UDV^\top \rangle = \sum_i d_i (u_1^\top u_i)(v_i^\top v_1)$ generally differs from zero. Thus Proposition 3.3 does not apply, and the spectral-norm LMO $\max\{\langle \nabla_X f, \xi \rangle : \xi \in T_X \mathcal{S}_R, \|\xi\|_2 \leq \tau\}$ has no closed form. The obstruction is structurally identical to that of the spectrahedron (Section P.4): a linear constraint on the tangent space that is incompatible with singular value replacement.

Connection to SSO.. Although SV-invariance fails, the LMO on $\mathcal{S}_R$ remains a well-posed convex problem, it is the maximization of a linear functional over the intersection of a spectral-norm ball and a hyperplane. This is precisely the problem solved by the Spectral Sphere Optimizer (SSO) [65], which performs steepest descent on $\mathcal{S}_R$ under the spectral norm for μ P-aligned [67] LLM training. In other words, SSO is an instance of our framework (4) with $G_X = I$ and $\varphi = \|\cdot\|_2$ on the manifold $\mathcal{S}_R$, it solves exactly the same intrinsic LMO, but without the benefit of a closed-form solution. The single hyperplane constraint requires one Lagrange multiplier λ , and SSO solves for it via bisection on the monotone function

$$h(\lambda) := \langle u_1 v_1^\top, \text{Ortho}(\nabla_X f + \lambda u_1 v_1^\top) \rangle = 0, \quad \xi^*(\lambda) = \text{Ortho}(\nabla_X f + \lambda u_1 v_1^\top).$$

Each bisection step requires one Ortho evaluation and approximately 5–7 bisection steps are needed in practice. If the tangent constraint $\langle u_1 v_1^\top, \xi \rangle = 0$ were dropped (i.e., no weight-norm preservation), the solution would reduce to $\xi^* = \text{Ortho}(\nabla_X f)$, standard Euclidean Muon. Thus the bisection cost is the computational price of the non-SV-invariant tangent constraint, which arises from simultaneously constraining both the update norm and the weight norm.

Takeaway. The spectral sphere illustrates that our framework applies beyond manifolds with SV-invariant tangent spaces: the LMO (4) is always well-posed, and SV-invariance determines only whether the solution is available in closed form or requires a solver. On the four manifolds of Section 4, SV-invariance holds and the LMO reduces to Ortho, sign, or rank-1 truncation. On the spectral sphere (and the spectrahedron of Section P.4), a single linear constraint breaks SV-invariance, but a one-dimensional bisection restores tractability at the cost of a few additional Ortho evaluations per step. The Frobenius instance ($\varphi = \|\cdot\|_F$) remains closed-form in all cases, as the LMO reduces to projecting the gradient onto the tangent space and normalizing.

P.6. Alternative SPD metrics: Log-Euclidean and Bures–Wasserstein. In Section 4 we use the affine-invariant (AI) metric on S_{++}^n . Here we analyze how two other common SPD metrics interact with the two structural properties of the framework: left-right structure (Proposition 3.1, enabling symmetry invariance) and SV-invariance of the scaled tangent space (Proposition 3.3, enabling closed-form reduction). All three metrics share the same tangent space $T_X S_{++}^n = \text{Sym}(n)$, which is SV-invariant (Section E). Thus Proposition 3.3 applies to all three, the LMO always reduces to an unconstrained Euclidean problem with the standard closed forms (sign(H), $H/\|H\|_F$, or rank-1 truncation). The metrics differ only in how the two propositions’ *other* aspects, symmetry group size and the explicitness of the scale/invert steps, play out.

Affine-invariant (AI) metric (recap). $G_X^{\text{AI}}(\xi) = X^{-1} \xi X^{-1}$, so $G_X^{1/2}(\xi) = X^{-1/2} \xi X^{-1/2}$ (left-right form with $P = Q = X^{-1}$).

- *Proposition 3.1 (left-right structure)*: holds with $\text{GL}(n)$ -invariance.
- *Proposition 3.3 (SV-invariance)*: $\mathcal{Z}_X = \text{Sym}(n)$, which is SV-invariant.
- *Scale/invert*: coordinate-free sandwich formulas $H = X^{1/2}(\nabla_X f)X^{1/2}$ and $\xi^* = X^{1/2}Z^*X^{1/2}$.

This is the ideal case: both propositions apply with the maximal symmetry group, and all three pipeline steps (scale, solve, invert) have explicit matrix-coordinate formulas.

Log-Euclidean metric. The Log-Euclidean metric [4] uses $\Phi : X \mapsto \log X$ as an isometry from (S_{++}^n, g^{LE}) to $(\text{Sym}(n), \|\cdot\|_F)$, with $g_X^{LE}(\xi, \eta) = \text{tr}(D_X \log(\xi)^\top D_X \log(\eta))$. In the log-coordinates $Y = \log X$, the metric is flat Euclidean ($G_Y = I$).

- *Proposition 3.1:* trivially satisfied ($G_Y = I$), but only $O(n)$ -invariant (under $X \mapsto QXQ^\top$ for orthogonal Q), not $\text{GL}(n)$ -invariant.
- *Proposition 3.3:* $\mathcal{Z}_Y = \text{Sym}(n)$, SV-invariant. Since $G_Y = I$, the reduction is trivial: no scaling or inversion needed.
- *Scale/invert:* identity (nothing to compute).

The spectral-norm LMO gives $\Delta Y^* = \tau \text{Ortho}(\nabla_Y f)$ in log-space, followed by $X_+ = \exp(Y - \eta \Delta Y^*)$. This is simply standard Euclidean Muon applied to $\log X$; the framework adds nothing beyond reparametrization. Both propositions hold, but vacuously, no non-trivial preconditioning or symmetry preservation emerges.

Bures–Wasserstein (BW) metric. The BW metric [9, 38] is $g_X^{BW}(\xi, \zeta) = \frac{1}{2} \text{tr}(L_\xi \zeta)$, where L_ξ is the unique symmetric solution of the Lyapunov equation $XL_\xi + L_\xi X = \xi$. The metric operator is $G_X^{BW} = \frac{1}{2} \mathcal{L}_X$, where $\mathcal{L}_X(\xi) := L_\xi$. In the eigenbasis $X = U\Lambda U^\top$ with eigenvalues $\lambda_1, \dots, \lambda_n$, entry (i, j) of $G_X^{BW}(\xi)$ scales as $\xi_{ij}/(\lambda_i + \lambda_j)$.

- *Proposition 3.1 (left-right structure):* the left-right assumption **fails**, the BW scaling $1/(\lambda_i + \lambda_j)$ does not factor as $f(\lambda_i) \cdot g(\lambda_j)$, so $G_X^{1/2}$ cannot be written as $P\xi Q^\top$ and the proof of Proposition 3.1 does not apply directly. However, the *conclusion* of Proposition 3.1 still holds for the symmetry group $O(n)$ via a direct argument: under $X \mapsto QXQ^\top$ with $Q \in O(n)$, the eigenbasis rotates $U \mapsto QU$ while eigenvalues λ_i are unchanged, so $G_X^{1/2}(Q\xi Q^\top) = Q G_X^{1/2}(\xi) Q^\top$. Since left- and right-multiplication by the same orthogonal Q preserves all unitarily invariant norms, $\varphi(G_X^{1/2} \tilde{\xi}) = \varphi(G_X^{1/2} \xi)$. Thus the intrinsic constraint is $O(n)$ -invariant, but not $\text{GL}(n)$ -invariant (because $G_X^{1/2}(N\xi N^\top) \neq N G_X^{1/2}(\xi) N^\top$ for general invertible N , since the Hadamard weights $1/\sqrt{\lambda_i + \lambda_j}$ change when the eigenvalues of X change under non-orthogonal congruence).
- *Proposition 3.3 (SV-invariance):* **holds**. The scaled tangent space $\mathcal{Z}_X = \{(G_X^{BW})^{1/2} \xi : \xi \in \text{Sym}(n)\} = \text{Sym}(n)$ (since $(G_X^{BW})^{1/2}$ is an invertible self-adjoint operator on $\text{Sym}(n)$), which is SV-invariant. Thus the LMO reduces to the unconstrained problem $Z^* = \tau \text{sign}(H)$ for the spectral norm, etc.
- *Scale/invert:* closed-form but requires the eigenbasis. In the eigenbasis $X = U\Lambda U^\top$, the operators $(G_X^{BW})^{1/2}$ and $(G_X^{BW})^{-1/2}$ act entrywise on $\hat{\xi} := U^\top \xi U$ as

$$[(G_X^{BW})^{1/2}(\xi)]_{ij} = \frac{\hat{\xi}_{ij}}{\sqrt{2(\lambda_i + \lambda_j)}}, \quad [(G_X^{BW})^{-1/2}(Z)]_{ij} = \sqrt{2(\lambda_i + \lambda_j)} \hat{Z}_{ij},$$

where $\hat{Z} = U^\top Z U$. Thus the full pipeline is: (i) diagonalize $X = U\Lambda U^\top$ once ($O(n^3)$); (ii) scale: $H_{ij} = \hat{G}_{ij}/\sqrt{2(\lambda_i + \lambda_j)}$ where $\hat{G} = U^\top (\text{grad } f) U$; (iii) solve: $Z^* = \tau \text{sign}(H)$ via eigendecomposition of H ; (iv) invert: $\hat{\xi}_{ij}^* = \sqrt{2(\lambda_i + \lambda_j)} (U^\top Z^* U)_{ij}$, then $\xi^* = U \hat{\xi}^* U^\top$. All steps are $O(n^3)$, the same asymptotic cost as the AI case.

The key difference from AI is therefore not computational cost but geometric: (a) the inversion formula is a Hadamard (entrywise) rescaling in the eigenbasis rather than a coordinate-free matrix sandwich $X^{1/2}(\cdot)X^{1/2}$, and (b) the symmetry group is $O(n)$ rather than $\text{GL}(n)$, so the intrinsic constraint is invariant under a smaller group.

Summary. We summarize the status of the two key structural properties across the three metrics:

	Affine-invariant	Log-Euclidean	Bures–Wasserstein
Symmetry group	$\text{GL}(n)$	$O(n)$	$O(n)$
Proposition 3.1 (left-right)	✓	trivial ($G = I$)	× (but $O(n)$ -inv. holds)
Proposition 3.3 (SV-inv.)	✓	✓	✓
Scale/invert formula	$X^{1/2}(\cdot)X^{1/2}$	identity	eigenbasis Hadamard
Cost	$O(n^3)$	$O(n^3)$	$O(n^3)$

All three metrics admit closed-form LMO solutions via Proposition 3.3 at the same asymptotic cost. The affine-invariant metric is distinguished by its $GL(n)$ -invariance and coordinate-free sandwich formulas. The Log-Euclidean metric reduces to standard Euclidean Muon in log-coordinates (no new spectral structure). The BW metric provides non-trivial preconditioning and closed-form solutions via eigenbasis Hadamard scaling; the intrinsic constraint is $O(n)$ -invariant (by direct conjugation, not via Proposition 3.1) but not $GL(n)$ -invariant, and the scale/invert formulas require the eigenbasis rather than coordinate-free sandwich products.